MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS, COUNTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, high speed, CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines and lead finisnes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	BCD decade counter, asynchronous reset, synchronous load
02	4-bit binary counter, asynchronous reset, synchronous load
03	BCD decade counter, synchronous reset and load
04	4-bit binary counter, synchronous reset and load
05	Synchronous 4-bit binary up/down counter, asynchronous load
06	Synchronous BCD decade up/down counter, asynchronous reset and load
07	Synchronous 4-bit binary up/down counter, asynchronous reset and load
08	Synchronous dual 4-bit decade counter, asynchronous reset
09	Synchronous dual 4-bit binary counter, asynchronous reset
10	Decade counter/divider, 10 decoded outputs, asynchronous reset
11	14-stage ripple-carry binary counter, asynchronous reset
12	7-stage ripple-carry binary counter, asynchronous reset
13	12-stage ripple-carry binary counter, asynchronous reset

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package (configuration one only)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package (configuration one only)
X	C-1 (16-terminal, .300" x .300"), square chip carrier package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage (V_{CC})	-0.5 V dc to +7.0 V dc
DC input voltage (V_{IN})	-0.5 V dc to V_{CC} +0.5 V dc
DC output voltage (V_{OUT})	-0.5 V dc to V_{CC} +0.5 V dc
Clamp diode current (I_{IK}, I_{OK})	±20 mA
DC output current per pin (I_{OUT})	±25 mA
DC V_{CC} or GND current per pin (I_{OC})	±50 mA
Storage temperature range (T_{STG})	-65°C to +150°C
Maximum power dissipation (P_D)	300 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases C, D, E, and F	(See MIL-M-38510, appendix C)
Case X	53°C/W
Case 2	60°C/W
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Input low (V_{IL}) maximum voltage	0.3 V at V_{CC} = 2.0 V 0.9 V at V_{CC} = 4.5 V 1.2 V at V_{CC} = 6.0 V
Input high (V_{IH}) minimum voltage	1.5 V at V_{CC} = 3.0 V 3.15 V at V_{CC} = 4.5 V 4.2 V at V_{CC} = 6.0 V

All devices:

Supply voltage (V_{CC})	2.0 V dc to 6.0 V dc
Output voltage	0 V dc to V_{CC}
Operating temperature	-55°C to +125°C
Input rise and fall times (t_r, t_f) maximum:	
V_{CC} = 2.0 V	1,000 ns
V_{CC} = 4.5 V	500 ns
V_{CC} = 6.0 V	400 ns
Setup time $\overline{\text{load}}$ before clock (t_{su}):	
Device types 01-04 ($\overline{\text{load}}$ to CLK)	45 ns (minimum)
Device type 05 (U/D to CLK)	62 ns (minimum)
Device types 03 and 04 ($\overline{\text{CLR}}$ to CLK)	45 ns (minimum)
Device types 01-04 (ENP, ENT)	60 ns (minimum)
Device type 05 (ENG to CLK)	62 ns (minimum)
Device type 10 (CE to CLK)	23 ns (minimum)
Device types 06 and 07 (A, B, C, and D to $\overline{\text{load}}$)	33 ns (minimum)
Device type 05 (A, B, C, and D to $\overline{\text{load}}$)	45 ns (minimum)
Hold time (t_H):	
Device types 01-04 (CLK to A, B, C, and D)	15 ns (minimum)
Device types 01-04 (CLK to $\overline{\text{load}}$)	8 ns (minimum)
Device types 03 and 04 (CLK to $\overline{\text{CLR}}$)	8 ns (minimum)
Device type 05 (CLK to U/D)	8 ns (minimum)
Device types 05-07 ($\overline{\text{load}}$ to A, B, C, and D)	8 ns (minimum)
Device type 10 (CLK to $\overline{\text{CE}}$)	23 ns (minimum)
Device type 05 (CLK to ENG)	8 ns (minimum)
Clock pulse width ($t_{p(clock)}$):	
Device types 01-04, 10	30 ns (minimum)
Device type 05	38 ns (minimum)
Device type 08	26 ns (minimum)
Device type 09	24 ns (minimum)
Device types 11-13	27 ns (minimum)
Device types 06-07	45 ns (minimum)
Reset pulse width ($t_{p(clear)}$):	
Device types 01, 02, 10-13	24 ns (minimum)
Device type 06	78 ns (minimum)
Device type 07	39 ns (minimum)
Device types 08 and 09	38 ns (minimum)
Clock enable pulse widtn ($t_{p(CE)}$):	
Device type 10	30 ns (minimum)

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Load pulse width ($t_{p(load)}$):	
Device types 05 and 07	30 ns (minimum)
Device type 30	78 ns (minimum)
t_{rem} (load, $\overline{CLR}$ to CLK):	
Device types 01 and 02	30 ns (minimum)
Device type 05	45 ns (minimum)
Device types 06 and 07	33 ns (minimum)
Device type 08	23 ns (minimum)
Device type 09	15 ns (minimum)
Device types 10-13	30 ns (minimum)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figures 1 and 2.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

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3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 40 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.10 through 3.1.14 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static test (test condition A) using circuit shown on figure 4, or equivalent. Ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 5, or equivalent. Ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

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d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I and C_C measurements) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and GND at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady-state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady-state life test is used, the circuit on figure 5, or equivalent, shall be used.

b. Electrostatic discharge sensitivity (ESDS) testing shall be performed in accordance with MIL-STD-883, method 3015. The option to categorize devices as ESD sensitive without performing the test is not allowed. Device types categorized as ESD sensitive shall be further tested using method 3015 modified as follows:

(1) Test method 3015, table I pin combinations 4(V+(A) to common (B)) and 5(V+(B) to common (A)) shall be deleted.

(2) The test sequence specified in method 3015 shall be repeated an additional four times instead of two.

(3) The category A limits specified on figure 3015-3, ESD sensitivity category, shall be 1,000 V to 2,000 V.

c. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
High level output voltage	V_{OH1} 2/	V_{IH} = 1.5 V V_{IL} = 0.3 V I_{OH} = -20 µA	All	2.0 V	1.95		V
	V_{OH2} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -20 µA	All	4.5 V	4.45		V
	V_{OH3}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -20 µA	All	6.0 V	5.95		V
	V_{OH4} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -4.0 mA	All	4.5 V	3.7		V
	V_{OH5}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -5.2 mA	All	6.0 V	5.2		V
Low level output voltage	V_{OL1} 2/	V_{IH} = 1.5 V V_{IL} = 0.3 V I_{OH} = 20 µA	All	2.0 V		0.05	V
	V_{OL2} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = 20 µA	All	4.5 V		0.05	V
	V_{OL3}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = 20 µA	All	6.0 V		0.05	V
	V_{OL4} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = 4.0 mA	All	4.5 V		0.4	V
	V_{OL5}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = 5.2 mA	All	6.0 V		0.4	V
Positive input clamp voltage	V_{IC^+}	V_{CC} = GND I_{IN} = 1 mA T_C = +25°C	All			1.5	V
Negative input clamp voltage	V_{IC^-}	V_{CC} = open I_{IN} = -1 mA T_C = +25°C	All			-1.5	V
Input current low	I_{IL}	V_{IN} = GND	All	6.0 V		-0.1	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
Input current high	I_{IH}	$V_{IN} = V_{CC}$	All	6.0 V		0.1	µA
Short circuit output current	I_{OS1} 2/	V_O = GND, V_I = GND	All	2.0 V	-2	-50	mA
	I_{OS2} 2/		All	4.5 V	-15	-150	
	I_{OS3} 2/		All	6.0 V	-25	-180	
	I_{OS4}		All	4.0 V	-10	-120	
Supply current quiescent	I_{CCH} I_{CCL}	V_I = 6.0 V, GND	All	6.0 V		20	µA
Input capacitance	C_{IN}	$T_C = +25°C$	01-07			10	pF
Clock, control capacitance	C_C		All			15	pF
Device equivalent capacitance	C_{PD} 2/	No load, $T_A = +25°C$	01-04	4.5 V		90	pF
			05-07	4.5 V		100	pF
			08,09	4.5 V		55	pF
			10	4.5 V		110	pF
			11,13	4.5 V		88	pF
			12	4.5 V		55	pF
Propagation delay clock to outputs	t_{PLH1} t_{PHL1}	C_L = 50 pF ±10%	01-04	4.5 V	8	48	ns
CLK to Q			05	4.5 V	7	52	ns
U or D to Q			06,07	4.5 V	11	64	ns
$\overline{CLK}$ A to QA			08	4.5 V	4	34	ns
$\overline{CLK}$ A to QC			08	4.5 V	9	68	ns
$\overline{CLK}$ B to QB			08	4.5 V	4	36	ns
$\overline{CLK}$ B to QC			08	4.5 V	5	49	ns
$\overline{CLK}$ B to QD			08	4.5 V	4	36	ns
CLK to QA			09	4.5 V	4	32	ns
CLK to QB			09	4.5 V	6	45	ns
CLK to QC			09	4.5 V	8	56	ns
CLK to QD			09	4.5 V	9	73	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
Propagation delay CLK to N - carry out	t_{PHL1} t_{PLH1}	C_L = 50 pF ±10%	10	4.5 V	6	54	ns
			11,12,13	4.5 V	5	49	ns
CLK to Q_N			12,13		10	98	ns
			12,13		15	147	ns
			11,12,13		20	196	ns
					25	245	ns
					30	294	ns
					35	343	ns
			11,13		40	392	ns
					45	441	ns
					50	490	ns
					55	539	ns
					60	588	ns
			11		65	637	ns
			11		70	686	ns
Propagation delay clock to carry out	t_{PLH2} t_{PHL2}		01-04	4.5 V	8	50	ns
CLK to RCO			05	4.5 V	5	35	ns
U/D to RCO			06,07	4.5 V	4	39	ns
Propagation delay CLK to min/max	t_{PLH3} t_{PHL3}		05	4.5 V	8	60	ns
Propagation delay ENT to RCO	t_{PLH4} t_{PHL4}		01-04	4.5 V	7	46	ns
$\overline{CE}$ to carry-out			10	4.5 V	6	54	ns
GNG to RCO			05	4.5 V	6	47	ns
Propagation delay $\overline{Load}$ to Q	t_{PLH5} t_{PHL5}		05	4.5 V	8	68	ns
			06,07	4.5 V	12	68	ns
Propagation delay A, B, C, and D to Q	t_{PLH6} t_{PHL6}		05	4.5 V	7	56	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
Propagation delay U/D to min/max	t_{PLH7} t_{PHL7}	C_L = 50 pF ±10%	05	4.5 V	6	44	ns
Propagation delay U/D to RCO	t_{PLH8} t_{PHL8}		05	4.5 V	8	54	ns
Propagation delay $\overline{CLR}$ to Q	t_{PHL9}		01,02	4.5 V	8	53	ns
			06,07	4.5 V	11	62	ns
Propagation delay CLR to Q_N	t_{PHL9}		08,09	4.5 V	5	39	ns
			11-13	4.5 V	6	56	ns
			12	4.5 V	5	49	ns
Propagation delay CLR to N - carry out	t_{PHL9} t_{PLH9}		10	4.5 V	6	54	ns
Propagation delay $\overline{CLR}$ to RCO	t_{PHL10}		01,02	4.5 V	8	51	ns
Propagation delay Qn to Qn+1	t_{PHL11} t_{PLH11}		11-13	4.5 V	4	29	ns
Output transition time	t_{TLH} t_{THL}		All	4.5 V	3	20	ns
Maximum frequency test	f_{MAX}	C_L = 50 pF ±10% 50% duty cycle	01-04, 09	4.5 V	21		MHz
			05	4.5 V	17		MHz
			06,07	4.5 V	14		MHz
			08	4.5 V	23		MHz
			10	4.5 V	17		MHz
			11-13	4.5 V	21		MHz

1/ Complete terminal conditions shall be as specified in table III.
2/ Guaranteed but not tested.
3/ Tested at V_{CC} = 4.5 V at +125°C for sample testing and V_{CC} = 4.5 V at +25°C for screening. Guaranteed at other V_{CC} voltages and temperatures, see table IA and IB as appropriate and the exception in 4.4.1d.
4/ See the formulae for determining switching times and maximum frequencies shown in tables IA and IB respectively.

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TABLE IA. Calculated dynamic figures at -55/+25 ambient temperature (°C).

V_{CC}	T_A = (°C)	
	+125	-55/+25
2.0 V 1/	5.00 X	5.00 Y
4.5 V	X = 1	0.75 X = Y
6.0 V 1/	0.85 X	0.85 Y

Normalized numbers
(+125°C equals 1)

1/ The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according 5/4.

TABLE IB. Calculated f_{MAX} figures at -55/+25 ambient temperature (°C).

V_{CC}	T_A = (°C)	
	+125	-55/+25
2.0 V 1/	0.2 X	0.2 Y
4.5 V	X = 1	1.33 X = Y
6.0 V 1/	1.18 X	1.18 Y

Normalized numbers
(+125°C equals 1)

1/ The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according 5/4.

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TABLE II. Burn-in and electrical test requirements.

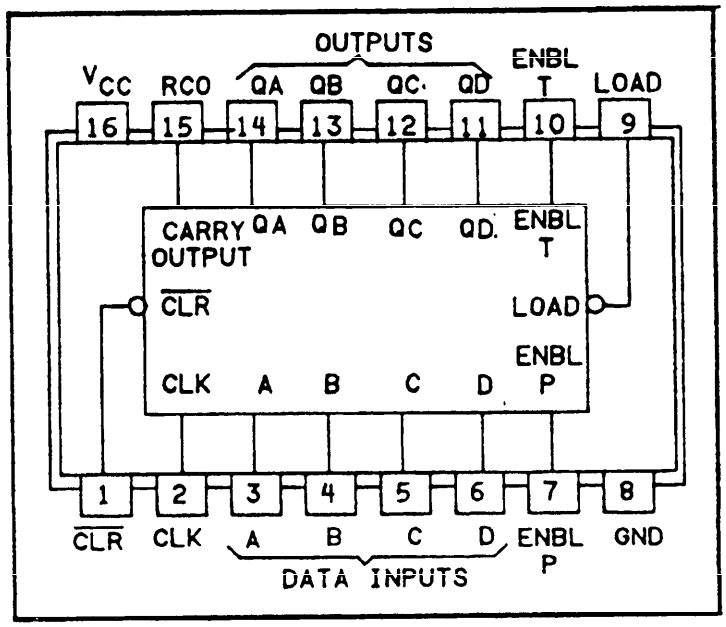
Line no.	Applicable tests and MIL-STD-883 test method	Class S device 1/				Class B device 1/			
		Reference paragraph	Table III subgroups 2/	Table IV delta limits 3/	Test circuit figure	Reference paragraph	Table III subgroups 2/	Table IV delta limits 3/	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2	Required		3		Not required		
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b 4.5.2	Required		3	4.2b 4.5.2	4/ Required		3
5	Same as line 1	4.2d	1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2	Required				Not required		
7	Same as line 1	4.2d	1	Δ					
8	Final electrical parameters (method 5004)		1*,2,3, 7,8,9				1*,2,7, 9 4/		
9	Group A test requirements (method 5005)	4.4.1	1,2,3,4, 7,8,9,10, 11			4.4.1	1,2,3,4, 7,8,9,10, 11		
10	Group B end-point electrical parameters (method 5005)	4.4.2	†1,2,3,7, 8,9,10,11 5/	Δ			†1 5/		
11	Group C end-point electrical parameters (method 5005)					4.4.3	1, 2	Δ	
12	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1, 2		

1/ Blank spaces indicate tests are not applicable.
2/ * indicates PDA applies to subgroup 1 (see 4.2.1c).
3/ Δ indicates delta limit shall be required only on table III, subgroup 1, where specified, and the delta values shall be computed with references to the previous interim electrical parameters (line 1).
4/ The device manufacturer may at his option, either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).
5/ † indicates also applies to electrostatic discharge sensitivity tests.

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Device types 01, 02, 03, and 04

Cases E, F, and X



Device types 01, 02, 03, and 04

Case 2

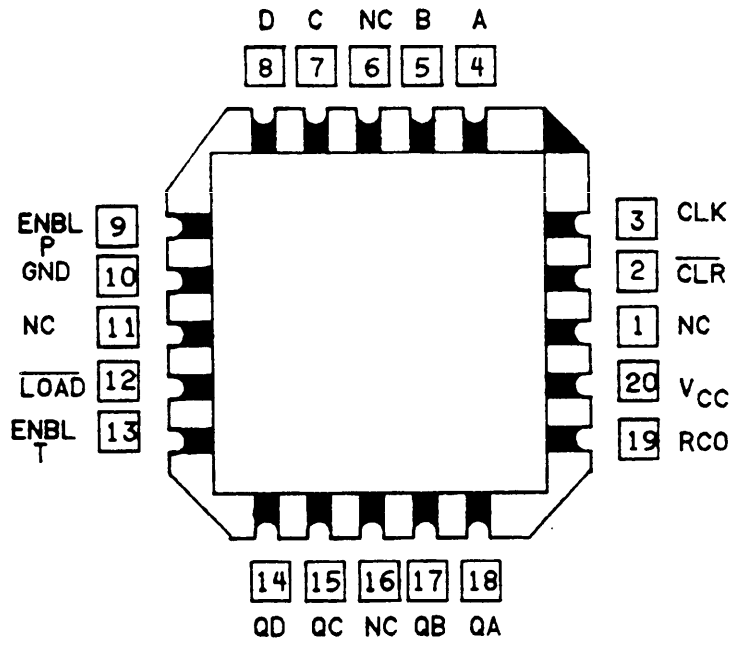
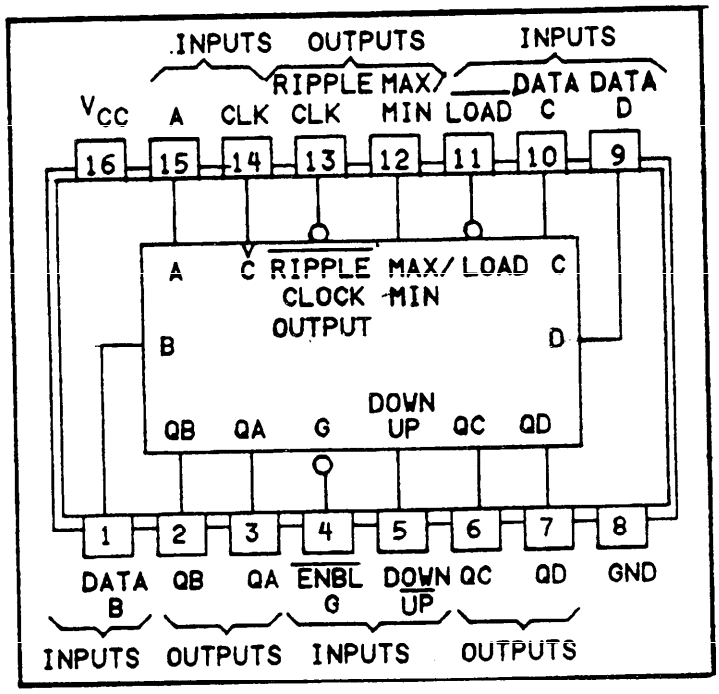


FIGURE 1. Terminal connections.

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Device type 05

Cases E, F, and X



Device type 05

Case 2

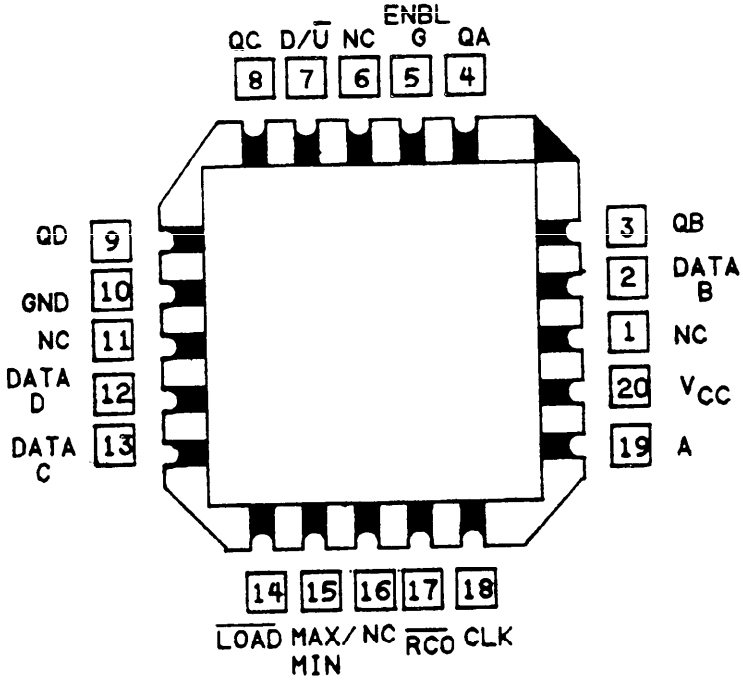
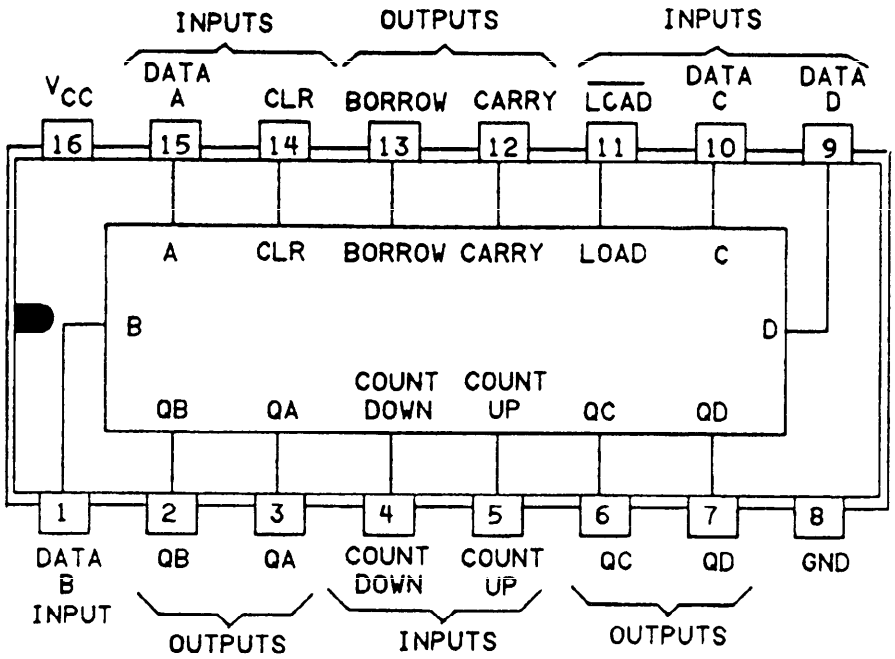


FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device types 06 and 07

Cases E, F, and X



Device types 06 and 07

Case 2

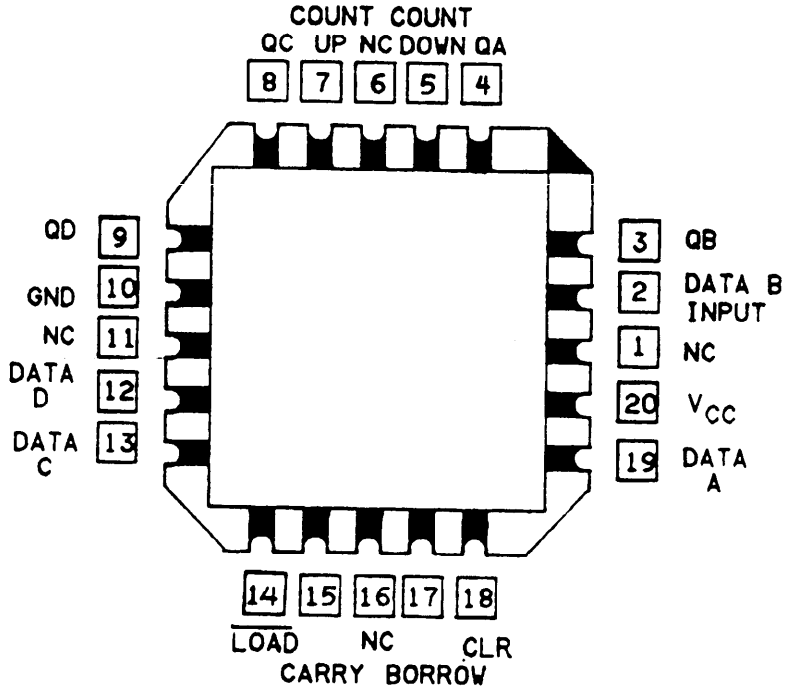
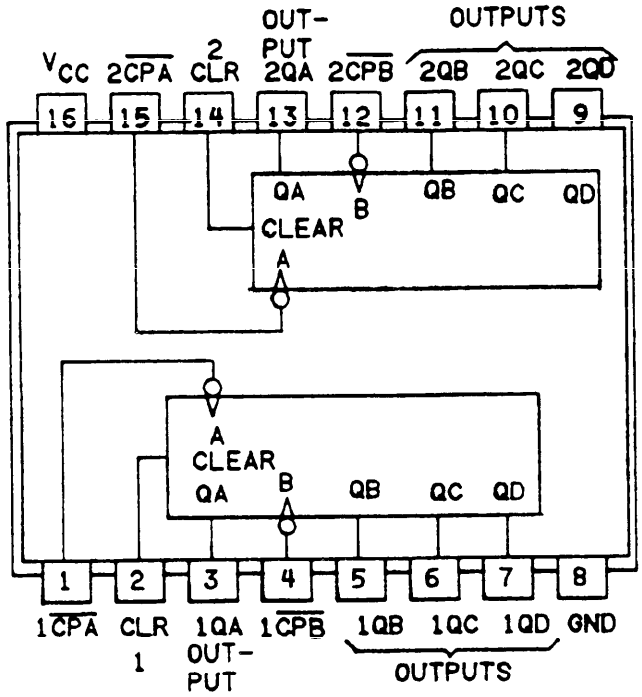


FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device type 08

Cases E, F, and X



Device type 06

Case 2

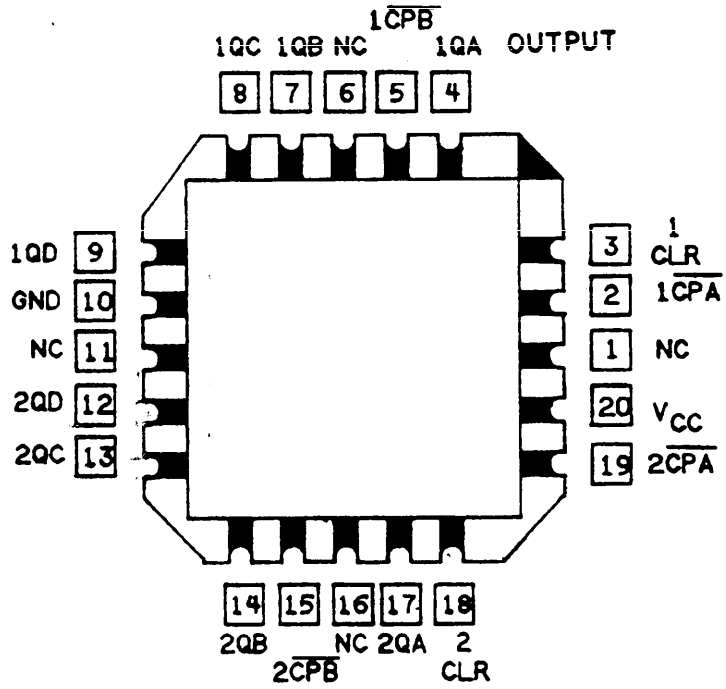
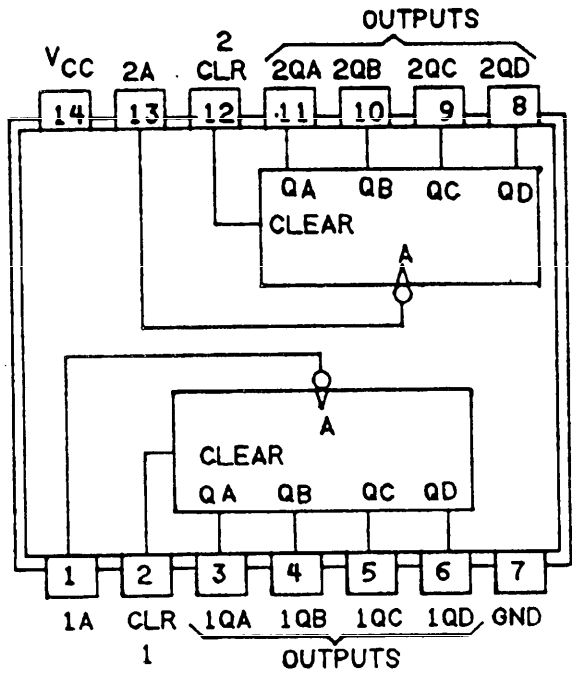


FIGURE 1. Terminal connections - Continued.

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Devices type 09

Cases C and D



Device type 09

Case 2

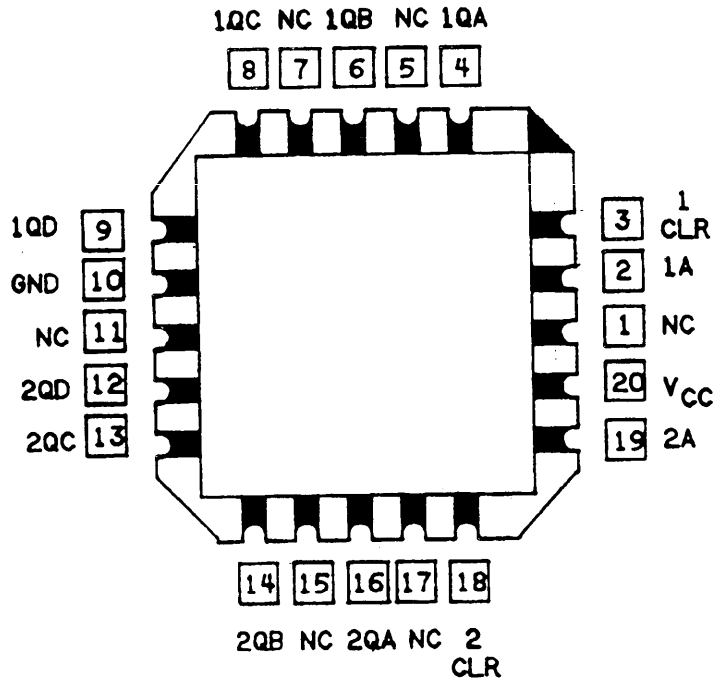
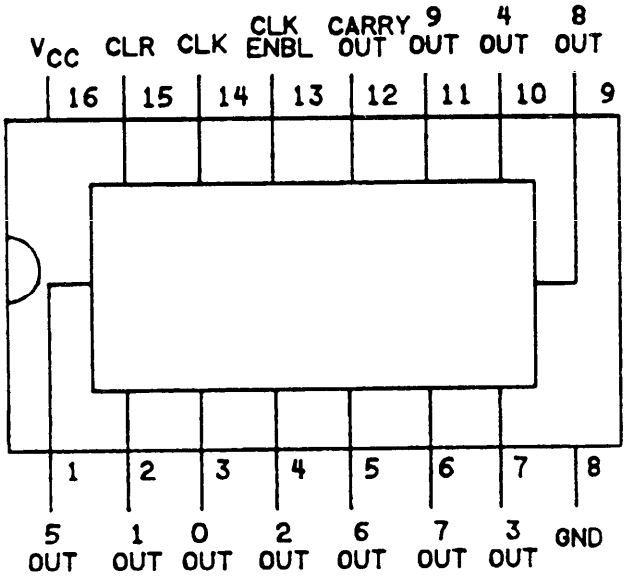


FIGURE 1. Terminal connections - Continued.

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Device type 10

Cases E, F, and X



Device type 10

Case 2

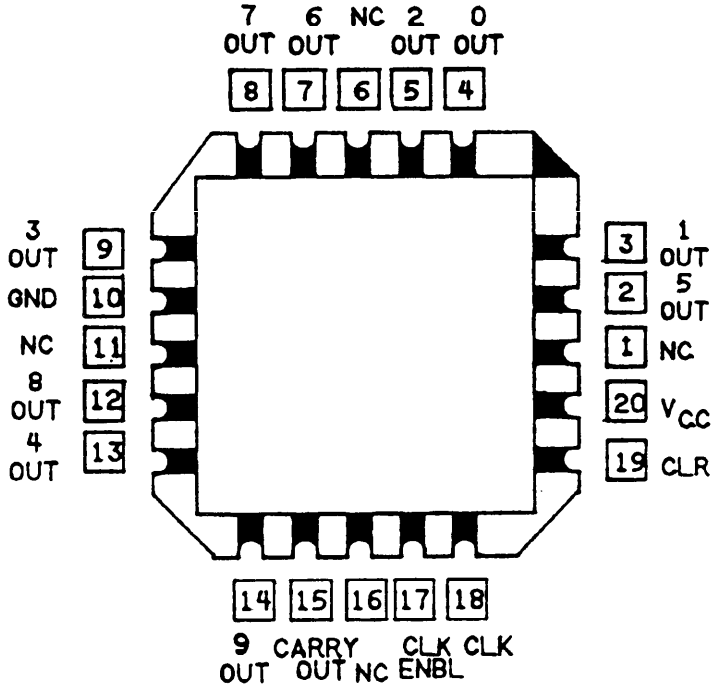
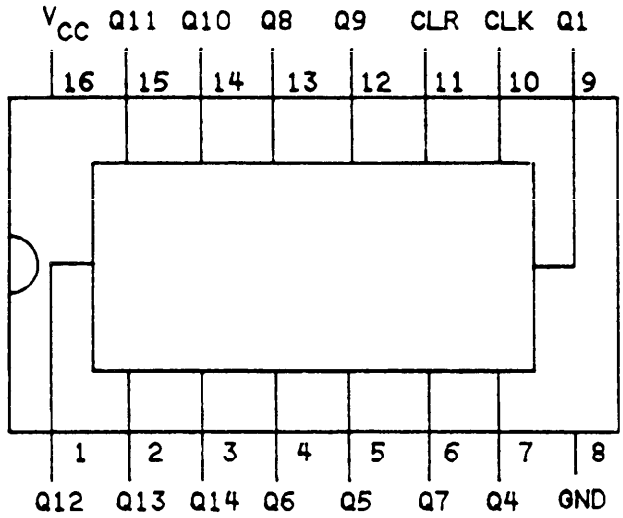


FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device type 11

Cases E, F, and X



Device type 11

Case 2

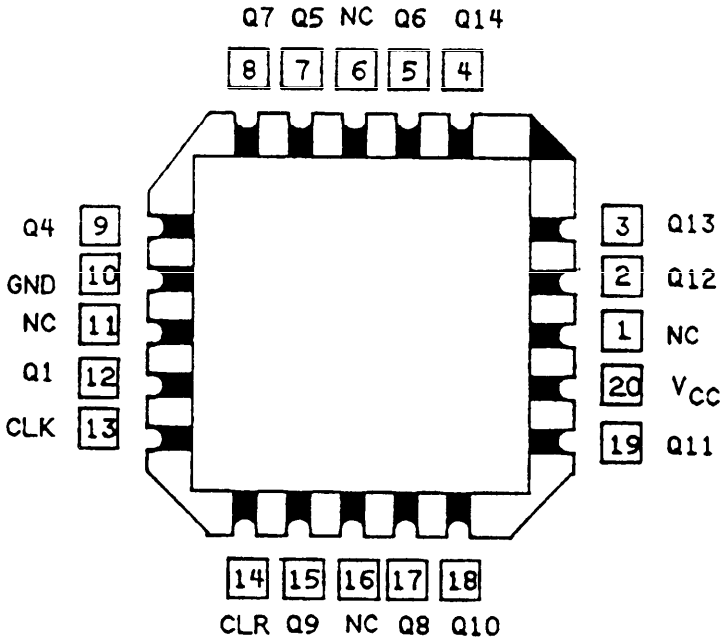
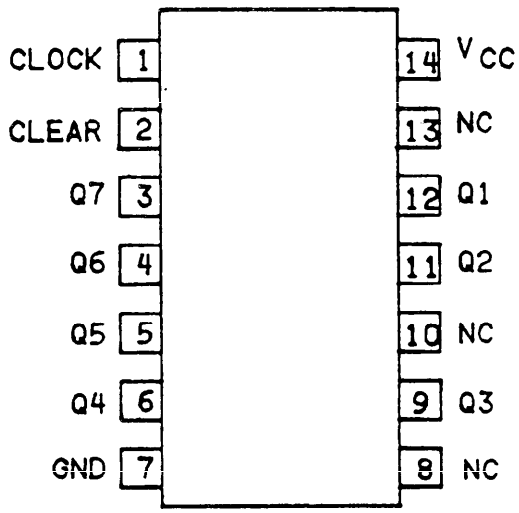


FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device type 12

Cases C and D



Device type 12

Case 2

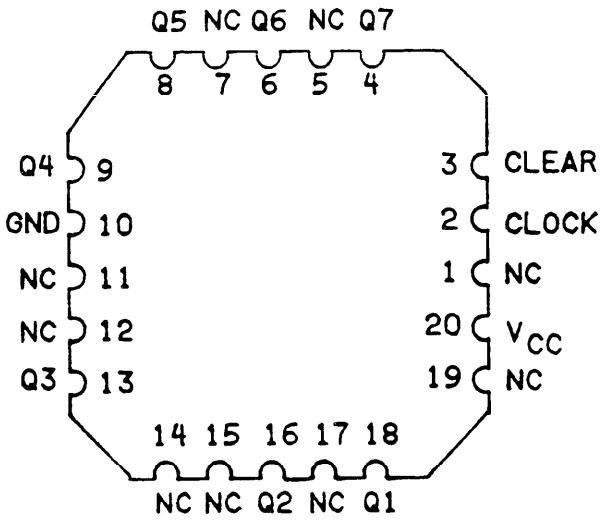
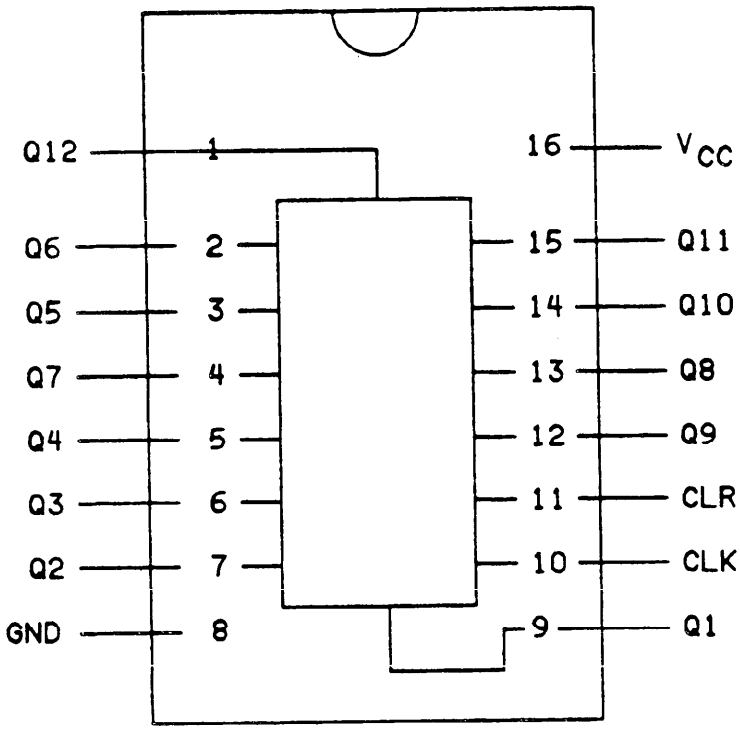


FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device type 13

Cases E, F, and X



Device type 13

Case 2

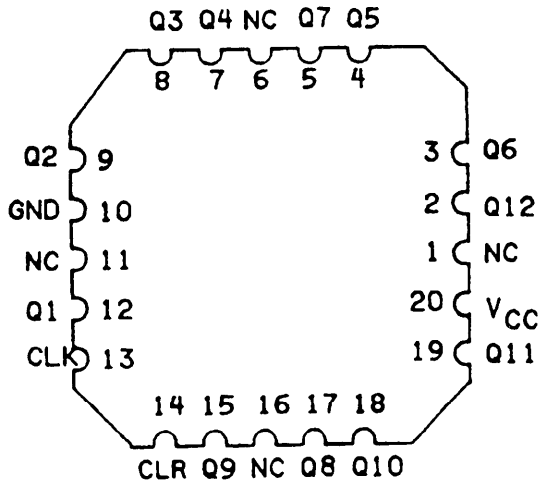


FIGURE 1. Terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 01

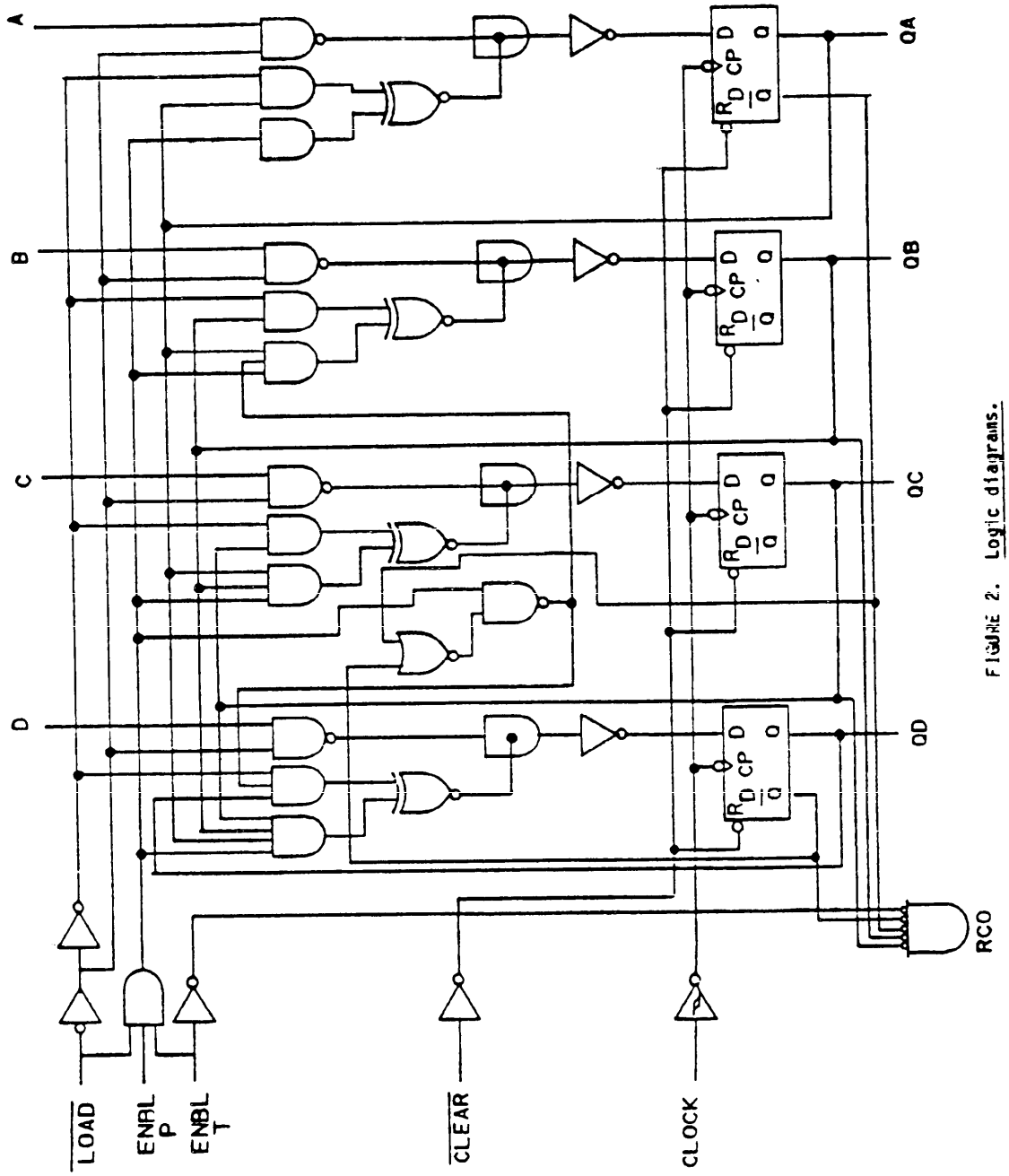


FIGURE 2. Logic diagrams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device type 02

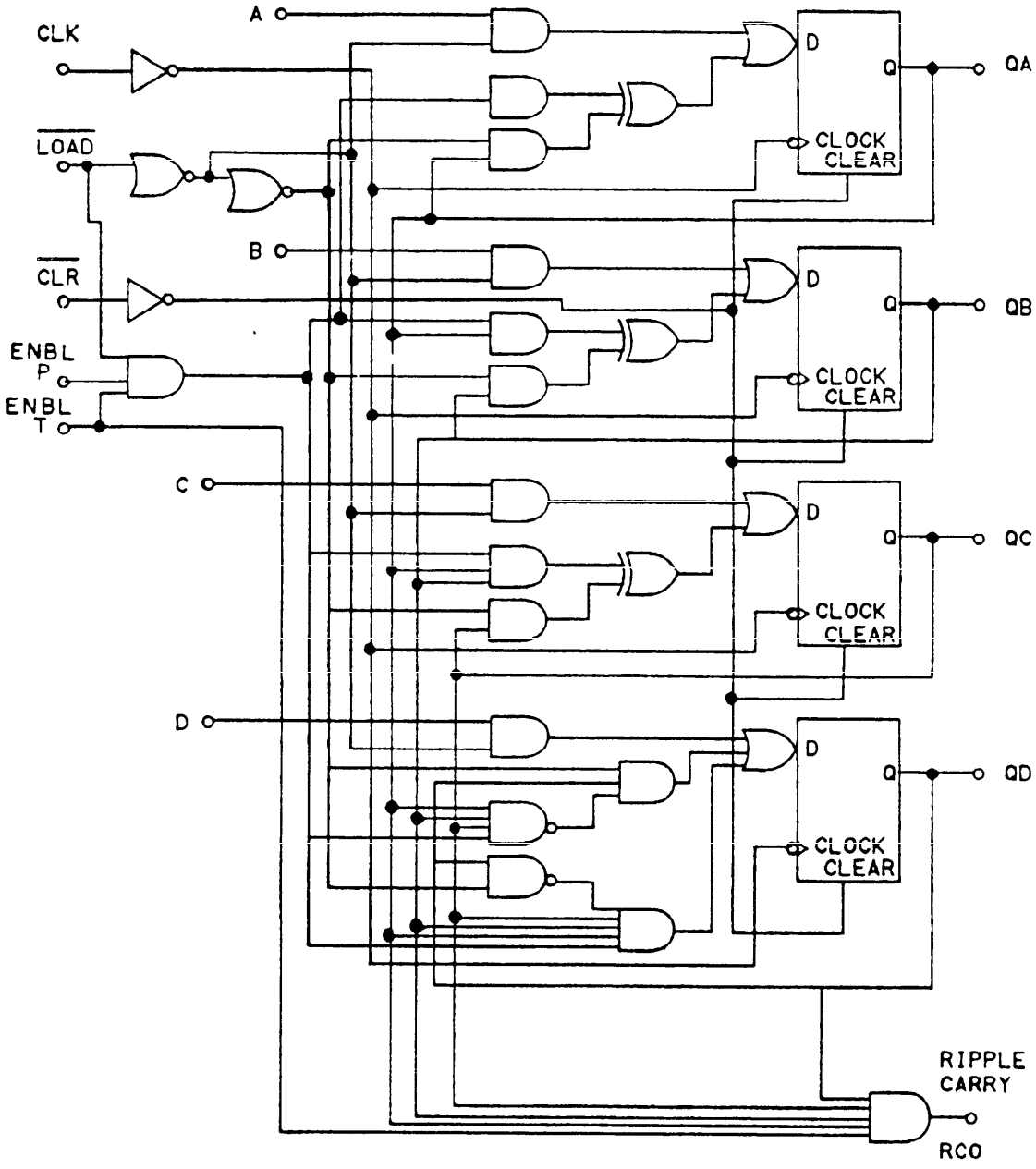


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device t pe 03

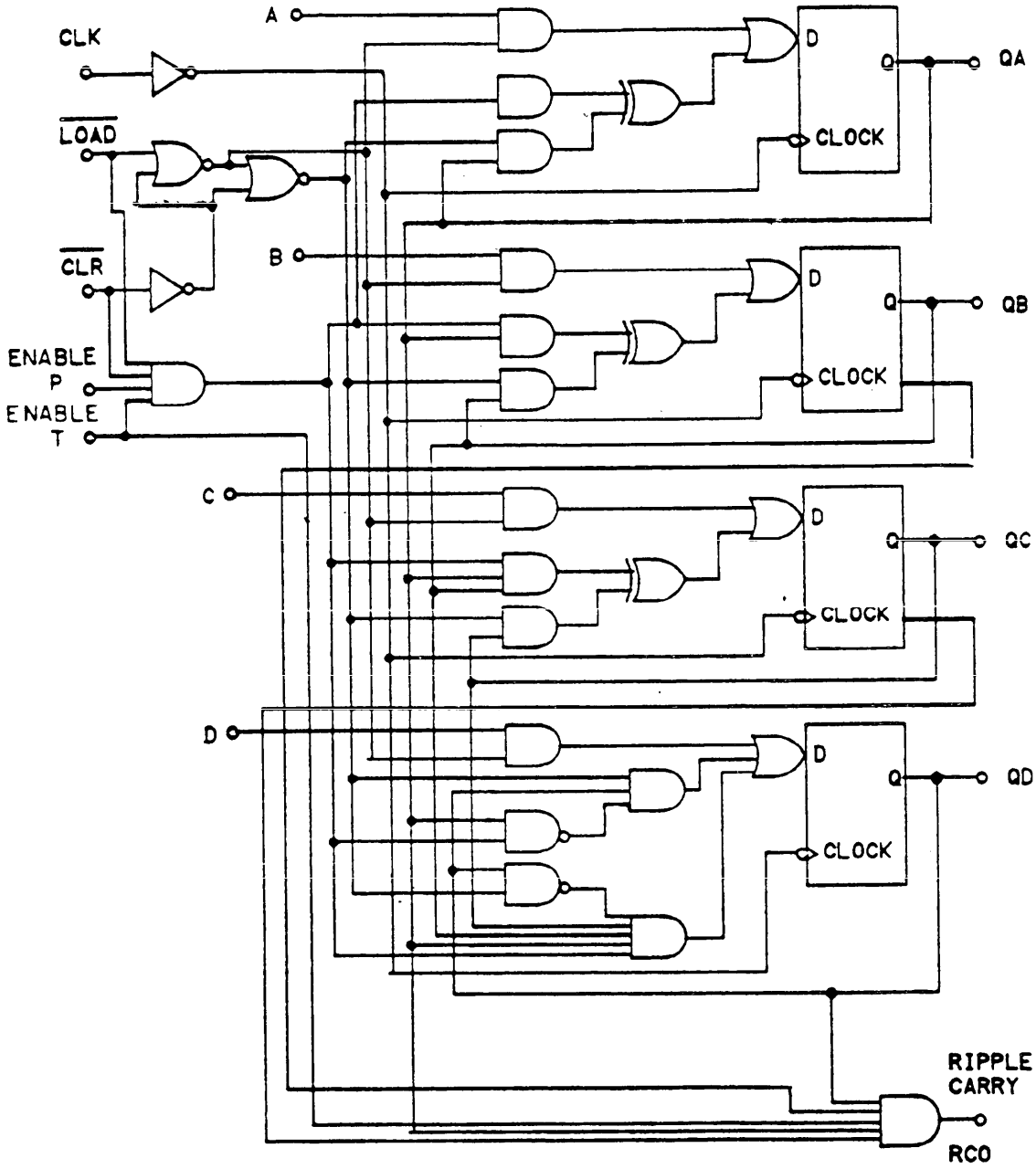


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 04

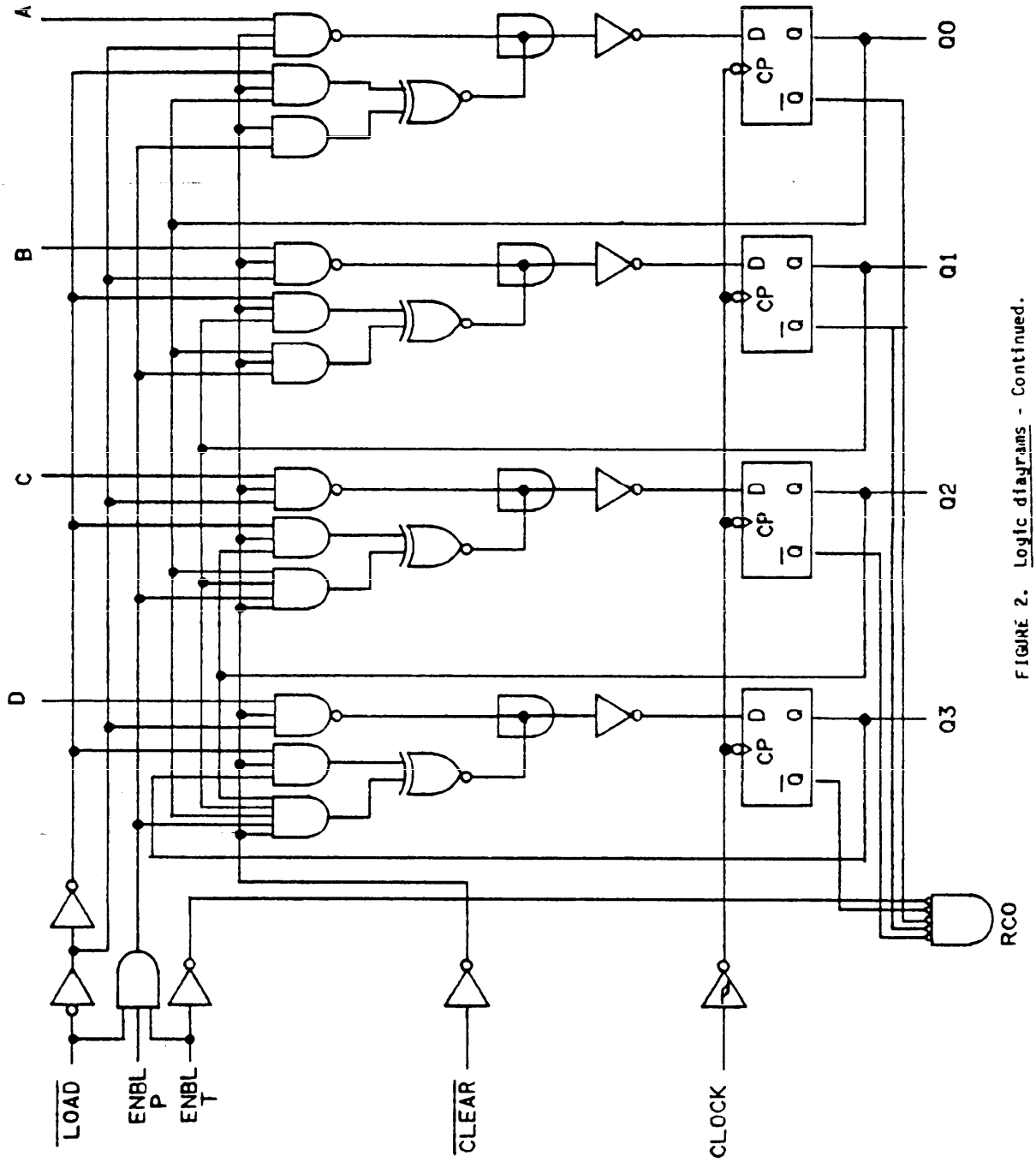


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 05

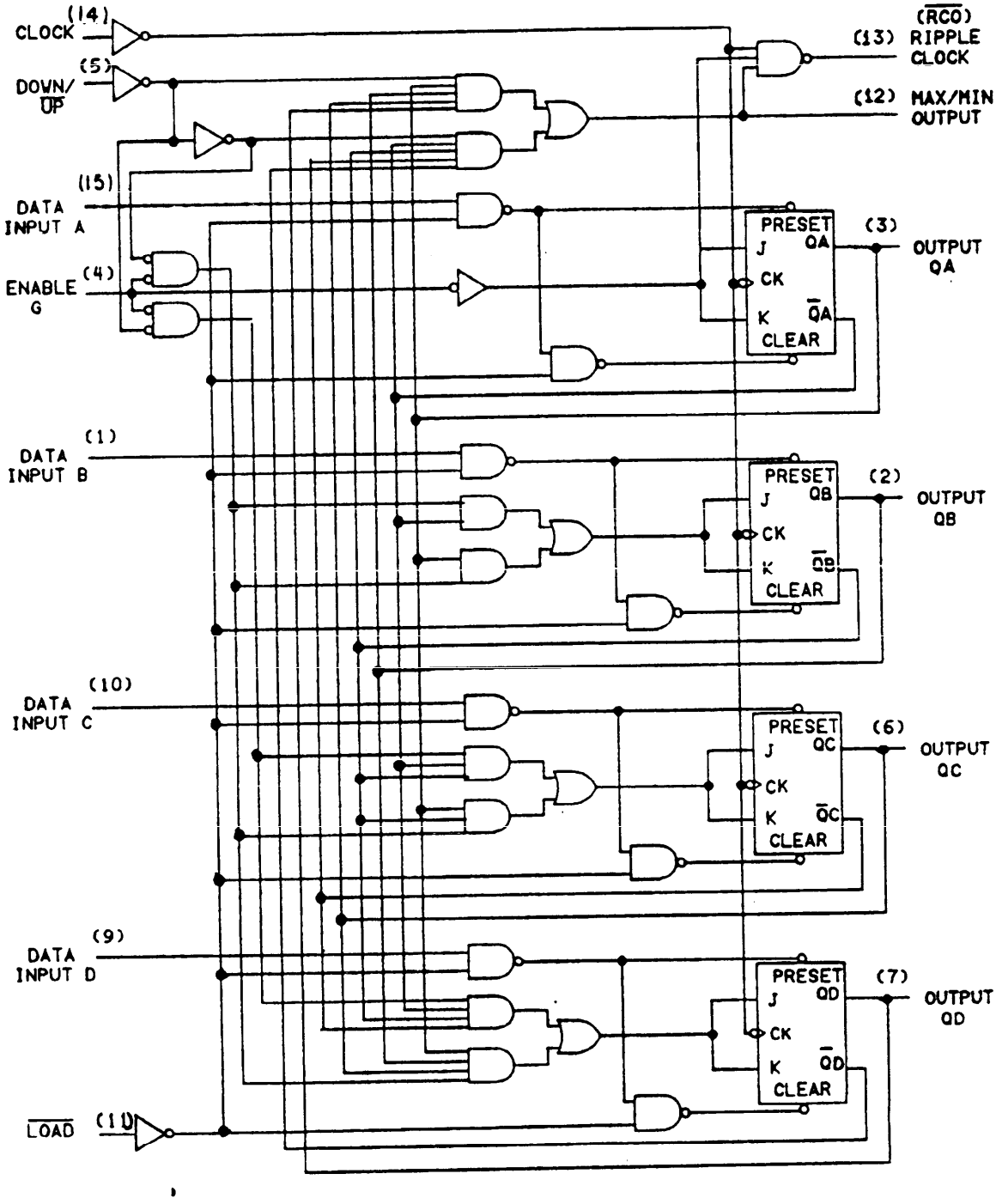


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 06

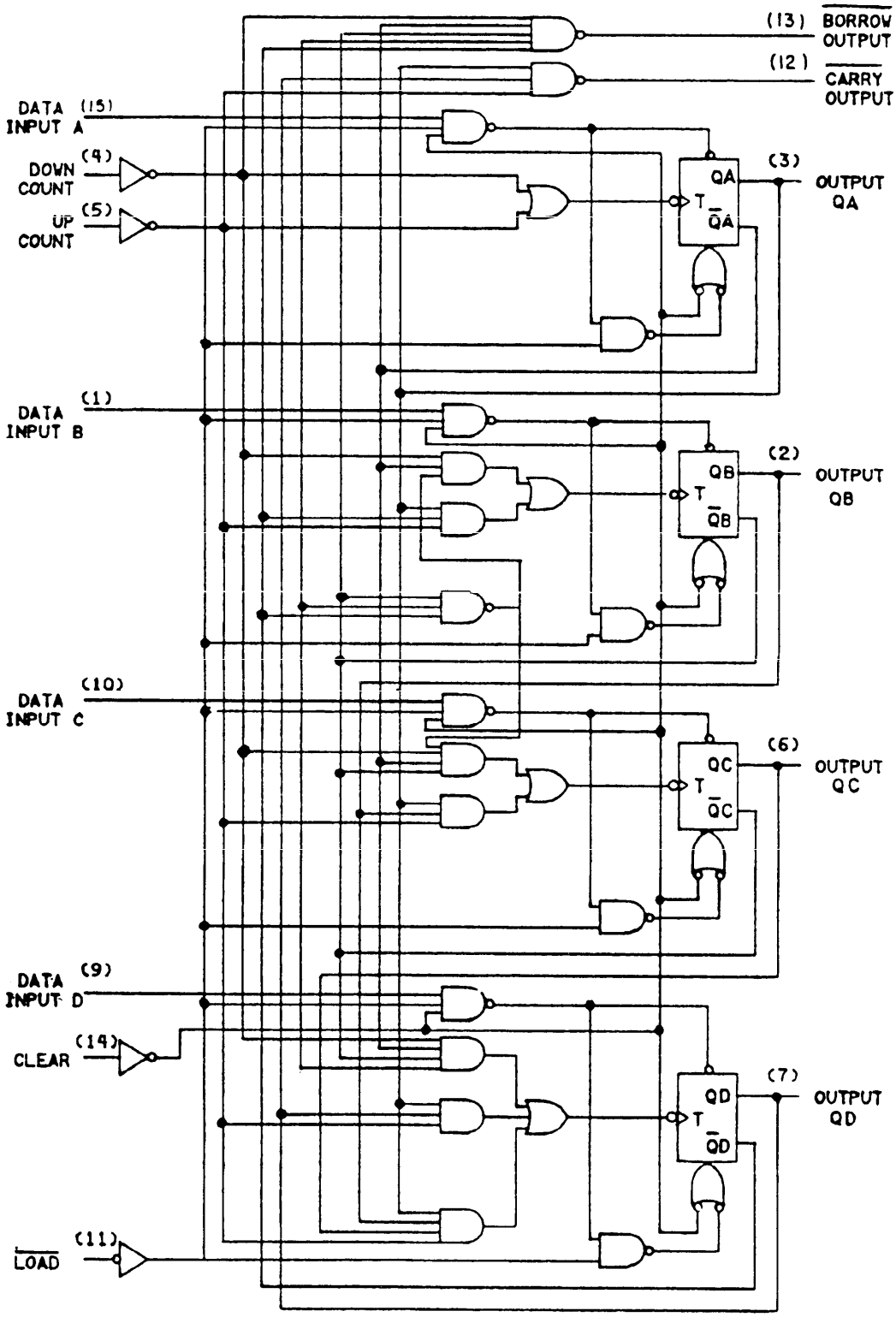


FIGURE 2. Logic diagrams - Continued.

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Device type 07

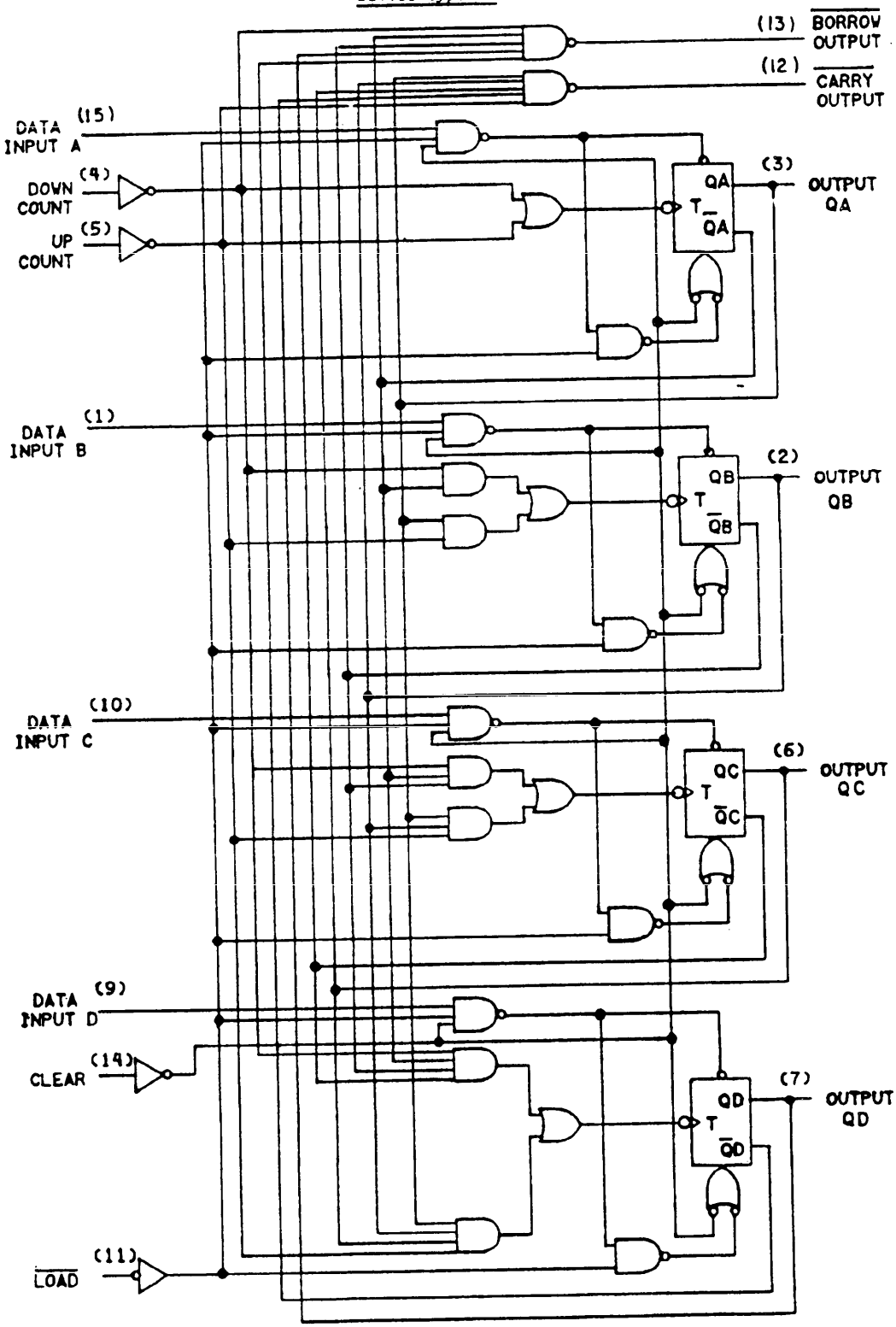


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 08

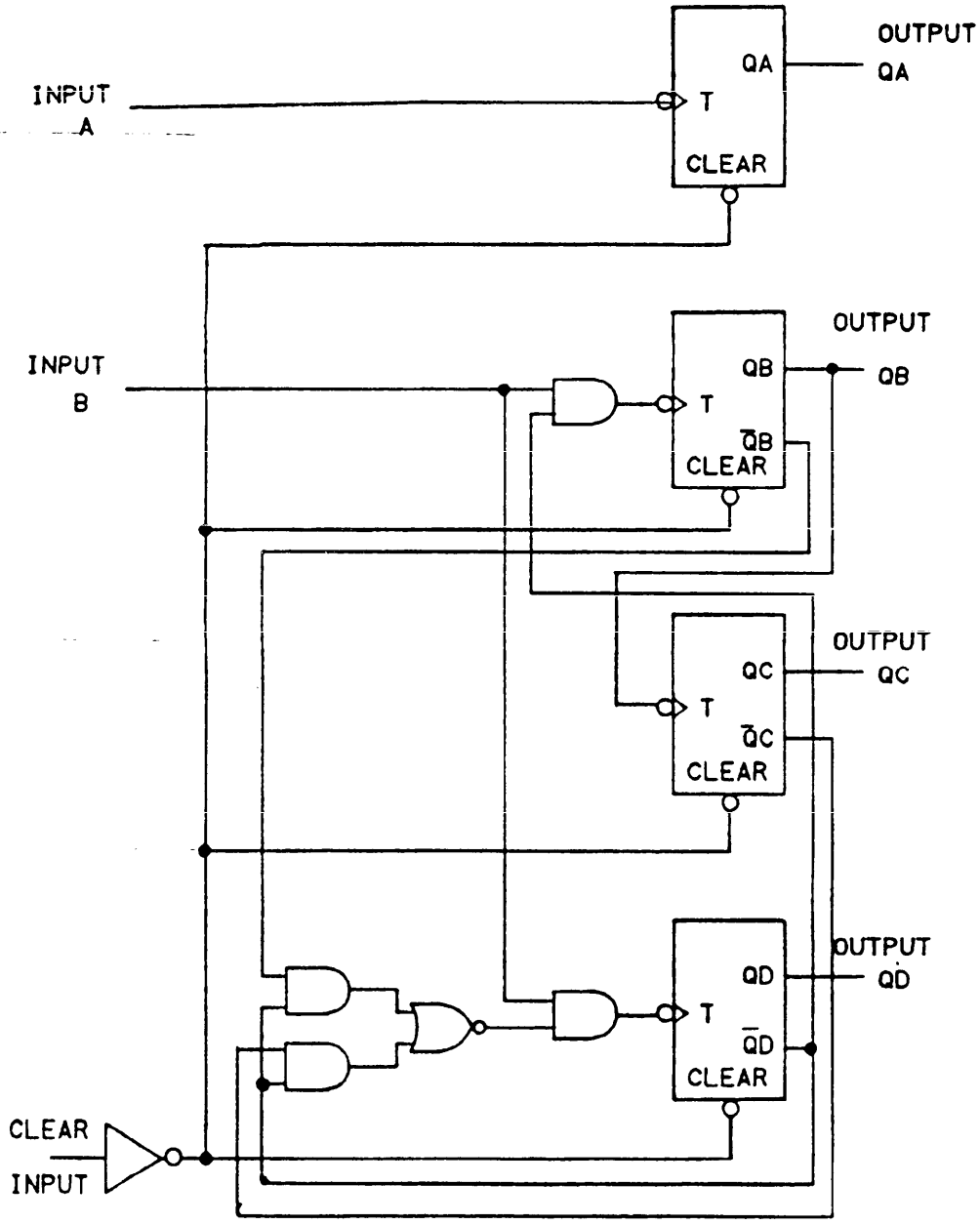


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device type 09

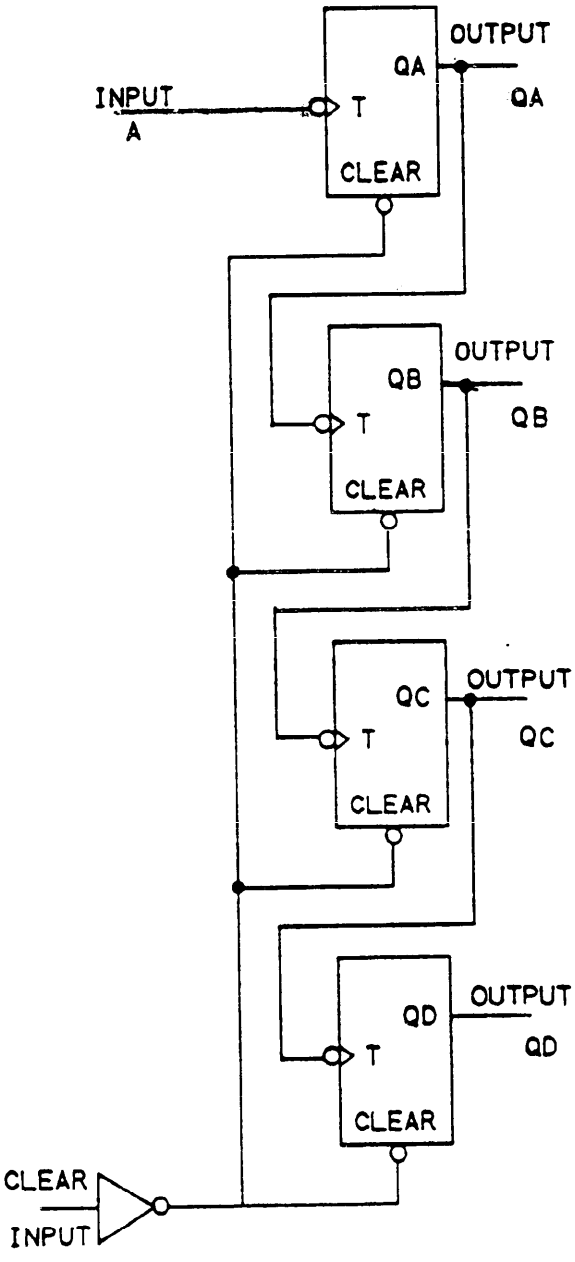


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 10

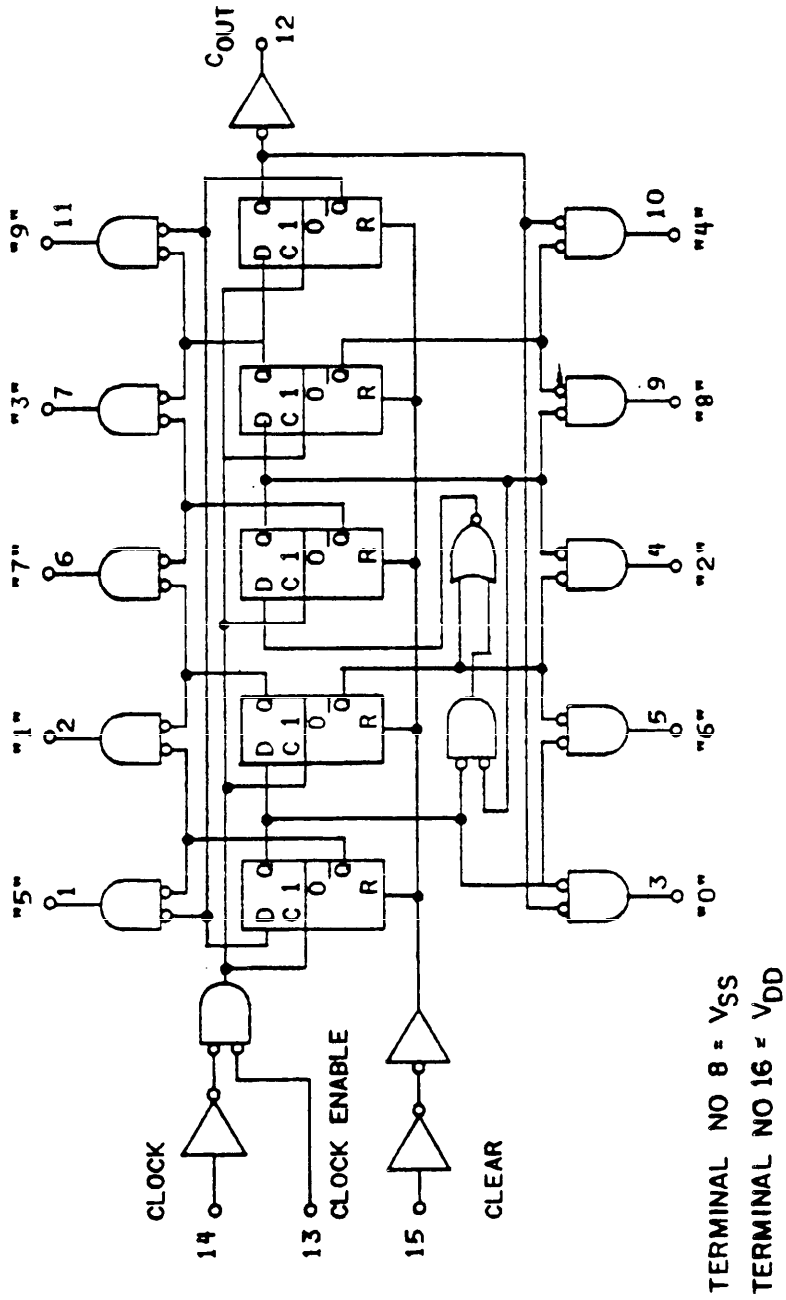


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 11

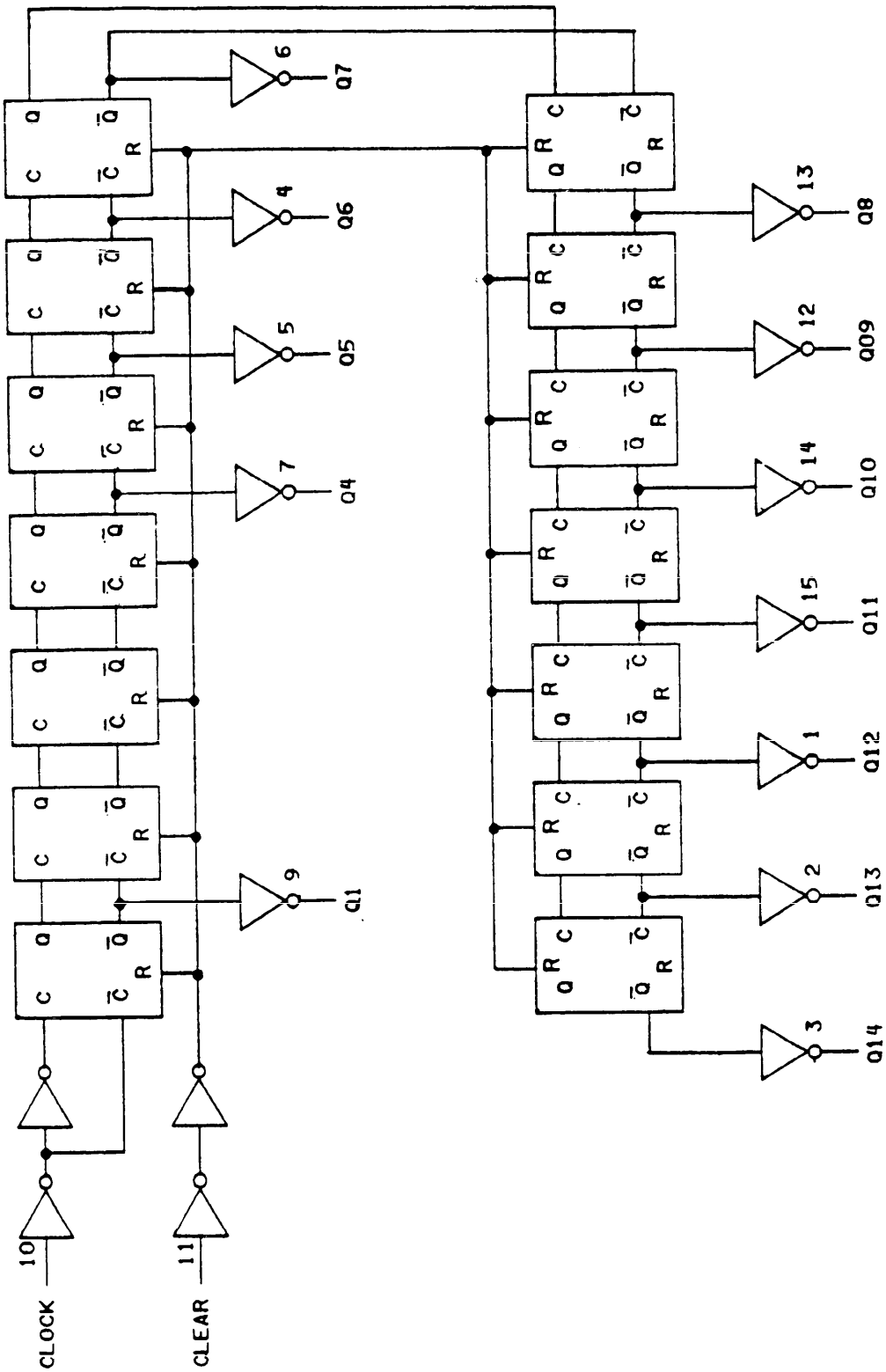


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:59Z
Check the source to verify that this is the current version before use.

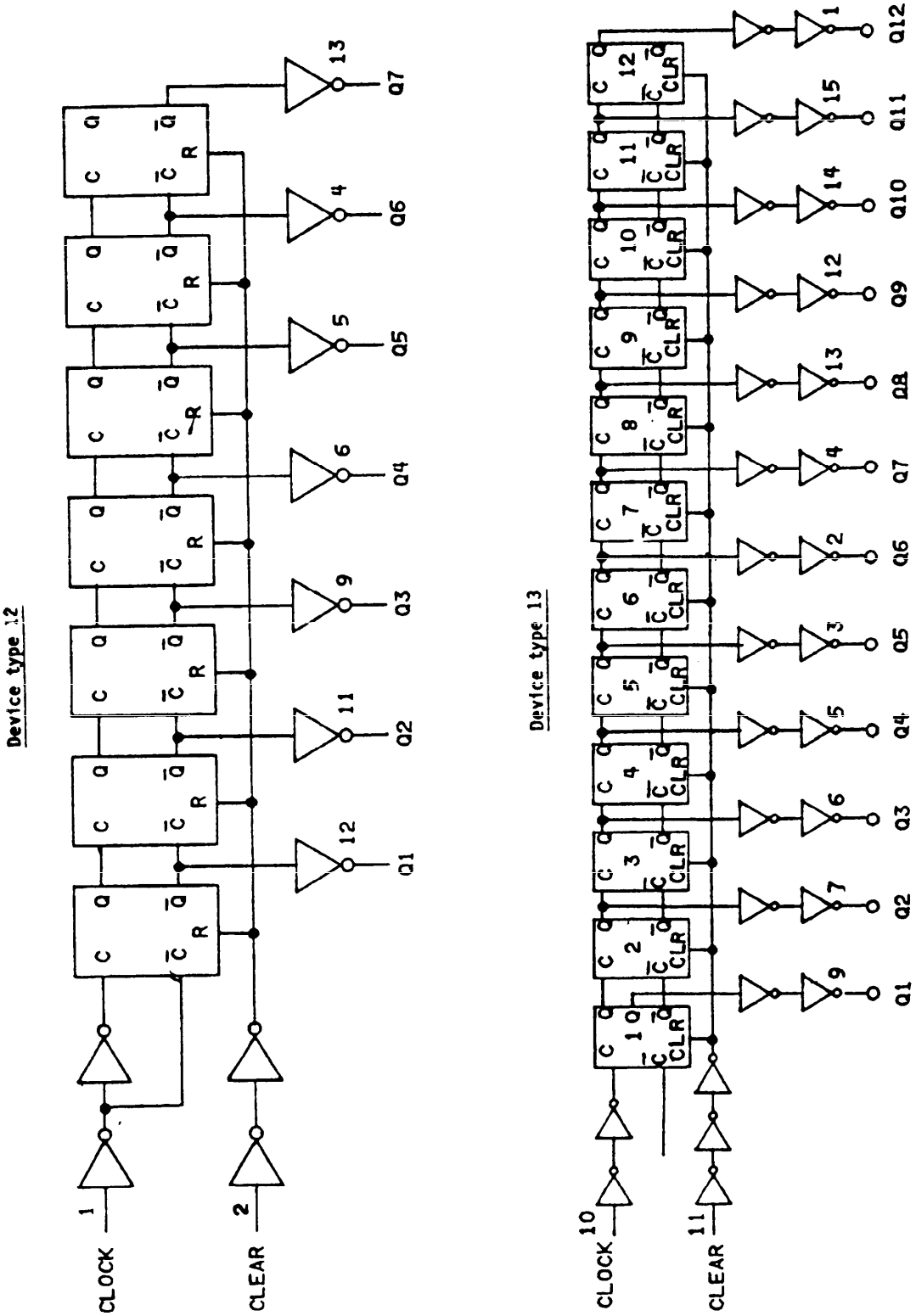


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:59Z
Check the source to verify that this is the currentversion before use.

Device type 01

Synchronous truth table

Inputs at time t_n									Outputs at time t_{n+1}				
CLOCK	ENABLE P	ENABLE T	$\overline{\text{LOAD}}$	A	B	C	D	$\overline{\text{CLEAR}}$	QA	QB	QC	QD	$\overline{\text{RCO}}$
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (See note 1)				H if count = 9 L if count < 9
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 9 L if count < 9
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
X	X	X	X	X	X	X	X	L	L	L	L	L	L

Device type 01

Asynchronous truth table

Inputs at time t_n									Outputs at time t_{n+1}				
CLOCK	ENABLE P	ENABLE T	$\overline{\text{LOAD}}$	A	B	C	D	$\overline{\text{CLEAR}}$	QA	QB	QC	QD	$\overline{\text{RCO}}$
X	X	X	X	X	X	X	X	L	L	L	L	L	L

FIGURE 3. Truth tables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device type 01

Up count sequence table

QA (LSB)	QB	QC	QD (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 02

Synchronous truth table

Inputs at time t_n									Outputs at time t_{n+1}				
CLOCK	ENABLE P	ENABLE T	$\overline{\text{LOAD}}$	A	B	C	D	$\overline{\text{CLEAR}}$	QA	QB	QC	QD	$\overline{\text{RCO}}$
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (See note 1)				H if count = 15 L if count < 15
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 15 L if count < 15
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
X	X	X	X	X	X	X	X	L	L	L	L	L	L

Device type 02

Asynchronous truth table

Inputs at time t_n									Outputs at time t_{n+1}				
CLOCK	ENABLE P	ENABLE T	$\overline{\text{LOAD}}$	A	B	C	D	$\overline{\text{CLEAR}}$	QA	QB	QC	QD	$\overline{\text{RCO}}$
X	X	X	X	X	X	X	X	L	L	L	L	L	L

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 02

Up count sequence table

QA (LSB)	QB	QC	QD (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	H

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 03

Synchronous truth table

Inputs at time t_n									Outputs at time t_{n+1}				
CLOCK	ENABLE P	ENABLE T	$\overline{LOAD}$	A	B	C	D	$\overline{CLEAR}$	QA	QB	QC	QD	$\overline{RCO}$
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (See note 1)				H if count = 9 L if count < 9
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 9 L if count < 9
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
CP	X	X	X	X	X	X	X	L	L	L	L	L	L

Up count sequence table

QA (LSB)	QB	QC	QD (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 04

Synchronous truth table

Inputs at time t_n									Outputs at time t_{n+1}				
CLOCK	ENABLE P	ENABLE T	$\overline{\text{LOAD}}$	A	B	C	D	$\overline{\text{CLEAR}}$	QA	QB	QC	QD	$\overline{\text{RCO}}$
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (See note 1)				H if count = 15 L if count < 15
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 15 L if count < 15
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
CP	X	X	X	X	X	X	X	L	L	L	L	L	L

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

Device type 04

Up count sequence table

QA (LSB)	QB	QC	QD (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	H

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.

FIGURE 3. Truth tables - Continued.

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fall-out region as specified on figure 1./

Device type 05

Mode select table

Inputs				Mode
$\overline{LOAD}$	$\overline{ENABLE}$ G	U/D	CLK	
H	L	L	⌐¬	Count up
H	L	H	⌐¬	Count down
L	X	X	X	Preset (Asyn)
H	H	X	X	No change (Hold)

Ripple carry truth table

Inputs		Outputs	
$\overline{ENABLE}$ G	CLK	Max/Min	$\overline{RCO}$
L	⌐¬	H	⌐¬
H	X	X	H
X	X	L	H

L = Low voltage level
H = High voltage level
X = Don't care
⌐¬ = Low-to-high clock transition
¬⌐ = Negative going clock pulse

NOTE: The up count and down count sequence for device type 05 is identical as that for device type 07.

FIGURE 3. Truth tables - Continued.

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Device type 06

Inputs at time t_n								Outputs at time t_{n+1}					
Count up	Count down	$\overline{LOAD}$	A	B	C	D	CLEAR	QA	QB	QC	QD	$\overline{CARRY}$	$\overline{BORROW}$
H	H	H	X	X	X	X	L	NC	NC	NC	NC	H	H
X	X	X	X	X	X	X	H	L	L	L	L	H	H
X	X	L	X	X	X	X	L	A	B	C	D	H	H
P	H	H	X	X	X	X	L	Previous count plus 1 (See note 1)				H	H
H	P	H	X	X	X	X	L	Previous count minus 1 (See note 2)				H	H
N	H	H	X	X	X	X	L	NC	NC	NC	NC	N if count = 9 H if count ≠ 9	
H	N	H	X	X	X	X	L	NC	NC	NC	NC		N if count = 0 H if count ≠ 0

NOTES:
1. See up count sequence table.
2. See down count sequence table.
3. L = V_{IL} for inputs, V_{OL} for outputs.
4. H = V_{IH} for inputs, V_{OH} for outputs.
5. X = V_{IH} or V_{IL}.
6. NC = No change.
7. NA = Not applicable.
8. P = Positive going pulse.
9. N = Negative going pulse.

FIGURE 3. Truth tables - Continued.

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Device type 06

Up count sequence table

QA (LSB)	QB	QC	QD (MSB)	$\overline{\text{CARRY}}$
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	L

Down count sequence table

QA (LSB)	QB	QC	QD (MSB)	$\overline{\text{BORROW}}$
H	L	L	H	H
L	L	L	H	H
H	H	H	L	H
L	H	H	L	H
H	L	H	L	H
L	L	H	L	H
H	H	L	L	H
L	H	L	L	H
H	L	L	L	H
L	L	L	L	L

FIGURE 3. Truth tables - Continued.

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Device type 07

Inputs at time t_n								Outputs at time t_{n+1}					
Count up	Count down	$\overline{\text{LOAD}}$	A	B	C	D	CLEAR	QA	QB	QC	QD	$\overline{\text{CARRY}}$	$\overline{\text{BORROW}}$
H	H	H	X	X	X	X	L	NC	NC	NC	NC	H	H
X	X	X	X	X	X	X	H	L	L	L	L	H	H
X	X	L	X	X	X	X	L	A	B	C	D	H	H
P	H	H	X	X	X	X	L	Previous count plus 1 (See note 1)				H	H
H	P	H	X	X	X	X	L	Previous count minus (See note 2)				H	H
N	H	H	X	X	X	X	L	NC	NC	NC	NC	N if count = 15 H if count ≠ 15	
H	N	H	X	X	X	X	L	NC	NC	NC	NC		N if count = 0 H if count ≠ 0

NOTES:
1. See up count sequence table.
2. See down count sequence table.
3. L = V_{IL} for inputs, V_{OL} for outputs.
4. H = V_{IH} for inputs, V_{OH} for outputs.
5. X = V_{IH} or V_{IL}.
6. NC = No change.
7. NA = Not applicable.
8. P = Positive going pulse.
9. N = Negative going pulse.

FIGURE 3. Truth tables - Continued.

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Device type 07

Up count sequence table

QA (LSB) (LSB)	QB	QC	QD (MSB)	$\overline{\text{CARRY}}$
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Down count sequence table

QA (LSB)	QB	QC	QD (MSB)	$\overline{\text{BORROW}}$
H	H	H	H	H
L	H	H	H	H
H	L	H	H	H
L	L	H	H	H
H	H	L	H	H
L	H	L	H	H
H	L	L	H	H
L	L	L	H	H
H	H	H	L	H
L	H	H	L	H
H	L	H	L	H
L	L	H	L	H
H	H	L	L	H
L	H	L	L	H
H	L	L	L	H
L	L	L	L	L

FIGURE 3. Truth tables - Continued.

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Device type 08

Each counter

BCD count sequence

(See note 1)

Count	Output			
	QD	QC	QB	QA
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

Bi-quinary (5-2)

(See note 2)

Count	Output			
	QA	QD	QC	QB
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	H	L	L	L
6	H	L	L	H
7	H	L	H	L
8	H	L	H	H
9	H	H	L	L

NOTES:
1. Output QA is connected to input B for BCD count.
2. Output QD is connected to input A for bi-quinary count.

Device type 09

Each counter

Count sequence

Count	Output			
	QD	QC	QB	QA
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H
10	H	L	H	L
11	H	L	H	H
12	H	H	L	L
13	H	H	L	H
14	H	H	H	L
15	H	H	H	H

FIGURE 3. Truth tables - Continued.

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Device type 10

Inputs				Outputs			
CLOCK	CLOCK ENABLE	CLEAR	D_{n-1}	Q_n	N_n	$\overline{Q}_n$	
X	H	L	X	Q_{n-1}	N_{n-1}	$\overline{Q}_{n-1}$	(No change)
X	X	H	X	L	L	H	
↓	X	L	X	Q_{n-1}	N_{n-1}	$\overline{Q}_{n-1}$	(No change)
↑	L	L	L	L	$N\text{-}1_{n-1}$	$\overline{9}_{n-1}$	
↑	L	L	H	H	$N\text{-}1_{n-1}$	$\overline{9}_{n-1}$	

On chip

Device type 11

CLOCK	CLEAR	Output state
↑	L	No change
↓	L	Advance to next stage
X	H	All outputs are low

Device type 12

CLOCK	CLEAR	Output state
↑	L	No change
↓	L	Advance to next stage
X	H	All outputs are low

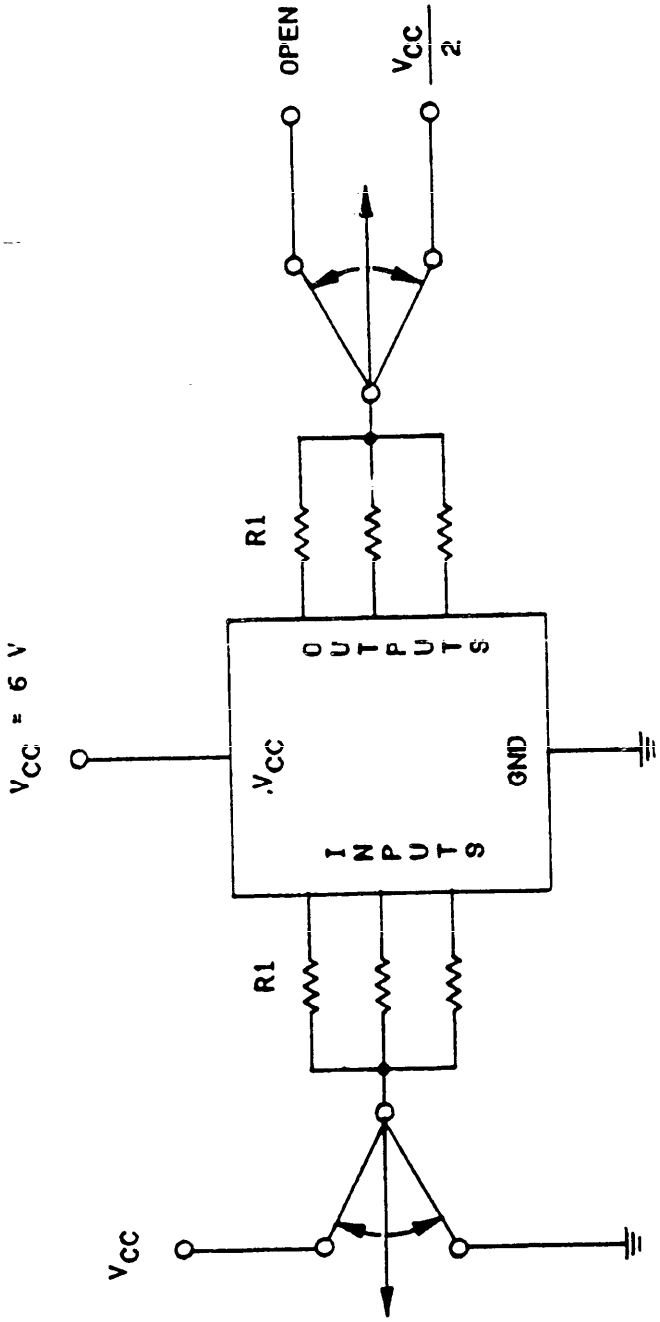
Device type 13

CLOCK	CLEAR	Output state
↑	L	No change
↓	L	Advance to next stage
X	H	All outputs are low

H = high level (steady-state)
L = low level (steady-state)
X = don't care

FIGURE 3. Truth tables - Continued.

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NOTES:
1. For static burn-in I, all inputs shall be connected to GND. Outputs may be open or connected to $V_{CC}/2$. Resistors R1 are optional on both inputs and open outputs, required on outputs connected to $V_{CC}/2$. R1 = 470Ω to 47 kΩ.
2. For static burn-in II, all inputs shall be connected through the R1 resistors to V_{CC}. Outputs may be open or connected to $V_{CC}/2$. Resistors are optional on open outputs, required on outputs connected to $V_{CC}/2$. R1 = 470Ω to 47 kΩ.
3. V_{CC} = 6.0 V ±0.5 V.

$\frac{V_{CC}}{2} = V_{CC} \frac{}{2}$ ±0.5 V.

FIGURE 4. Static burn-test circuits.

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Device types 01, 02, 03, and 04

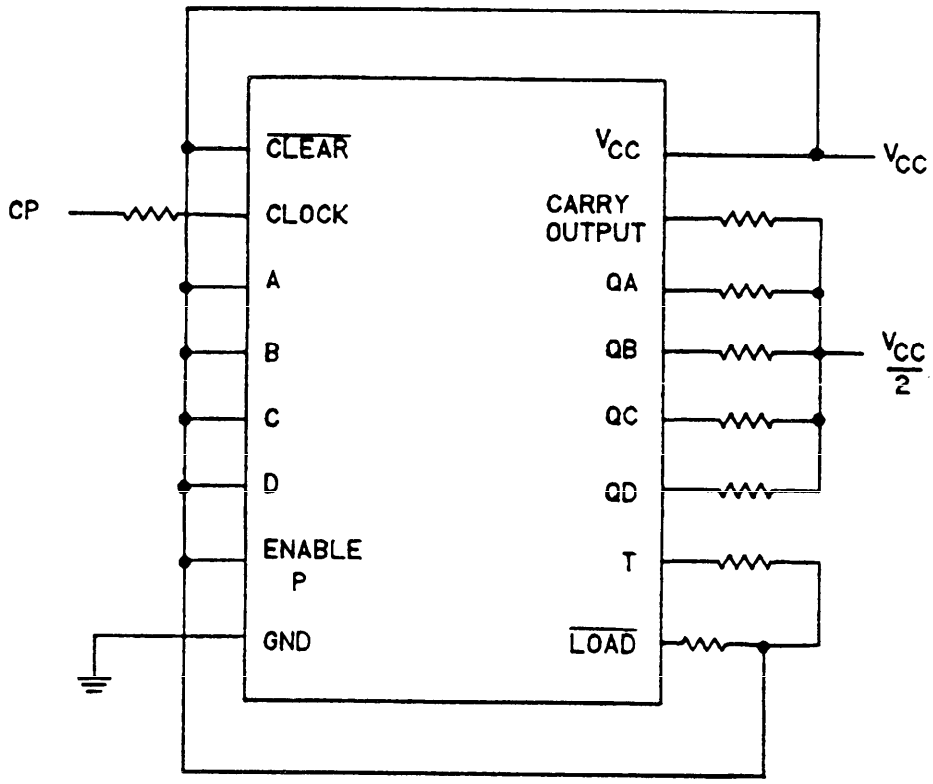


FIGURE 5. Dynamic burn-in and life test circuits.

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Device type 05

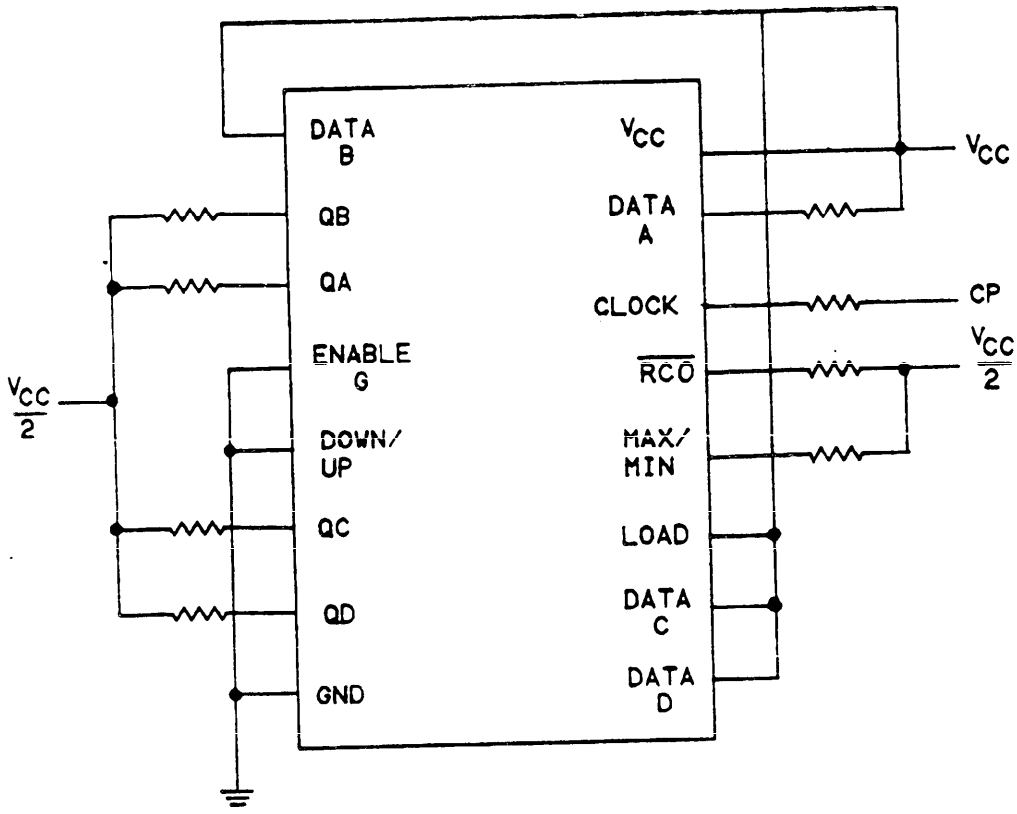


FIGURE 5. Dynamic burn-in and life test circuits - Continued.

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Device types 06 and 07

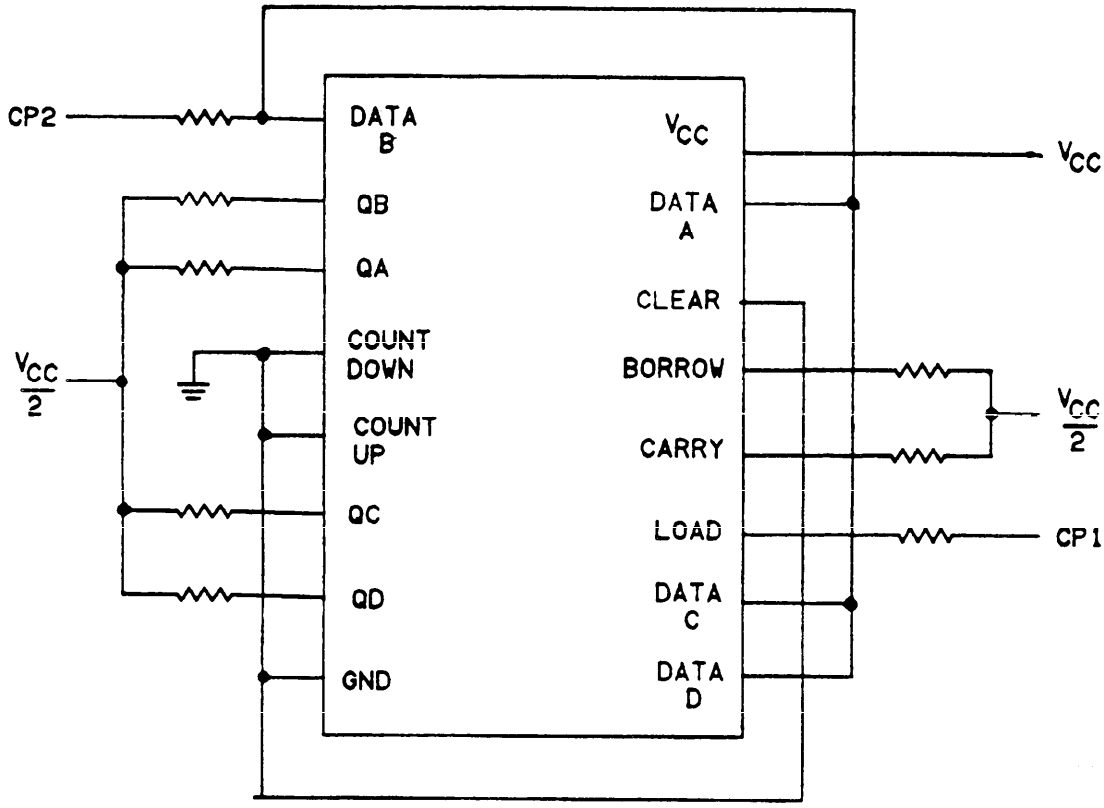
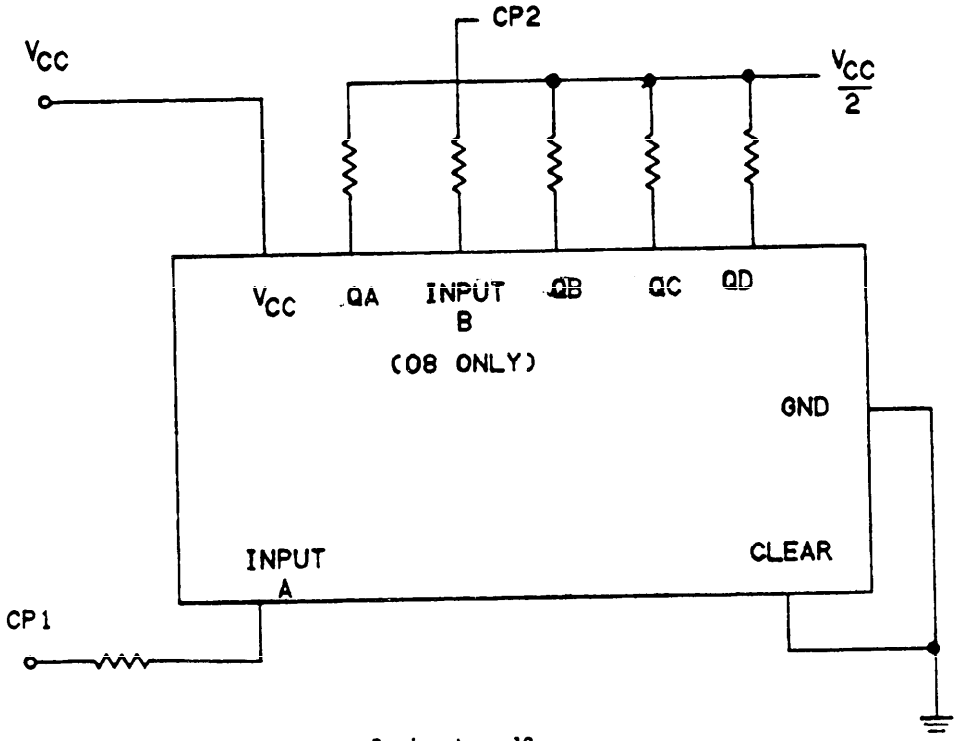


FIGURE 5. Dynamic burn-in and life test circuits - Continued.

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Device type 08 and 09



Device type 10

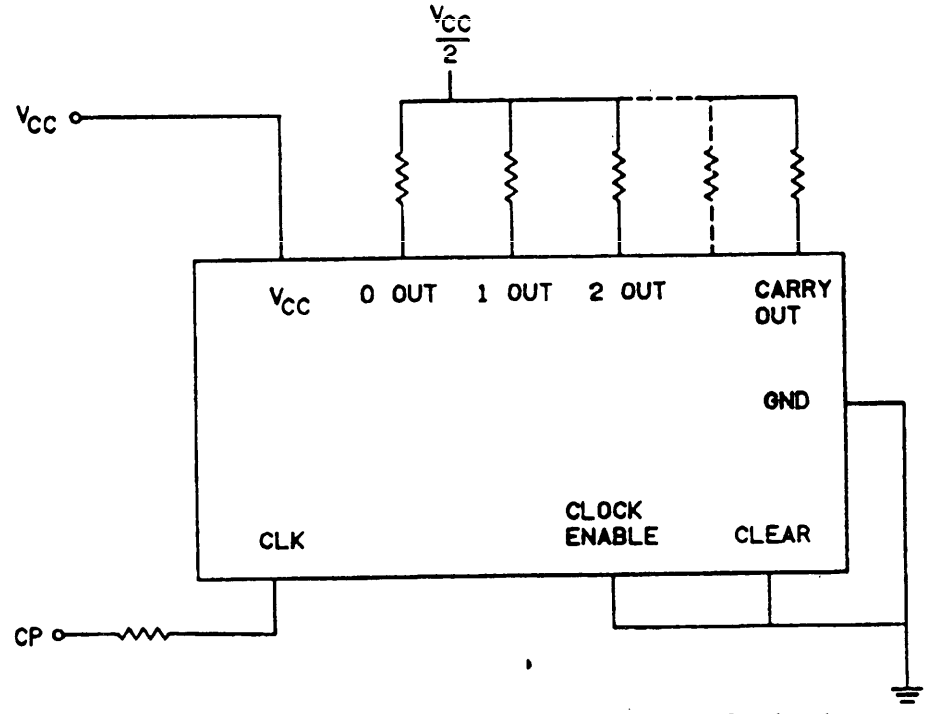
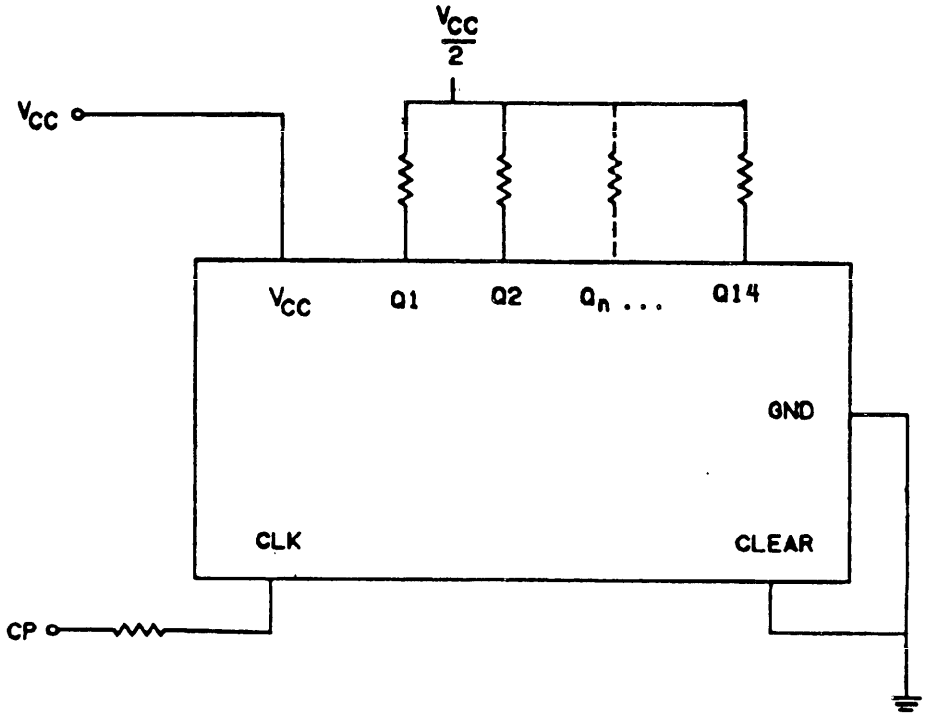


FIGURE 5. Dynamic burn-in and life test circuits - Continued.

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Device types 11 through 13



NOTES:
1. Input resistors = 470Ω to 47 kΩ ±20 percent.
2. Output resistors = 1 kΩ ±5 percent.
3. V_{CC} = 6.0 V ±0.5 V.
4. $\frac{V_{CC}}{2} = \frac{V_{CC}}{2}$ ±0.5 V.
5. CP = CP1, CP2 = $\frac{CP1}{2}$, CP = 25 kHz to 1 MHz ±50 percent square wave; duty cycle = 50 ±15 percent, V_{IH} = 4.5 minimum to V_{CC} volts maximum; V_{IL} = 0 V ±.5 V; t_{THL}, t_{TLH}, ≤ 500 ns.

FIGURE 5. Dynamic burn-in and life test circuits - Continued.

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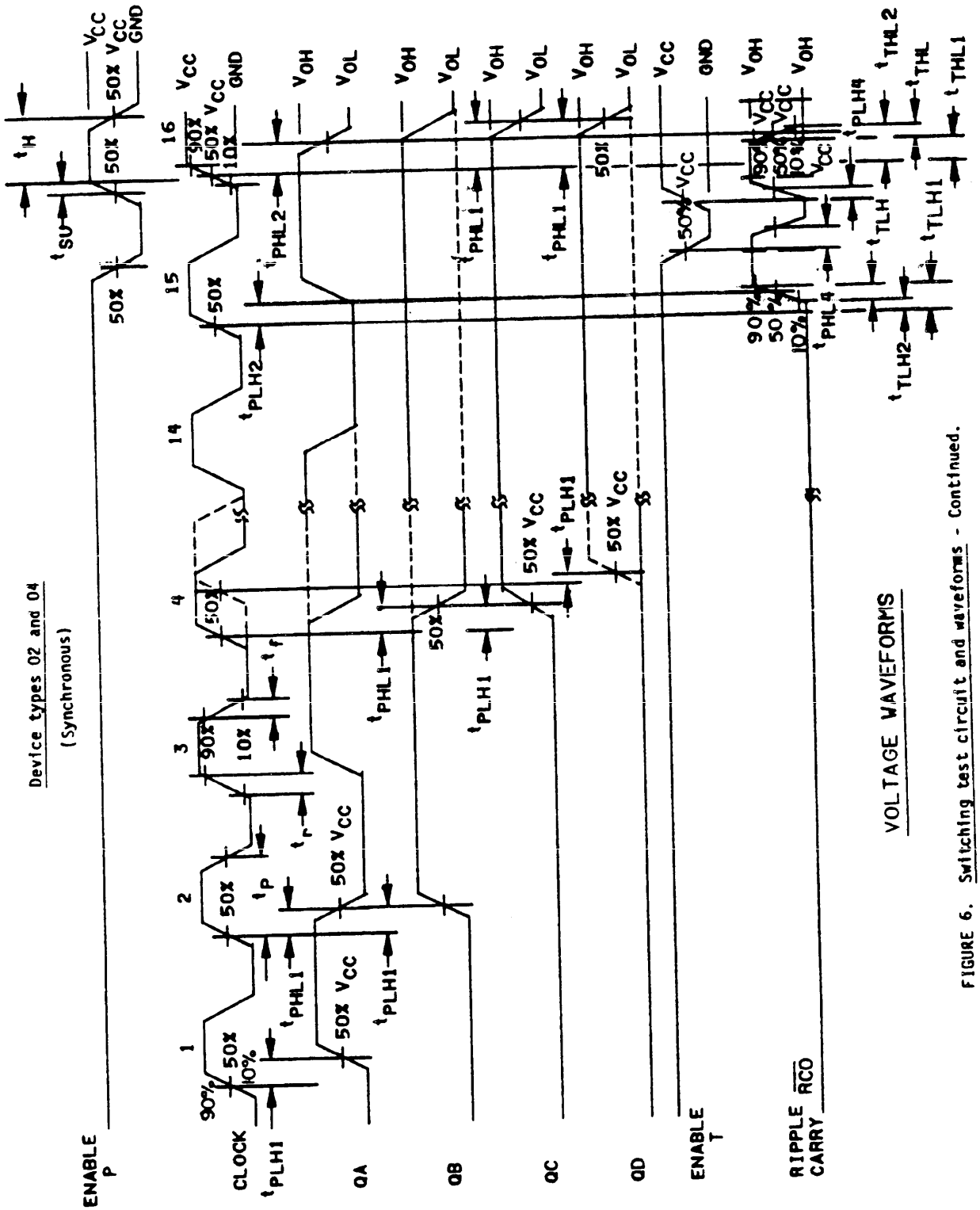


FIGURE 6. Switching test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:59Z
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NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (clock) ≥ 30 ns.
2. ENABLE input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (ENABLE T) = t_p (ENABLE P) ≥ 68 ns; $t_{SETUP} \leq 60$ ns; $t_{HOLD} \leq 8$ ns;
3. The clock input characteristics for f_{MAX} are as follows: $t_r = t_f \leq 6$ ns; t_p (clock) ≥ 30 ns; PRR $\geq$ 21 MHz.
4. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
5. Voltage measurements are to be made with respect to network ground terminal.
6. $t_{TLH} = t_{TLH1} - t_{TLH2}$; $t_{THL} = t_{THL2} - t_{THL1}$.

FIGURE 6. Switching test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

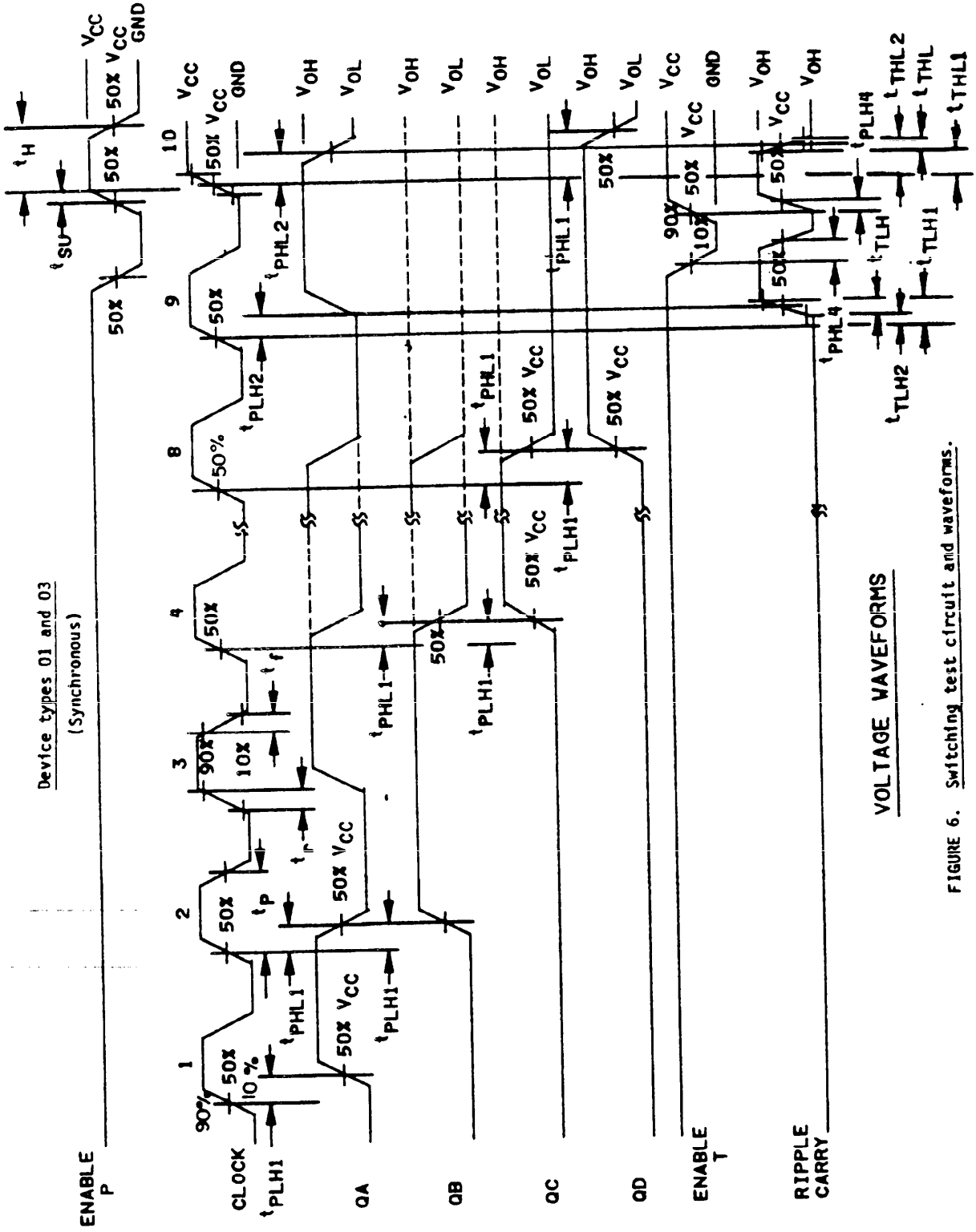


FIGURE 6. Switching test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:59Z
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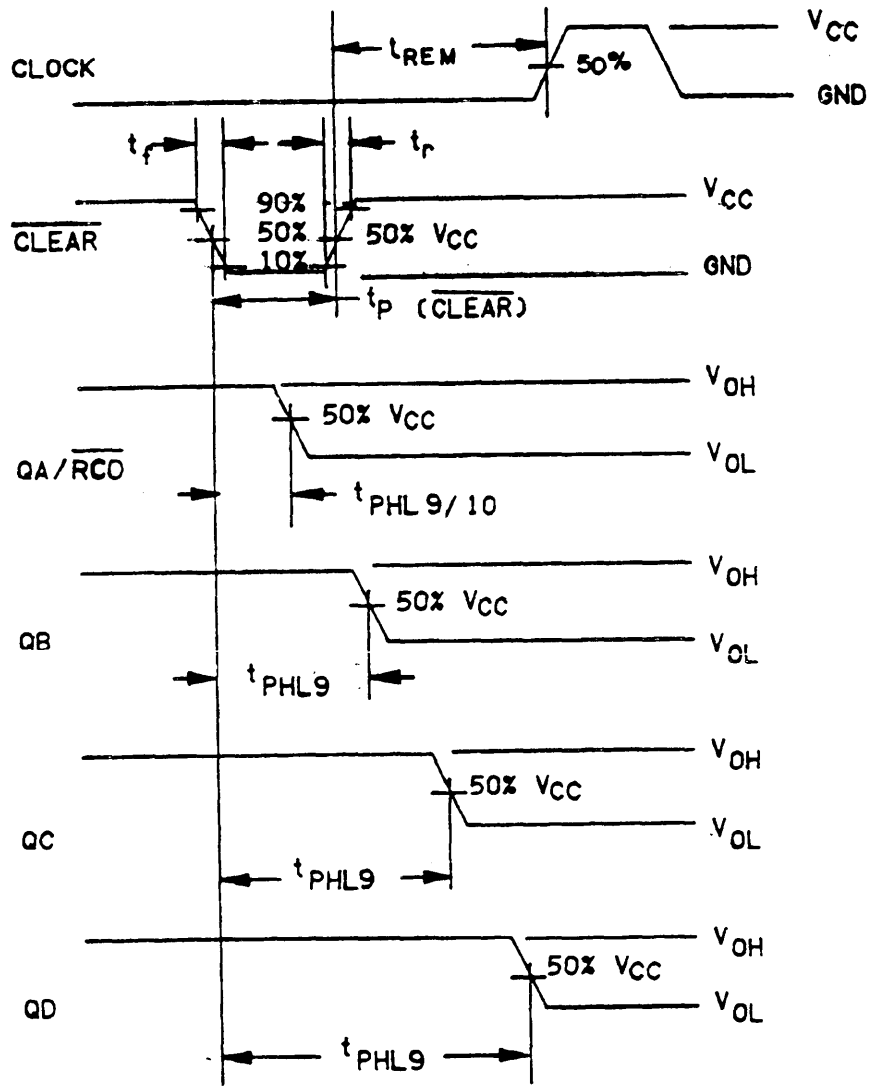
NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
   t_p (clock) $\leq$ 30 ns.
2. ENABLE input pulse characteristics: $t_r = t_f \leq 6$ ns.
   t_p (ENABLE T) = t_p (ENABLE P) $\leq$ 68 ns; $t_{SETUP} \leq$ 60 ns; $t_{HOLD} \leq$ 8 ns;
3. The clock input characteristics for f_{MAX} are as follows:
   $t_r = t_f \leq 6$ ns; t_p (clock) $\leq$ 30 ns; PRR $\geq$ 21 MHz.
4. C_L = 50 pF ±10 percent (including probe and jig capacitance).
5. Voltage measurements are to be made with respect to network ground terminal.
6. $t_{TLH} = t_{TLH1} - t_{TLH2}$; $t_{THL} = t_{THL2} - t_{THL1}$.

FIGURE 6. Switching test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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Device types 01 and 02

(CLEAR)



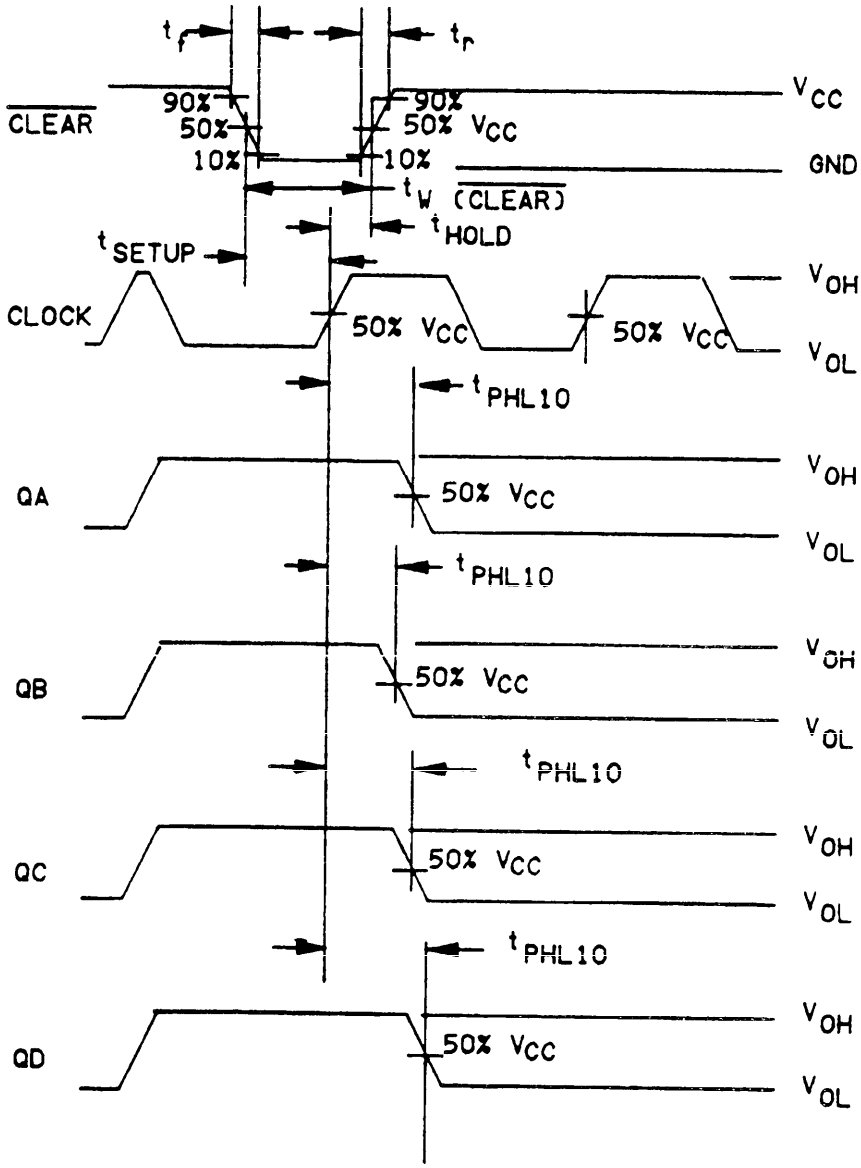
NOTES:
1. $\overline{CLEAR}$ device types 01 and 02 (asynchronous) pulses are active low and dominates regardless of the state of the clock.
2. $\overline{CLEAR}$ input pulse characteristics are as follows: $t_r = t_f \leq 6$ ns; clear tp ($\overline{CLR}$) $\leq$ 24 ns; tp (CLK) $\leq$ 30 ns; $t_{REM} \leq$ 38 ns;
3. C_L = 50 pF ±10 percent (including probe and jig capacitance).
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit and waveforms - Continued.

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Device types 03 and 04

($\overline{\text{CLEAR}}$)



NOTES:
1. $\overline{\text{CLEAR}}$ device types 03 and 04 (synchronous) pulses are active low and dominates regardless of the state of the clock.
2. $\overline{\text{CLEAR}}$ input pulse characteristics are as follows: $t_r = t_f \leq 6$ ns; t_p ($\overline{\text{CLR}}$) ≤ 56 ns; $t_r \leq 8$ ns; $t_{SU} \leq 48$ ns; PRR ≤ 1 MHz.
3. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
4. t_p (CLK) ≤ 30 ns; PRR ≤ 1 MHz.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

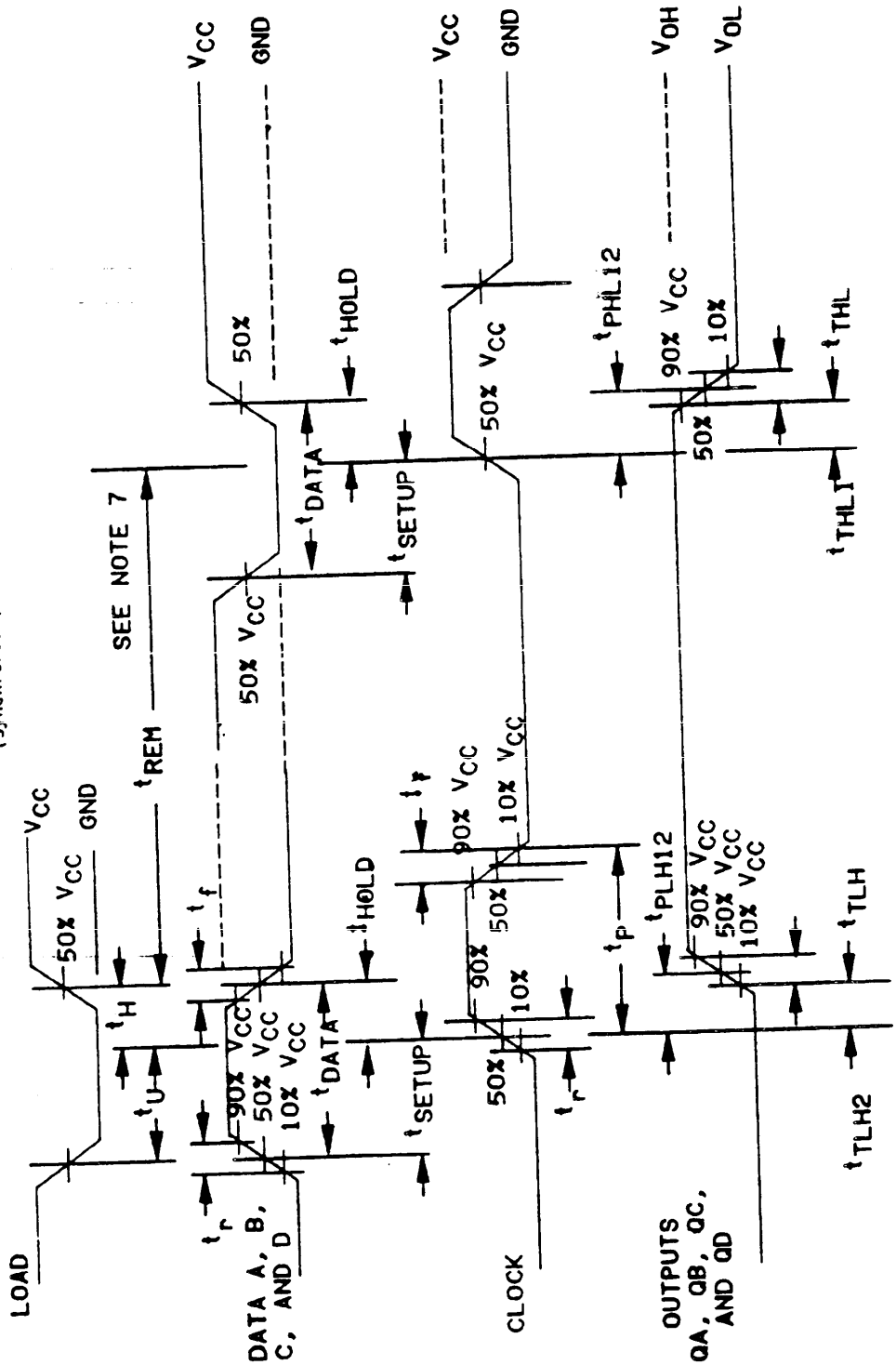


FIGURE 6. Switching test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:59Z
Check the source to verify that this is the current version before use.

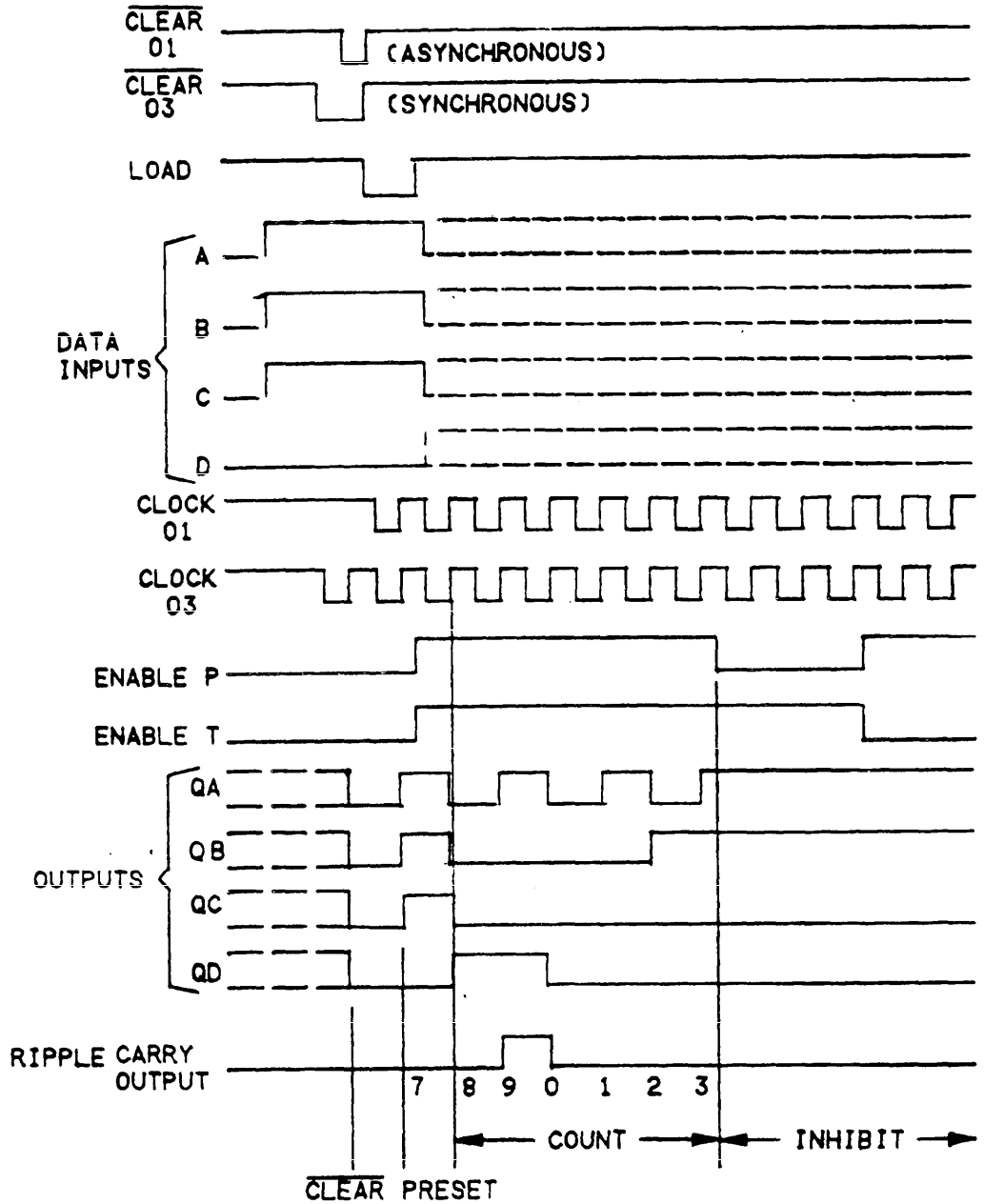
NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns.
   t_p (clock) $\leq$ 30 ns.
2. Data input pulse characteristics: $t_r = t_f \leq 6$ ns,
   t_p (data) $<$ 60 ns; $t_{SETUP} \leq$ 45 ns; $t_{HOLD} \leq$ 15 ns,
3. The clock input characteristics for f_{MAX} are as follows:
   $t_r = t_f \leq 6$ ns; t_p (clock) $\leq$ 24 ns; PRR $\geq$ 21 MHz.
4. C_L = 50 pF ±10 percent (including probe and jig capacitance).
5. Voltage measurements are to be made with respect to network ground terminal.
6. $t_{TLH} = t_{TLH1} - t_{TLH2}$; $t_{THL} = t_{THL2} - t_{THL1}$.
7. Device types 01 and 02 $t_{REM} \leq$ 30 ns; Load input characteristics:
   $t_r = t_f \leq 6$ ns; t_p (load) $\leq$ 53 ns; $t_{SU} \leq$ 45 ns; $t_H \leq$ 8 ns;

FIGURE 6. Switching test circuit and waveforms - Continued.

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Device types 01 and 03

SYNCHRONOUS DECADE COUNTERS TYPICAL $\overline{\text{CLEAR}}$, PRESET, COUNT AND INHIBIT SEQUENCES



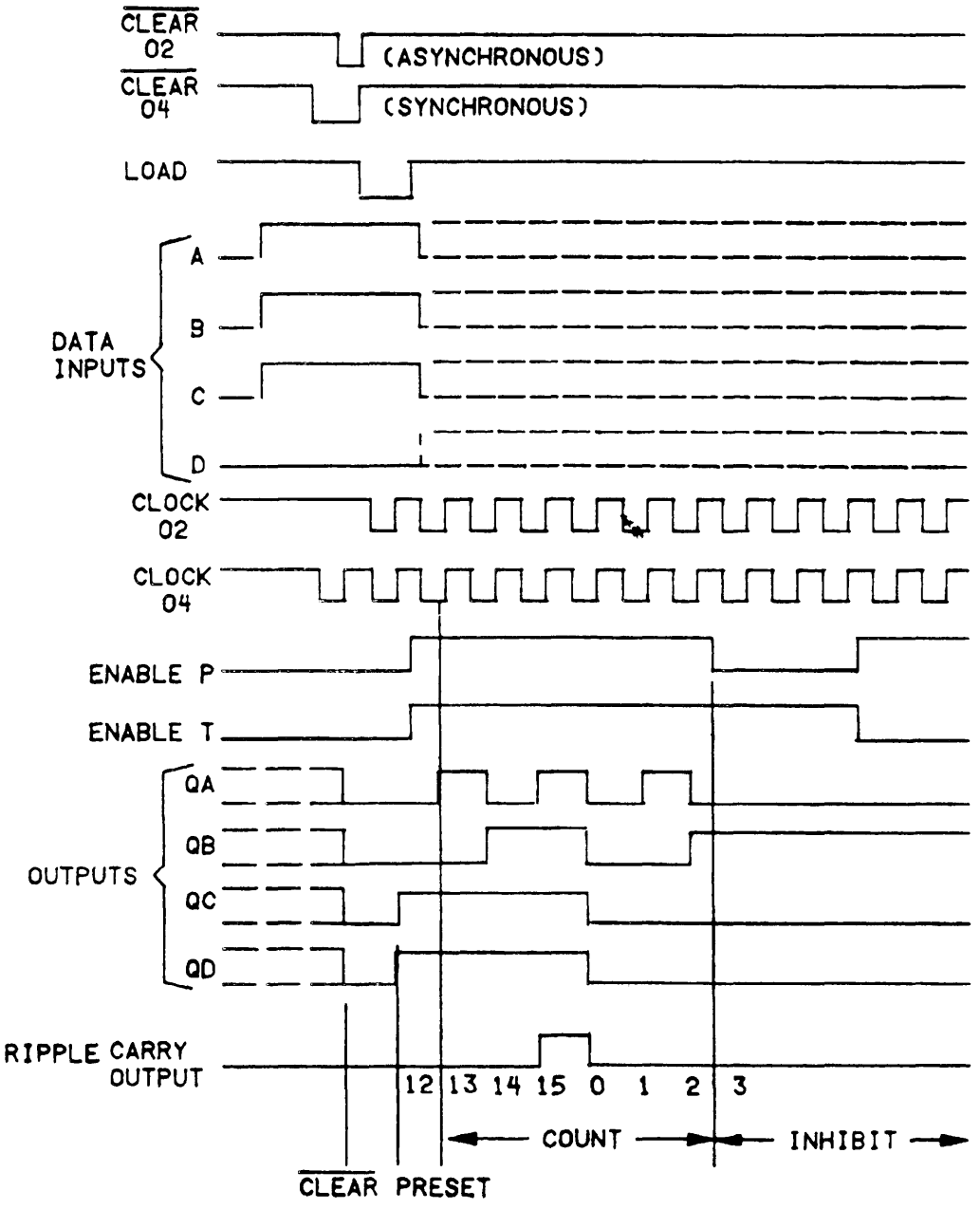
Sequence:
1. $\overline{\text{CLEAR}}$ outputs to zero.
2. Preset BCD seven.
3. Count to eight, nine, zero, one, two and three.
4. Inhibit.

FIGURE 6. Switching test circuit and waveforms - Continued.

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Check the source to verify that this is the current version before use.

Device types 02 and 04

SYNCHRONOUS BINARY COUNTERS TYPICAL $\overline{\text{CLEAR}}$, PRESET, COUNT AND INHIBIT SEQUENCES

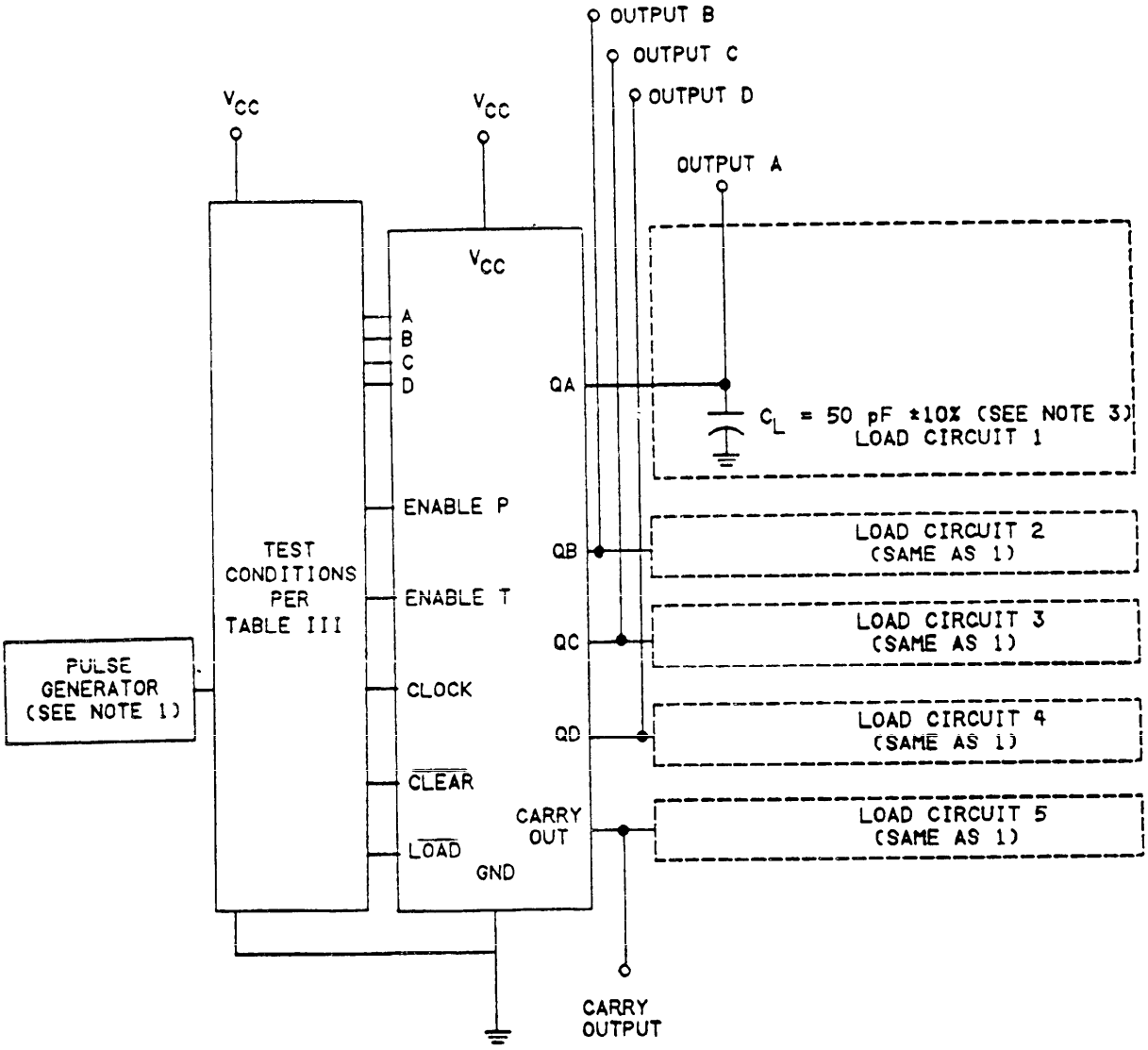


Sequence:
1. **$\overline{\text{CLEAR}}$ outputs to zero.**
2. **Preset to binary.**
3. **Count to thirteen, fourteen, fifteen zero, one, and two.**
4. **Inhibit.**

FIGURE 6. Switching test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.

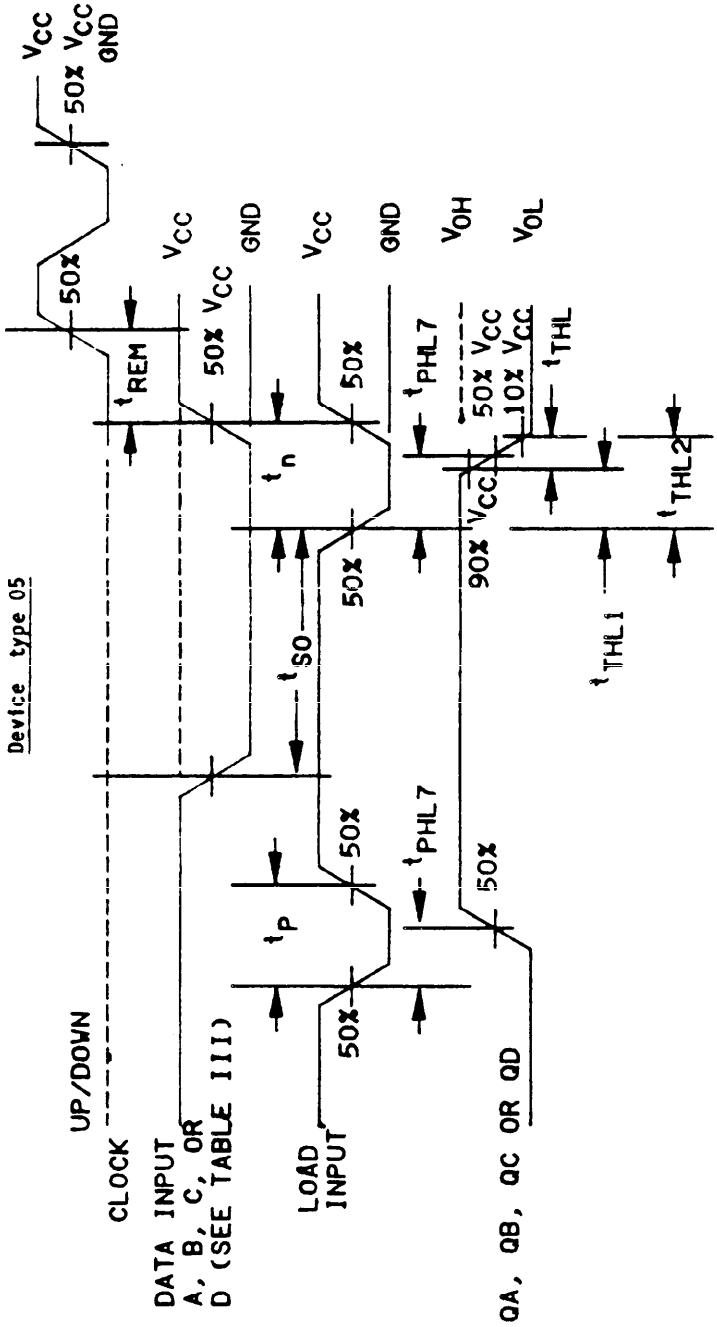
Device types 01, 02, 03, and 04



NOTES:
1. The pulse generator has the following characteristics: $V_{gen} = V_{CC}$ $t_r \leq 6$ ns, $t_f \leq 6$ ns, PRR $\leq$ 1 MHz.
2. C_L Includes probe and jig capacitance.
3. Voltage values are with respect to ground terminal.
4. f_{MAX}: $t_r = t_f \leq 6$ ns.

FIGURE 6. Switching test circuit and waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
Check the source to verify that this is the current version before use.



NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns, tp (clock) ≤ 30 ns.
2. Data input pulse characteristics: $t_r = t_f \leq 6$ ns; tp (data) ≤ 55 ns; $t_{SETUP} \leq 45$ ns; $t_{HOLD} \leq 0$ ns.
3. The clock input characteristics for f_{MAX} are as follows: $t_r = t_f \leq 6$ ns; tp (clock) ≤ 29 ns; PRR ≥ 17 MHz.
4. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
5. $t_{TLH} = t_{TLH2} - t_{TLH1}$; $t_{THL} = t_{THL2} - t_{THL1}$.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit and waveforms - Continued.

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Check the source to verify that this is the currentversion before use.

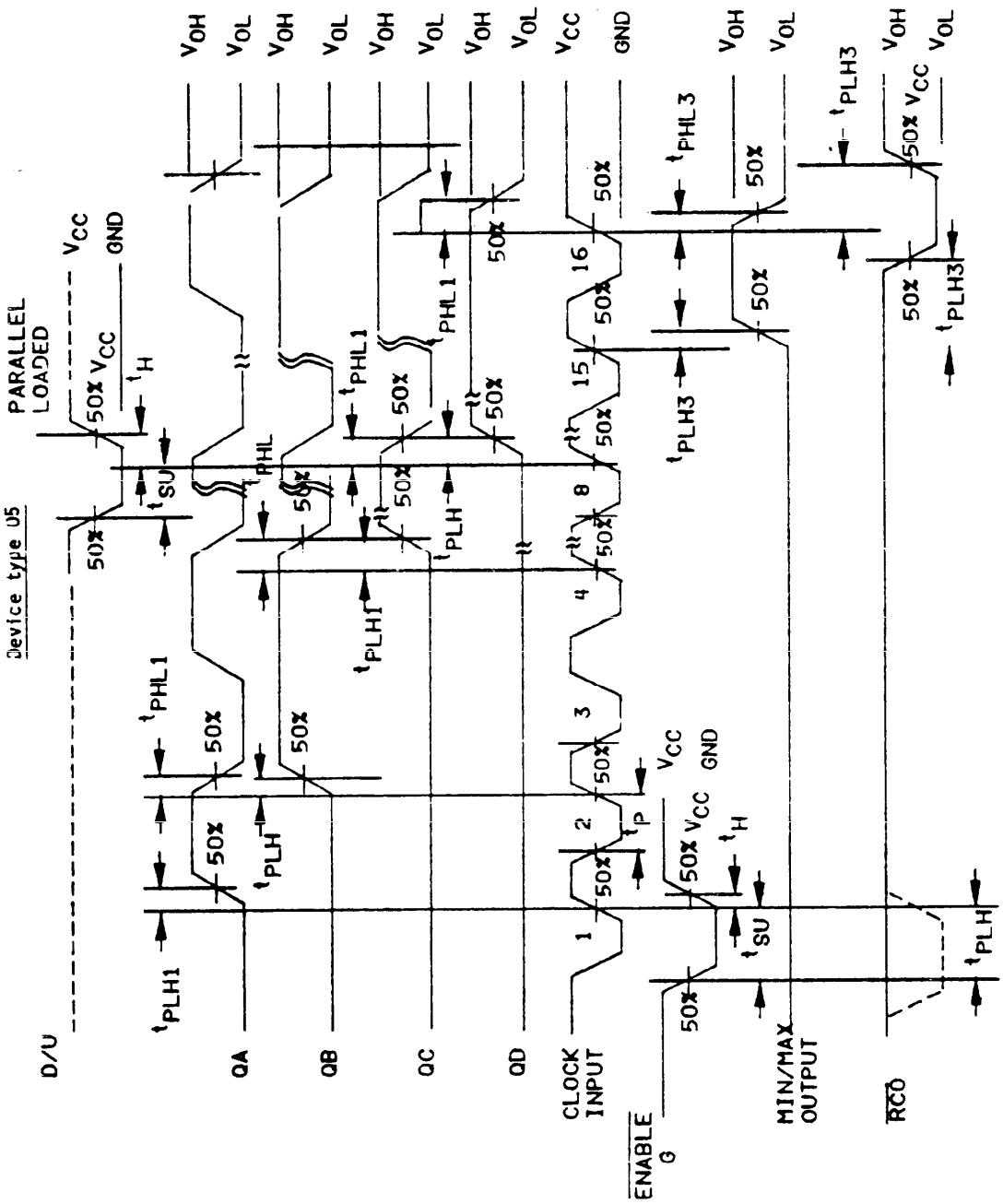
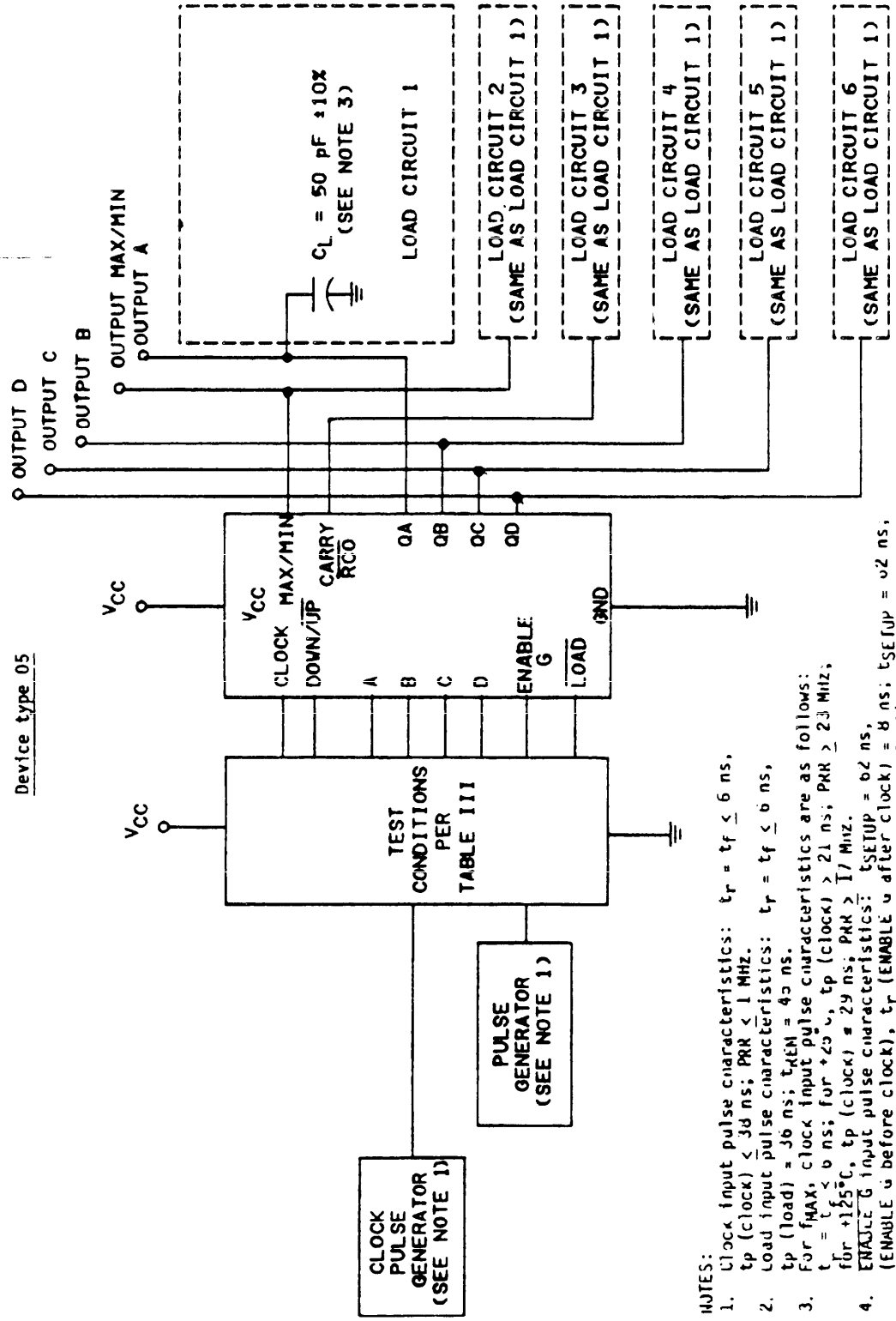


FIGURE 6. Switching test circuit and waveforms - Continued.

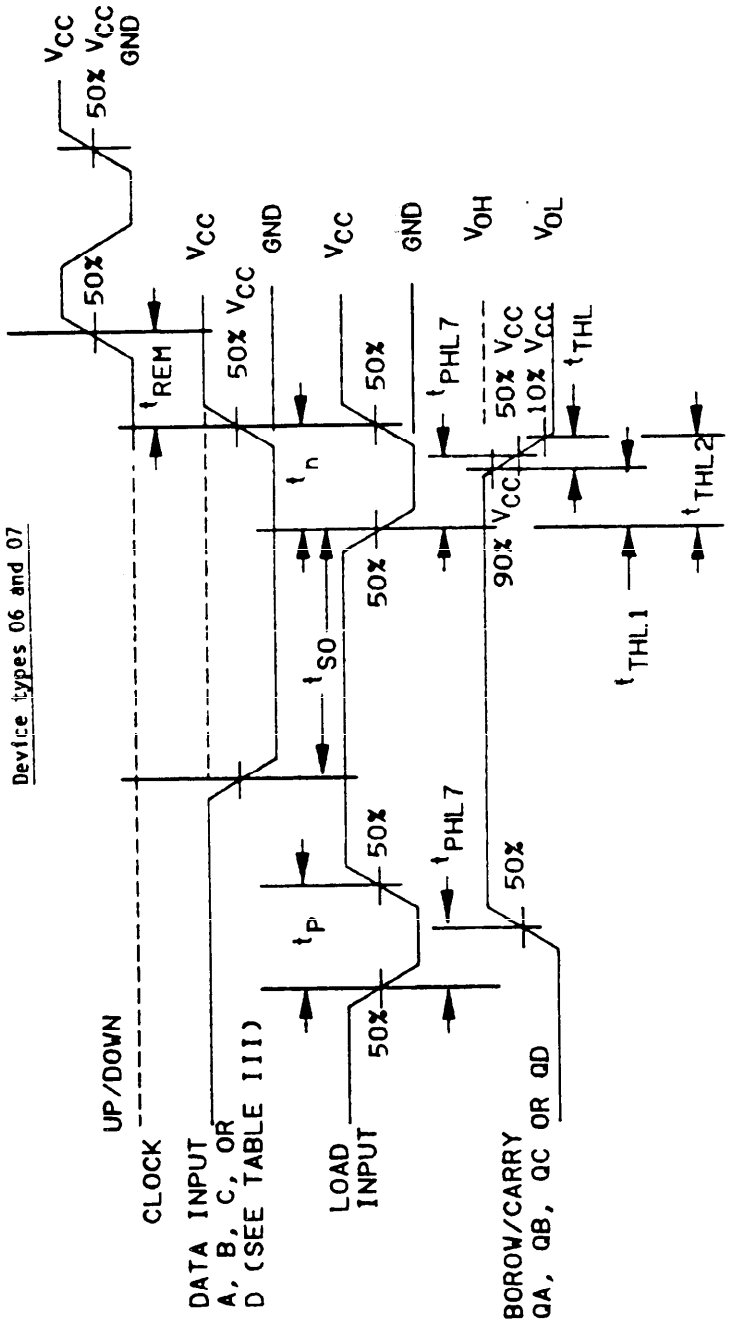
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns, tp (clock) ≤ 38 ns; PRR ≤ 1 MHz.
2. Load input pulse characteristics: $t_r = t_f \leq 6$ ns, tp (load) = 36 ns; t_{WEH} = 45 ns.
3. For f_{MAX}, clock input pulse characteristics are as follows: $t_r = t_f \leq 6$ ns; for +25°C, tp (clock) ≥ 21 ns; PRR ≥ 23 MHz; for +125°C, tp (clock) ≥ 29 ns; PRR ≥ 17 MHz.
4. ENABLE G input pulse characteristics: t_{SETUP} = 62 ns, (ENABLE G before clock), t_r (ENABLE G after clock) = 8 ns; t_{SETUP} = 62 ns; (down/up before clock), t_r = 8 ns; (down/up after clock).
5. C_L = 50 pF ±10 percent (including probe and jig capacitance).

FIGURE 6. Switching test circuit and waveforms - Continued.

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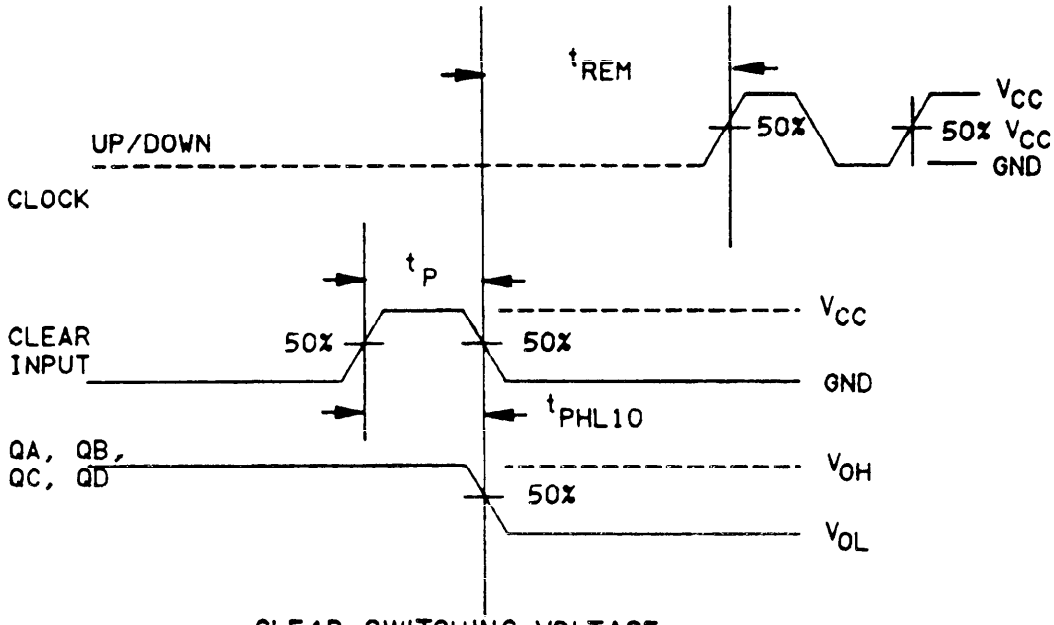


NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns; tp (clock) ≤ 45 ns.
2. Data input pulse characteristics: $t_r = t_f \leq 6$ ns; tp (data) ≤ 41 ns; $t_{SETUP} \leq 33$ ns; $t_{HOLD} \leq 3$ ns;
3. The clock input characteristics for f_{MAX} are as follows: $t_r = t_f \leq 6$ ns; tp (clock) ≤ 36 ns; PRR ≥ 14 MHz.
4. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
5. Voltage measurements are to be made with respect to network ground terminal.
6. $t_{TLH} = t_{TLH1} - t_{TLH2}$; $t_{THL} = t_{THL2} - t_{THL1}$.

FIGURE 6. Switching test circuit and waveforms - Continued.

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Device types 06 and 07



CLEAR SWITCHING VOLTAGE WAVEFORM

FIGURE 6. Switching test circuit and waveforms - Continued.

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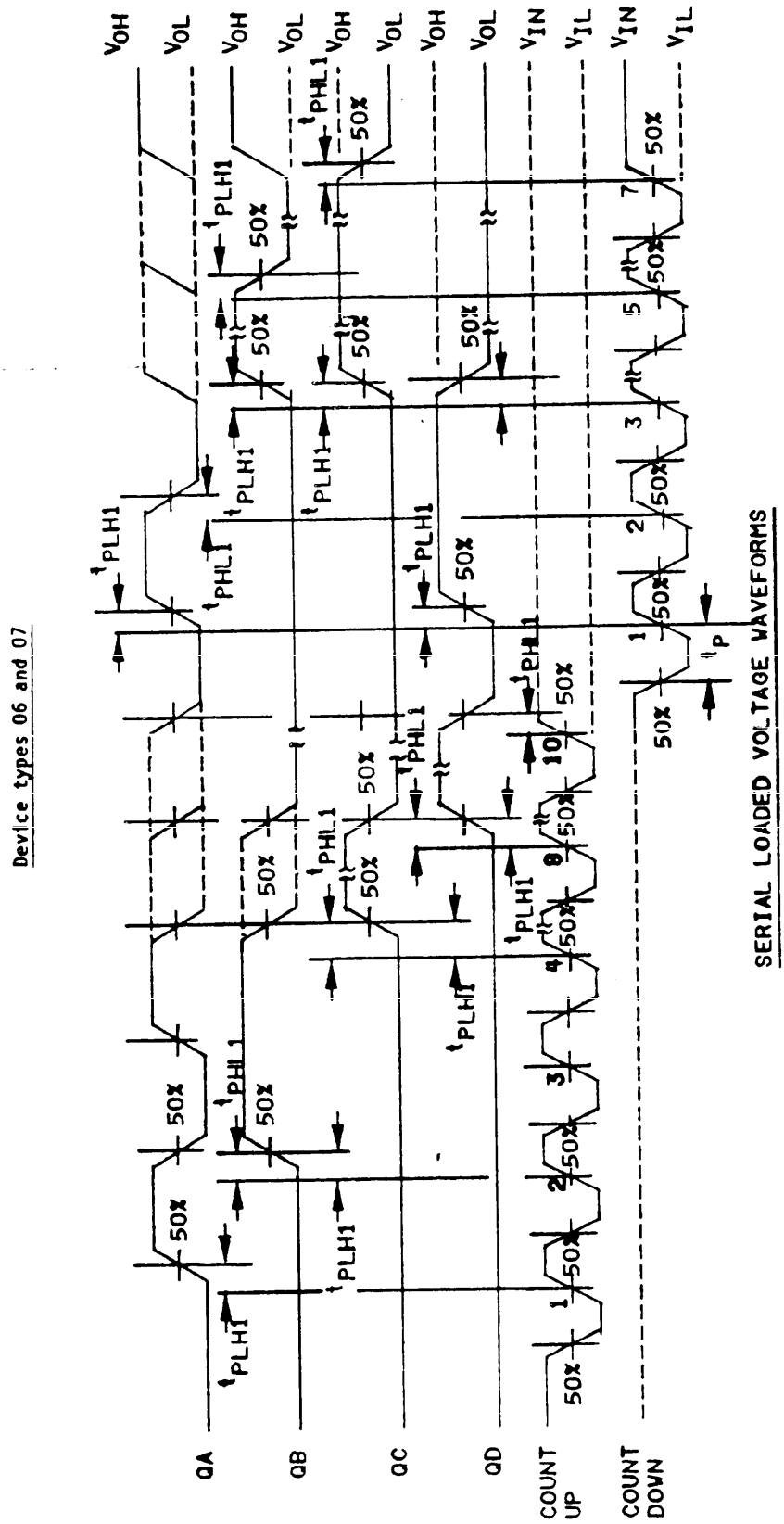
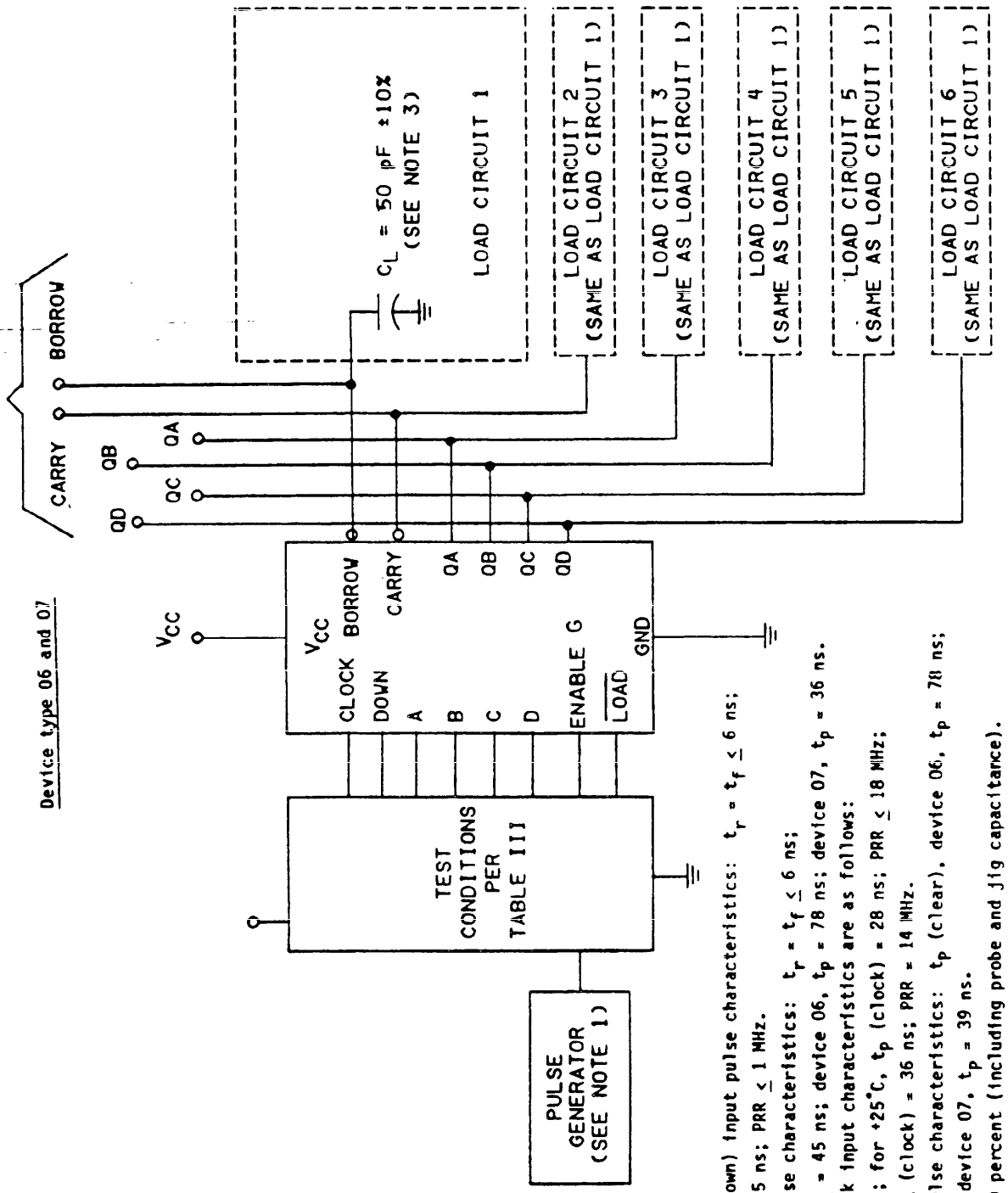


FIGURE 6. Switching test circuit and waveforms - Continued.

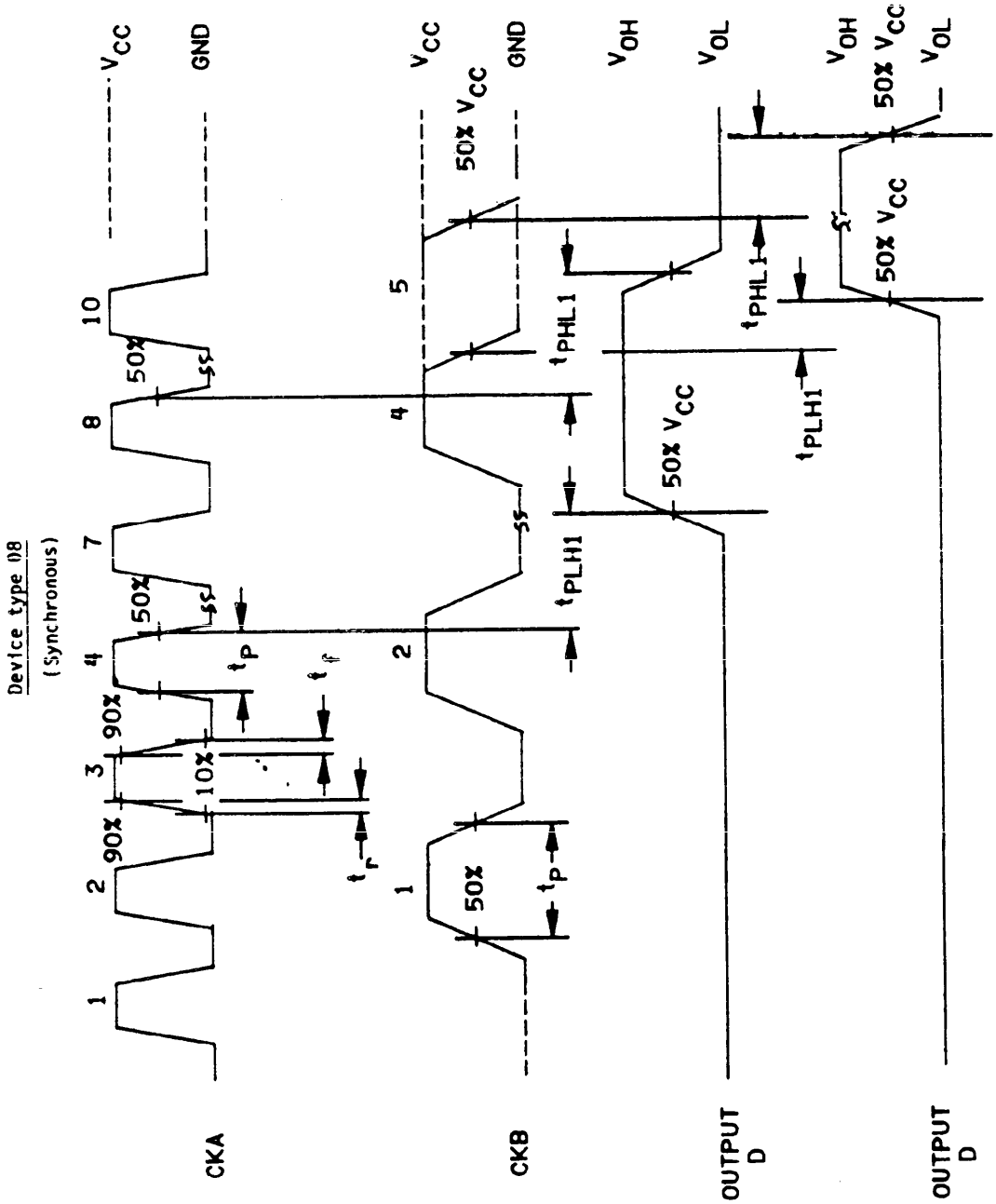
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:59Z
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NOTES:
1. Clock (up or down) input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (clock) $\leq$ 45 ns; PRR $\leq$ 1 MHz.
2. Load input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (load) t_{REM} = 45 ns; device 06, t_p = 78 ns; device 07, t_p = 36 ns.
3. For f_{MAX}, clock input characteristics are as follows: $t_r = t_f \leq 6$ ns; for +25°C, t_p (clock) = 28 ns; PRR $\leq$ 18 MHz; for +125°C, t_p (clock) = 36 ns; PRR = 14 MHz.
4. Clear input pulse characteristics: t_p (clear), device 06, t_p = 78 ns; t_{REM} = 33 ns; device 07, t_p = 39 ns.
5. C_L = 50 pF ±10 percent (including probe and jig capacitance).

FIGURE 6. Switching test circuit and waveforms - Continued.

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COUNT VOLTAGE WAVEFORMS

FIGURE 6. Switching test circuit and waveforms - Continued.

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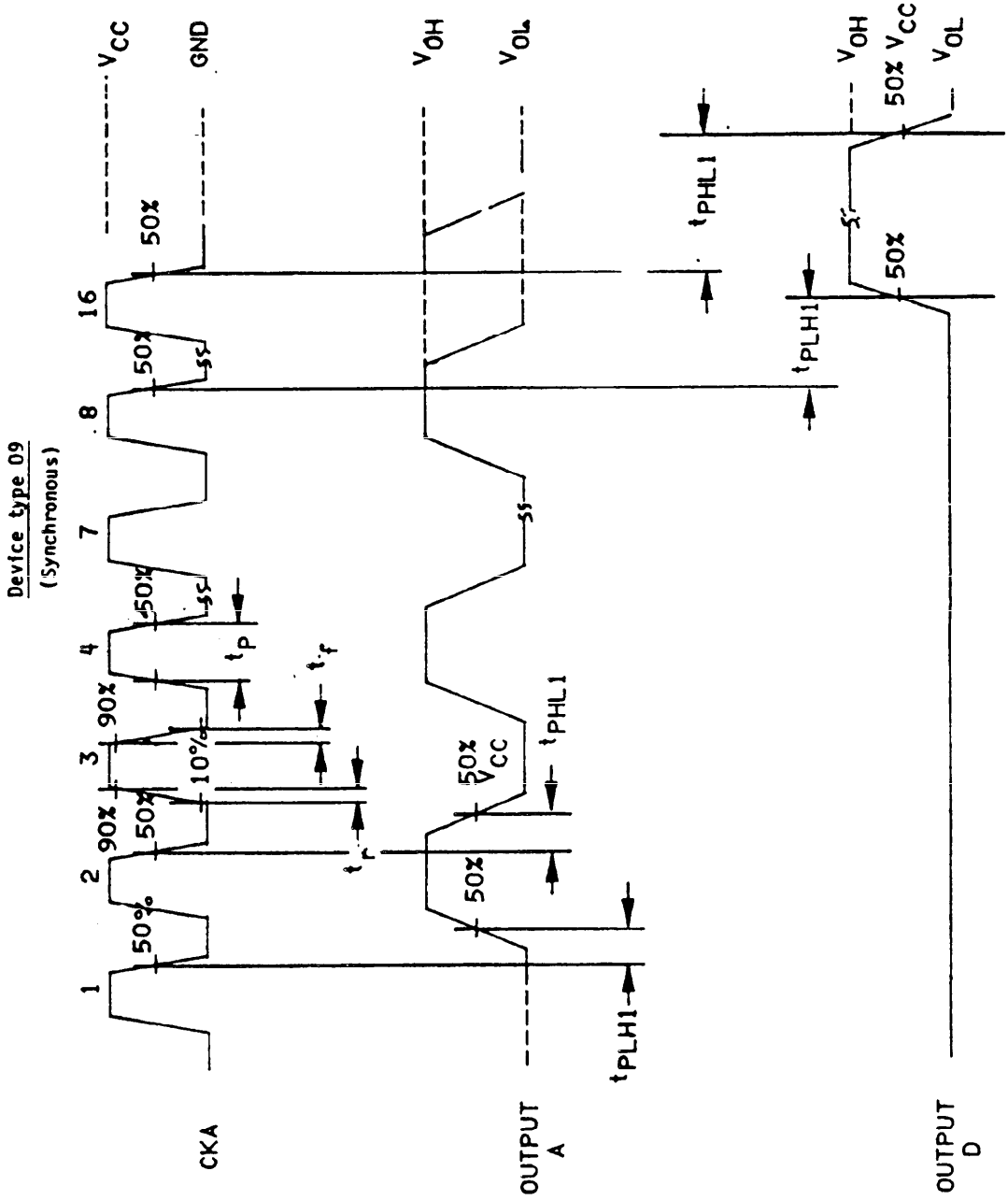
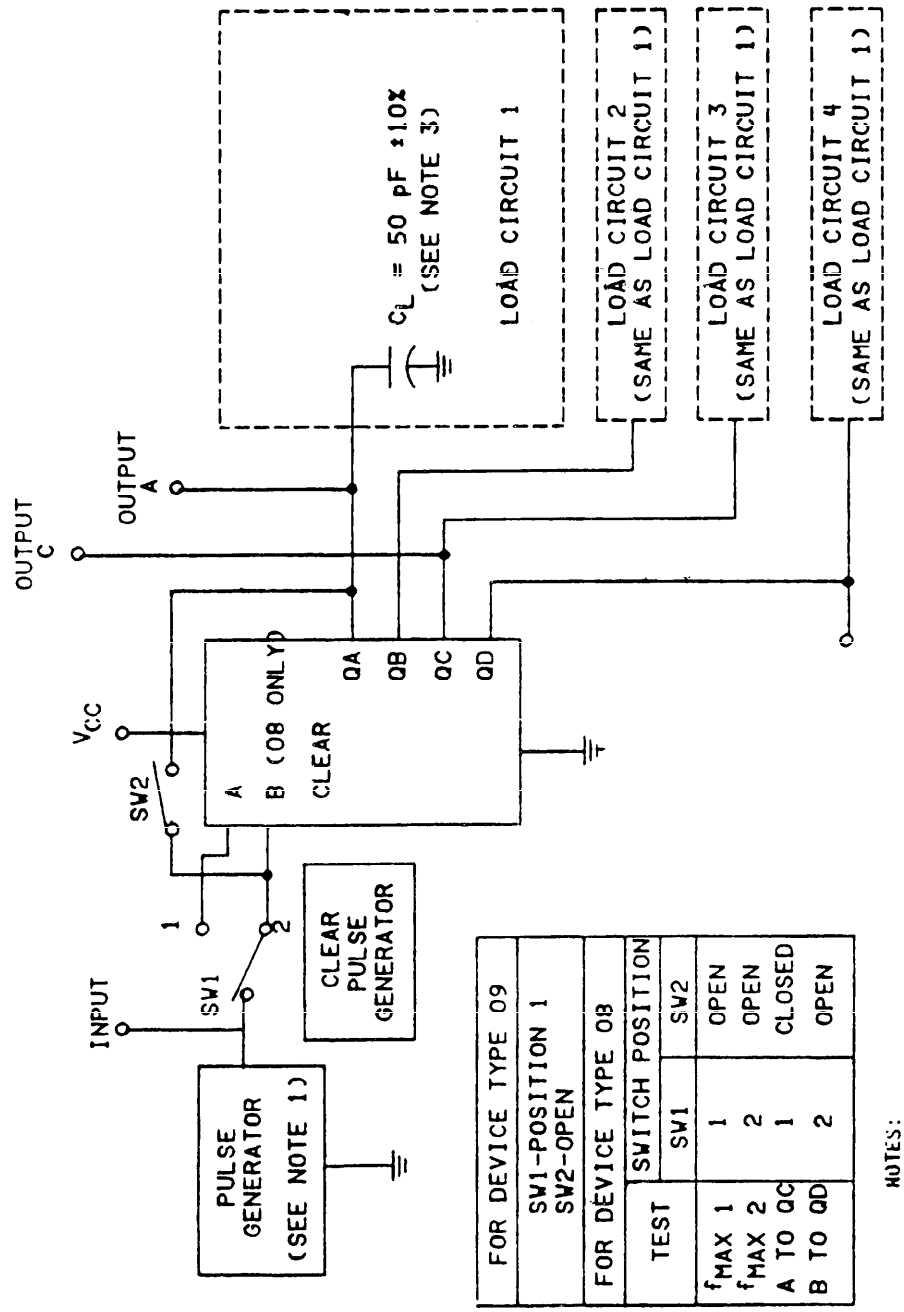


FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 08 and 09



FOR DEVICE TYPE 09			
SW1-POSITION 1 SW2-OPEN			
FOR DEVICE TYPE 08			
TEST	SWITCH POSITION		
	SW1	SW2
f_{MAX} 1 f_{MAX} 2	1 2	OPEN OPEN
A TO QC	1	CLOSED
B TO QD	2	OPEN

NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns; PRR < 1 MHz. Device type 08, tp (clock) $\leq$ 20 ns; device type 09, tp (clock) $\leq$ 24 ns.
2. For f_{MAX}, clock input pulse characteristics are as follows: $t_r = t_f \leq 6$ ns; device type 08, tp (clock) $\leq$ 22 ns; PRR $\geq$ 23 MHz; device type 09, tp (clock) $\leq$ 24 ns; PRR $\geq$ 21 MHz.
3. Clear input pulse characteristics: tp(clear) $\leq$ 38 ns; device type 08, $t_{REM} \leq$ 33 ns; device type 09, $t_{REM} \leq$ 15 ns.
4. C_L = 50 pF ±10 percent (including probe and jig capacitance).

FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 10

(Synchronous)

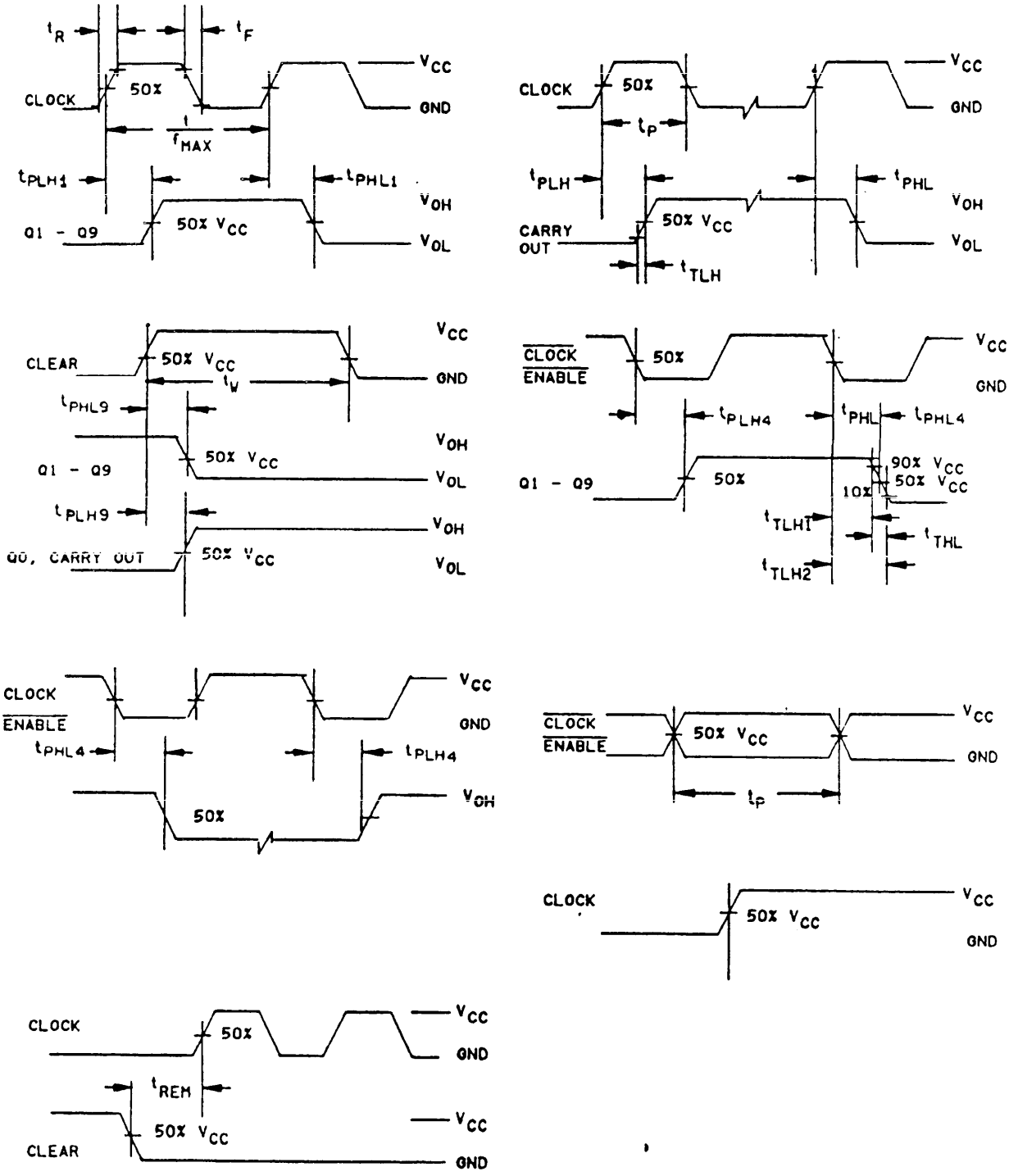
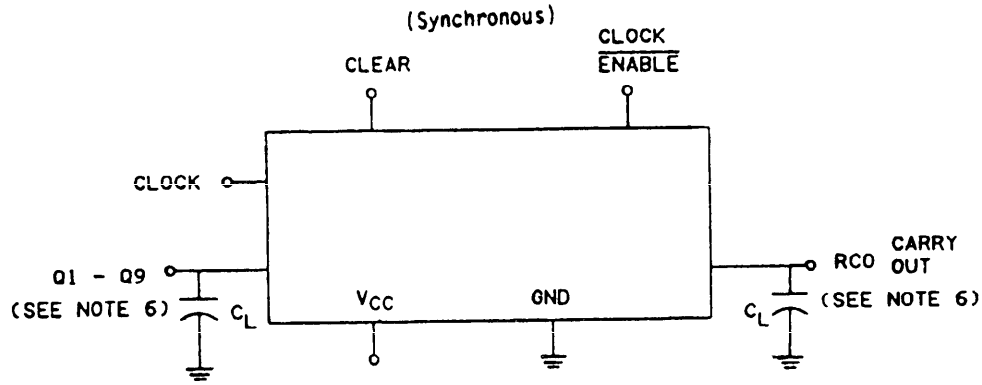


FIGURE 6. Switching test circuit and waveforms - Continued.

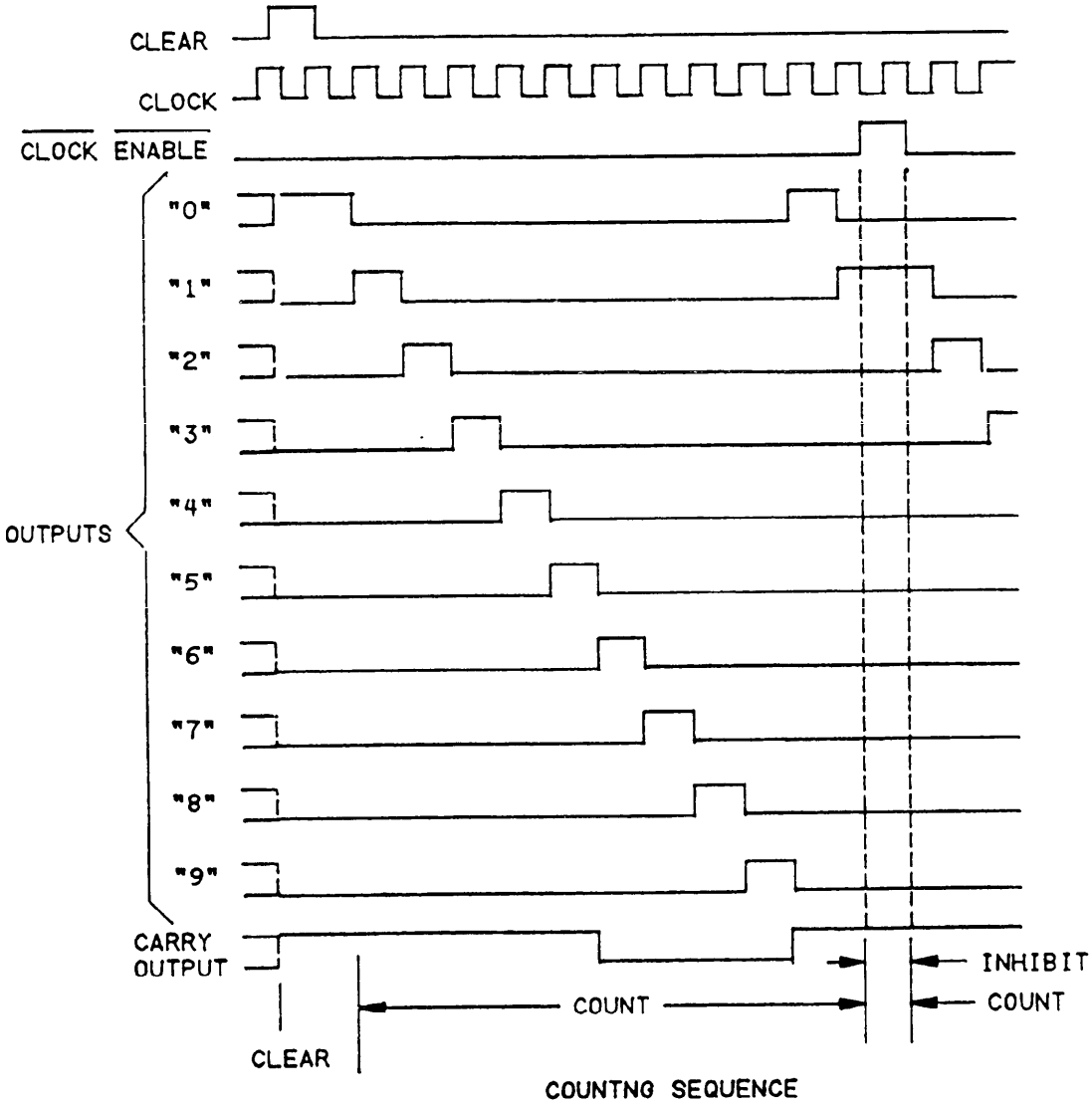
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Device type 10

(Synchronous)



TEST CIRCUIT



COUNTNG SEQUENCE

FIGURE 6. Switching test circuit and waveforms - Continued.

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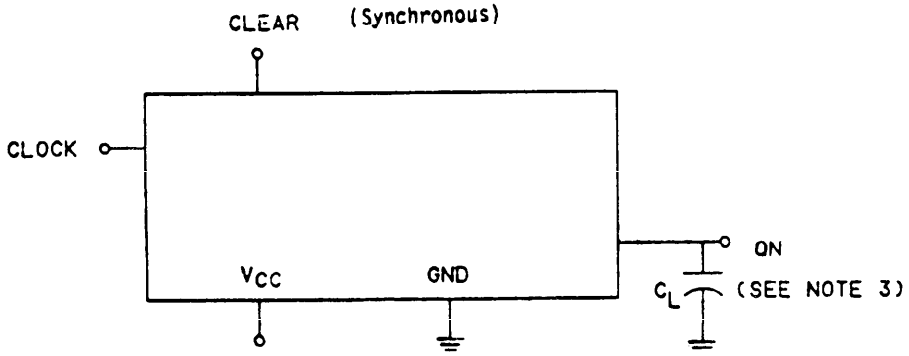
Device type 10

NOTES:
1. $\overline{\text{Clock enable}}$ pulse is active low and dominates regardless of the state of the clock.
2. Clear input pulse characteristics are as follows: $t_r = t_f \leq 6$ ns; t_p (clear) ≤ 24 ns; PRR ≤ 1 MHz, $t_{REM} \leq 30$ ns;
3. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (clock) ≤ 30 ns; PRR $= \leq 1$ MHz.
4. $\overline{\text{Clock enable}}$ pulse characteristics are as follows: $t_r = t_f \leq 6$ ns; t_p ($\overline{\text{enable}}$) ≤ 30 ns; PRR ≤ 1 MHz.
5. The clock input characteristics for f_{MAX} are as follows: $t_r = t_f \leq 6$ ns; t_p (clock) ≤ 29 ns; PRR ≤ 17 MHz.
6. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
7. Voltage measurements are to be made with respect to network ground terminal.
8. $t_{TLH} = t_{TLH1} - t_{TLH2}$; $t_{THL} = t_{THL2} - t_{THL1}$.

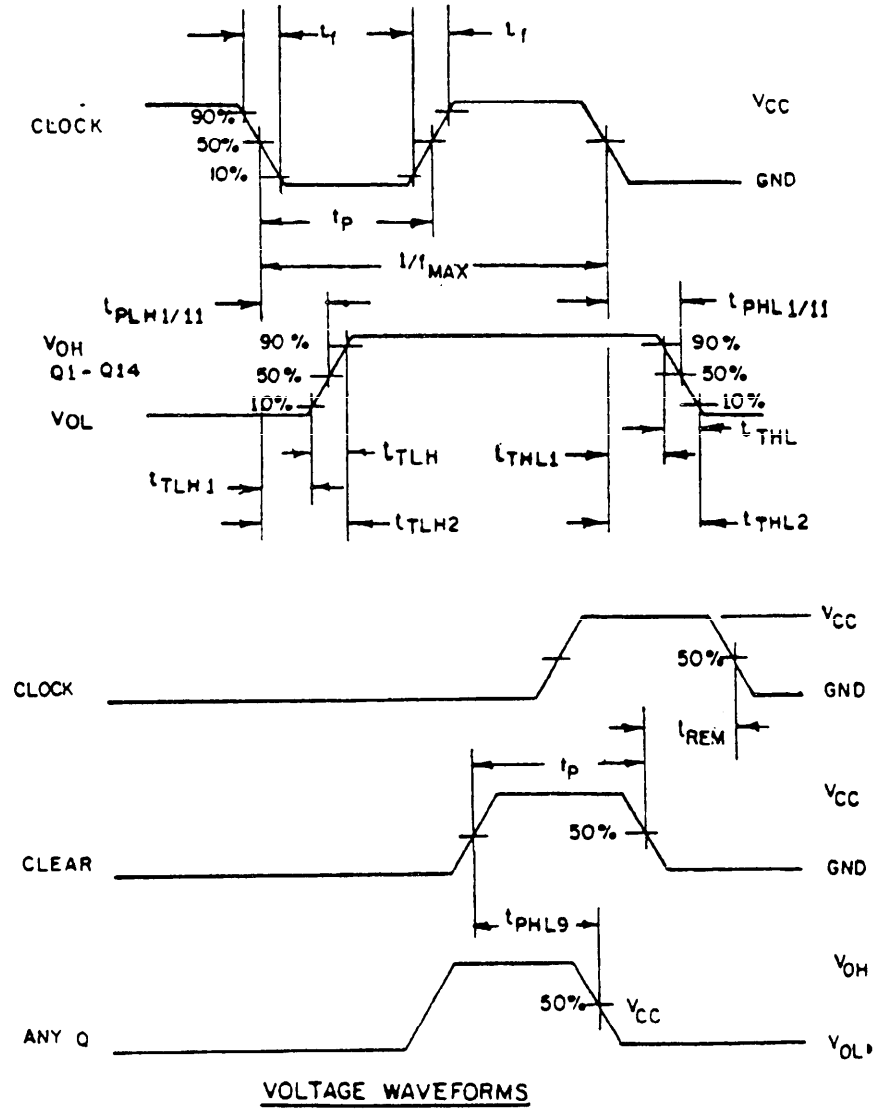
FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 11



TEST CIRCUIT



VOLTAGE WAVEFORMS

FIGURE 6. Switching test circuit and waveforms - Continued.

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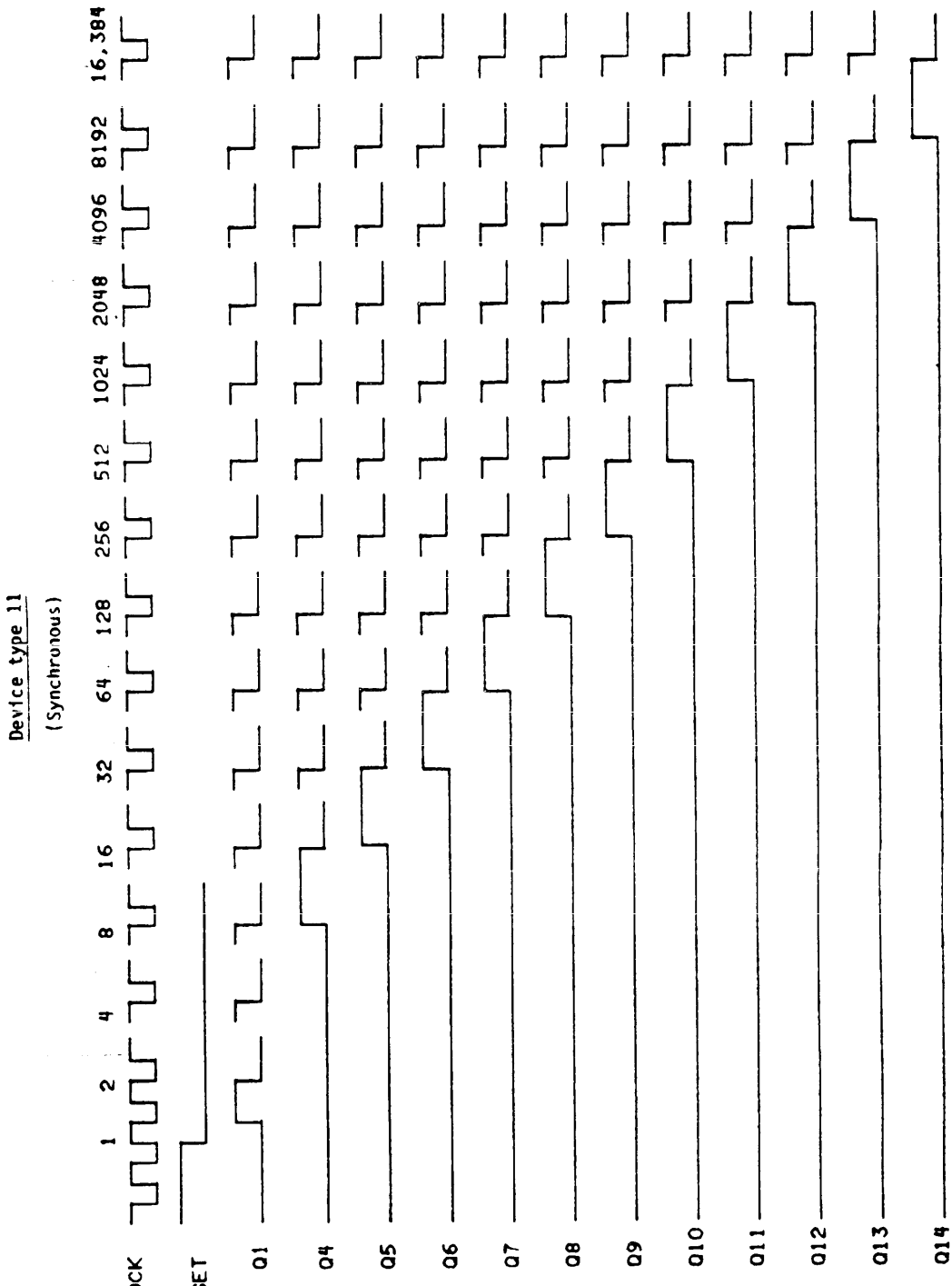


FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 11

NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (clock) ≤ 27 ns.
2. Clear input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (clear) ≤ 24 ns; $t_{REM} \leq 30$ ns.
3. The clock input characteristics for f_{MAX} are as follows: $t_r = t_f \leq 6$ ns; t_p (clock) ≤ 24 ns; PRR ≥ 21 MHz.
4. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
5. $t_{TLH} = t_{TLH2} - t_{TLH1}$; $t_{THL} = t_{THL2} - t_{THL1}$.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 12

(Synchronous)

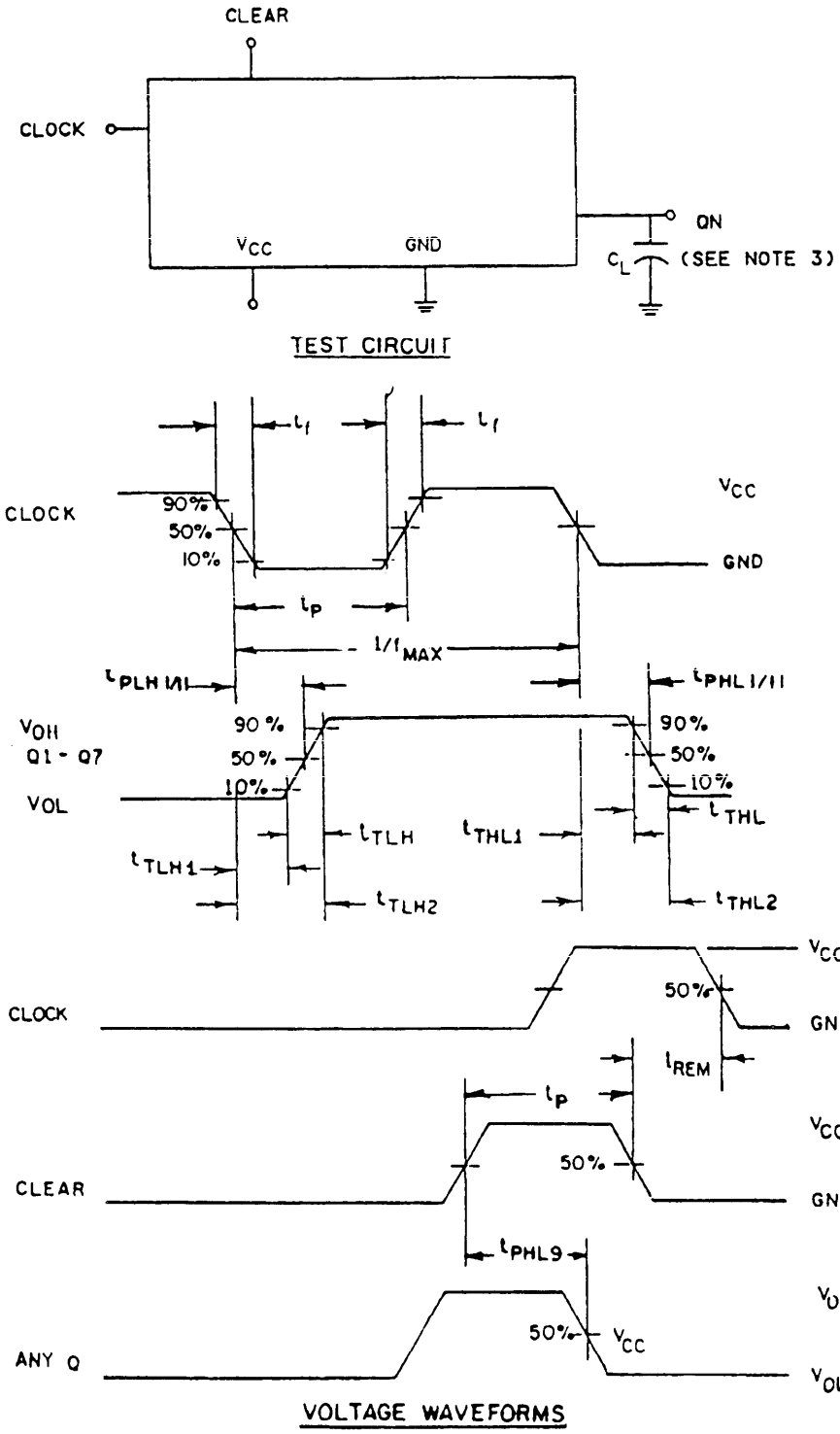
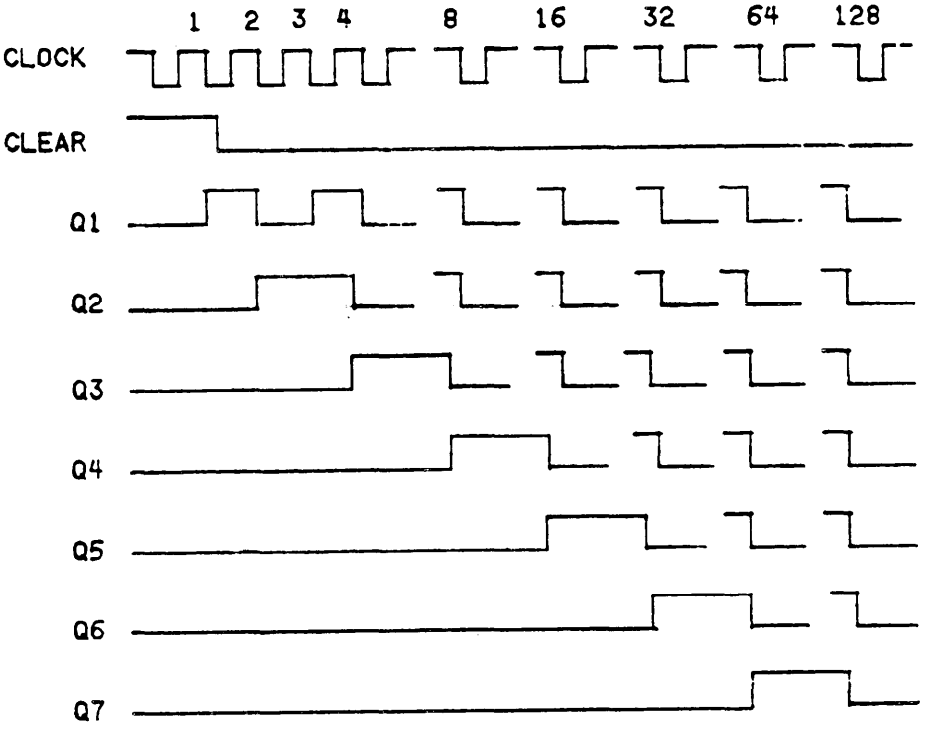


FIGURE 6. Switching test circuit and waveforms - Continued.

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producing potential (noise) drops within the equipment. Conversely, no conducted circulating currents are

Device type 12



COUNTNG SEQUENCE

NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (clock) ≤ 27 ns.
2. Clear input pulse characteristics: $t_r = t_f \leq 6$ ns; t_p (clear) ≤ 24 ns, $t_{REM} \leq 30$ ns.
3. The clock input characteristics for f_{MAX} are as follows: $t_r = t_f \leq 6$ ns; t_p (clock) ≤ 24 ns; PRR ≥ 21 MHz.
4. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
5. $t_{TLH} = t_{TLH2} - t_{TLH1}$; $t_{THL} = t_{THL2} - t_{THL1}$.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 13

(Synchronous)

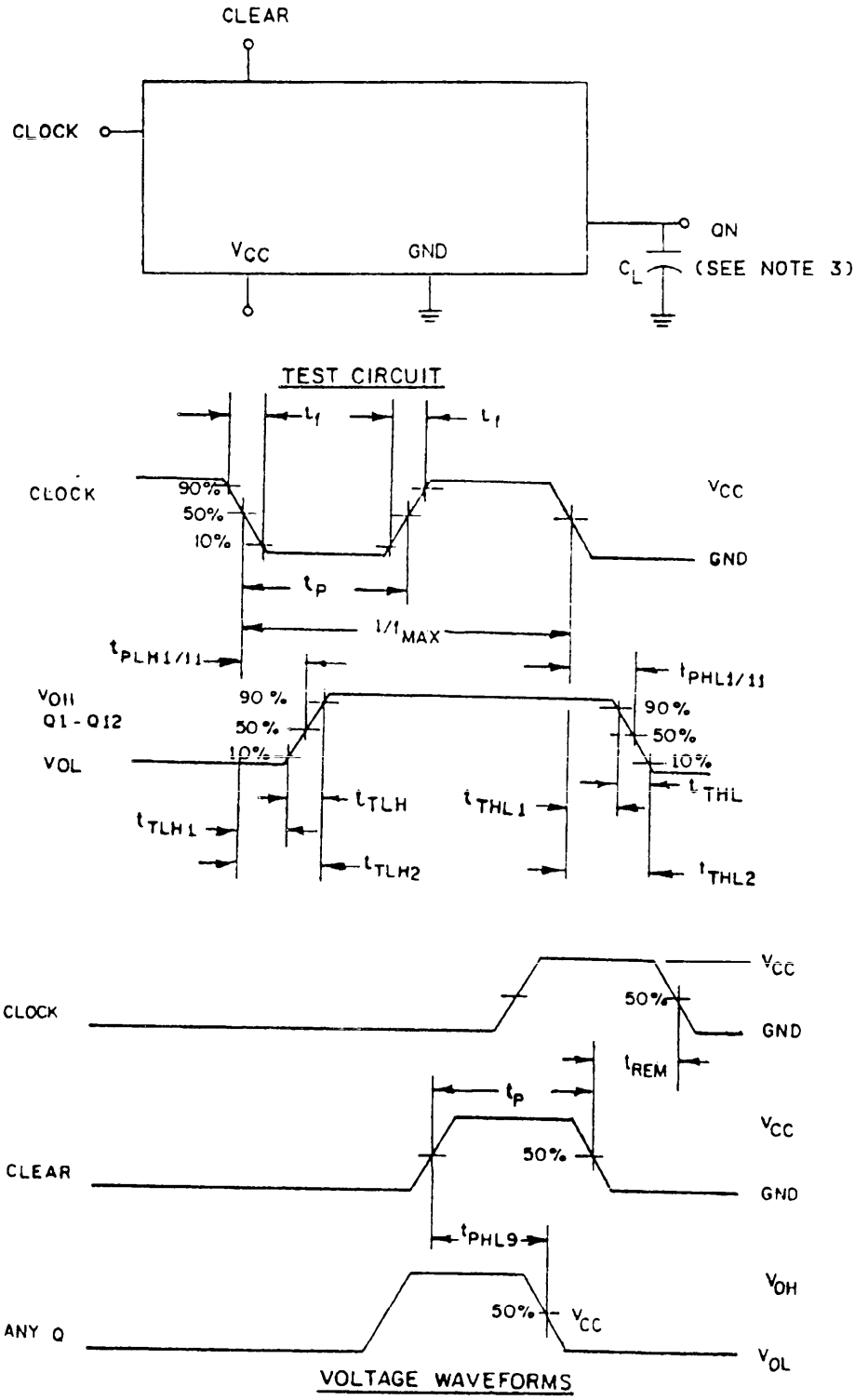


FIGURE 6. Switching test circuit and waveforms - Continued.

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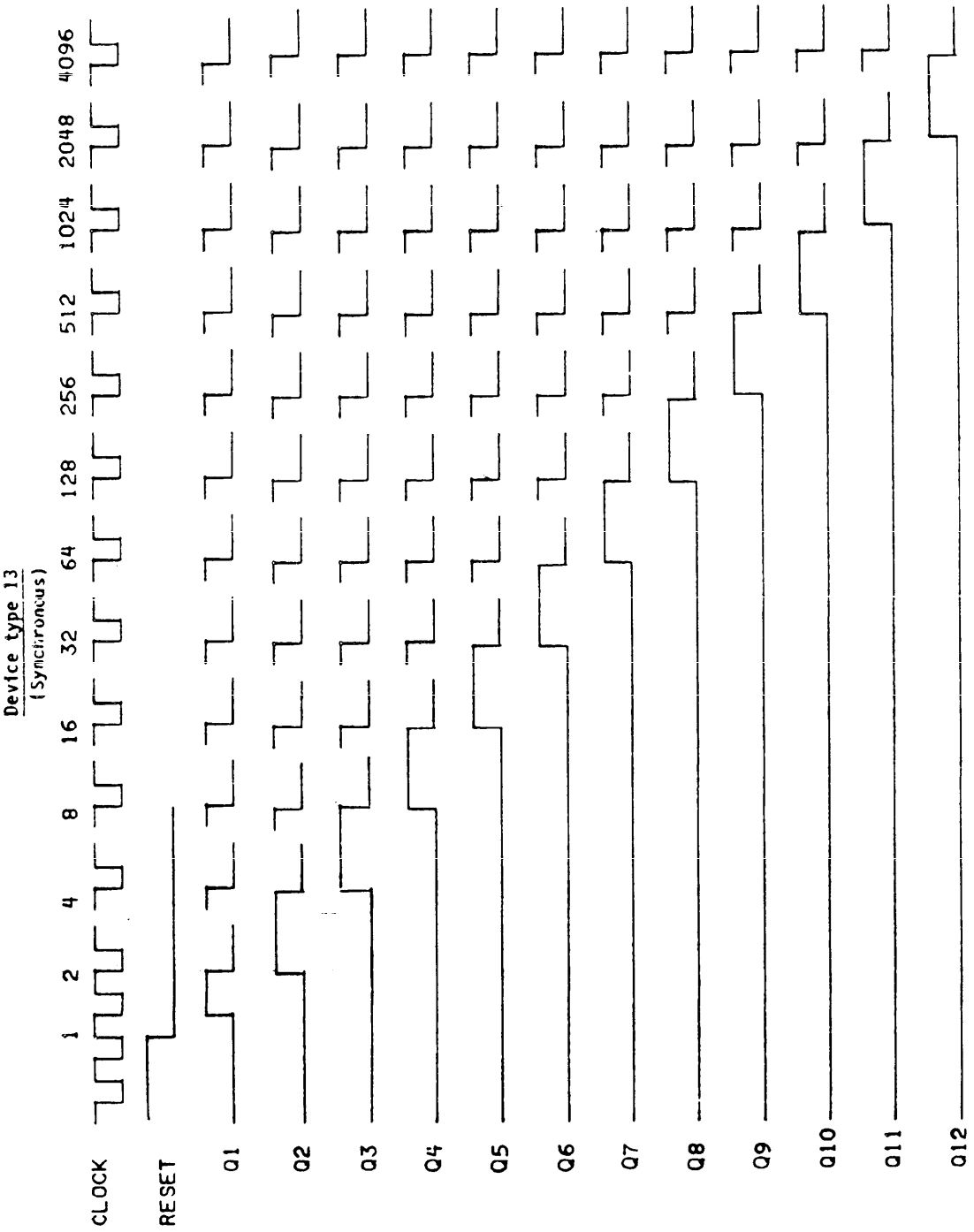


FIGURE 6. Switching test circuit and waveforms - Continued.

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Device type 13

NOTES:
1. Clock input pulse characteristics: $t_r = t_f \leq 6$ ns, t_p (clock) ≤ 27 ns.
2. Clear input pulse characteristics: $t_r = t_f \leq 6$ ns, t_p (clear) ≤ 24 ns; $t_{REM} \leq 30$ ns.
3. The clock input characteristics for f_{MAX} are as follows: $t_r = t_f \leq 6$ ns; t_p (clock) ≤ 24 ns; PRR ≥ 21 MHz.
4. $C_L = 50$ pF ±10 percent (including probe and jig capacitance).
5. $t_{TLH} = t_{TLH2} - t_{TLH1}$; $t_{THL} = t_{THL2} - t_{THL1}$.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit and waveforms - Continued.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Case C_s, F_s, 2	Terminal conditions 17																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CCK	CLK	A	B	C	D	ENABLE P	GND	CCKO	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$ 1/		1 2 3 4 5 6 7 8 9	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1/	1 mA	1 mA						GND	CCK- CLK A B C D ENABLE P CCKO ENABLE T		1.5						V
$V_{IC(neg)}$		10 11 12 13 14 15 16 17 18	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND	-1 mA	-1 mA						1/	CCK CLK A B C D ENABLE P CCKO ENABLE T		-1.5						V
I_{CCH}	3005	19	6.0 V	GND	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V		GND	6.0 V						6.0 V	V_{CC}		0.2		20			μA
I_{CCL}	3005	20	GND	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	GND		6.0 V	GND							V_{CC}		0.2		20			μA
V_{OH3}	3006 2/	21 22 23 24 25	4.2 V	16	4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V			GND	4.2 V	-10 μA	-20 μA	-20 μA	-20 μA	-20 μA		RCO	5.95		5.95		5.95		V
V_{OH5}	3006 2/	26 27 28 29 30			4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V				4.2 V	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA		RCO	5.48		5.2		5.48		V
V_{OL3}	3007 4/	31 32 33 34 35			1.2 V	1.2 V	1.2 V	1.2 V				1.2 V	10 μA	20 μA	20 μA	20 μA	20 μA		RCO		0.05		0.05		0.05	V
V_{OL5}	3007 4/	36 37 38 39 40			1.2 V	1.2 V	1.2 V	1.2 V		GND		1.2 V	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA		RCO		0.26		0.4		0.26	V
I_{OS4}	3011 5/	41 42 43 44 45	4.0 V		4.0 V 4.0 V	4.0 V 6/	4.0 V 6/	4.0 V 4.0 V	4.0 V		GND	4.0 V	GND	GND	GND	GND	GND	4.0 V	RCO	-10	-120	-10	-120	10	-120	mA

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case E_2, X	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLR	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QB	QC	QD	QA	RCO	V_{CC}								
I_{IH}	3010	46	6.0 V	GND	GND	GND	GND	GND	GND	GND	GND	GND						6.0 V	CLK		0.05		0.1			µA
		47	GND	6.0 V	6.0 V	6.0 V	GND	GND											CLR							
		48		GND	GND	GND	6.0 V												A							
		49			6.0 V	GND	GND												B							
		50			GND	6.0 V		6.0 V											C							
		51				GND	6.0 V	GND											D							
		52					GND		6.0 V										ENABLE P							
		53							GND		6.0 V								LOAD							
		54									GND	6.0 V							ENABLE T							
I_{IL}	3009	55	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V		6.0 V								CLK		-0.05		-0.1			
		56		GND	6.0 V														CLR							
		57		6.0 V	GND														A							
		58			6.0 V	GND													B							
		59				6.0 V	GND												C							
		60					6.0 V	GND											D							
		61							GND		6.0 V								ENABLE P							
		62							6.0 V		6.0 V								LOAD							
		63									GND	GND	1/	1/	1/	1/			ENABLE T							
																				Subgroup 4 T_C = +25°C Min	Max					
C_C	3012	64	0/	0/						GND	0/	0/						GND	CLK		15					pF
		65																	CLR							
		66																	LOAD							
		67							0/										ENABLE T							
		68																	ENABLE P							
C_{IN}	3012	69			0/	0/													A		10					pF
		70					0/												B							
		71						0/											C							
		72																	D							
																			Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	
Truth table tests 10/ 11/	3014	73	B	A	A	A	A	A	A	A	A	A	L	L	L	L	L	0.5 V	All outputs 12/ 13/							
		74	A	B	A	A	A	A	A	A	A	A				H										
		75			B	A	A	A	A	A	A	A					L									
		76				B	A	A	A	A	A	A					H									
		77					B	A	A	A	A	A						L								
		78						B	A	A	A	A						H								
		79							B	A	A	A							L							
		80								B	A	A							H							
		81									B	A							L							
		82										B							H							
		83											B	B	B	B	B	B	L							
		84																	H							
		85																	L							
		86																	H							
		87																	L							
		88																	H							
		89																	L							
		90																	H							
		91																	L							
		92																	H							
		93																	L							
		94																	H							
		95																	L							
		96																	H							

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case E,F,X	Case 2	Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal 9/	Test limits Subgroup 7 TC = +25°C Min	Max	Subgroup 8 TC = +125°C Min	Max	Subgroup 9 TC = -55°C Min	Max	Unit
						2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20								
						CE1	CLK	A	B	C	D	ENABLE P	CR0	CR0	ENABLE T	QC	QB	QA	RCO	Vcc											
Truth table tests 10/ 11/	3014			97																					All outputs 12/ 13/							0.5 V
				98																												
				99																												
				100																												
				101																												
				102																												
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See footnotes at end of table III.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD 883 method	Case E,F,K	Terminal conditions 17/																	Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 14/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 14/		
		Test no.	CEN	CLK	A	B	C	D	ENABLE P	GND	ICRO	ENABLE T	QD	QC	QB	QA	CCO	V_{CC}		Min	Max	Min	Max	Min	Max	
f_{MAX} 15/	3003	168	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		4.5 V	QA	20		21		20		MHz
t_{PLH1}	3003 Fig. 6	169	•	•					•	•	•	•				OUT		•	CLK to QA	0	36	0	48	0	36	ns
		170	•	•					•	•	•	•						•	CLK to QB	•	•	•	•	•	•	•
		171	•	•					•	•	•	•		OUT	OUT			•	CLK to QC	•	•	•	•	•	•	•
		172	•	•					•	•	•	•	ONI					•	CLK to QD	•	•	•	•	•	•	•
t_{PHL1}	3003 Fig. 6	173	•	•					•	•	•	•				OUT		•	CLK to QA	•	•	•	•	•	•	•
		174	•	•					•	•	•	•						•	CLK to QB	•	•	•	•	•	•	•
		175	•	•					•	•	•	•		OUT	OUT			•	CLK to QC	•	•	•	•	•	•	•
		176	•	•					•	•	•	•	ONI					•	CLK to QD	•	•	•	•	•	•	•
t_{PLH2}	3003 Fig. 6	177	•	•					•	•	•	•					OUT	•	CLK to RCO	•	38	•	50	•	38	•
t_{PHL2}	3003 Fig. 6	178	•	•					•	•	•	•					OUT	•	CLK to RCO	•	38	•	50	•	38	•
t_{PLH4}	3003 Fig. 6	179	•	7/					•	•	•	IN					OUT	•	ENABLE T to RCO	7	34	7	46	7	34	•
t_{PHL4}	3003 Fig. 6	180	•	7/					•	•	•	IN					OUT	•	ENABLE T to RCO	7	34	7	46	7	34	•
t_{PHL9}	3003 Fig. 6	181	IN	16/	4.5 V	4.5 V	4.5 V	4.5 V		•	GND					OUT		•	CEN to QA	0	40	0	53	0	40	•
		182	•	•						•	•							•	CEN to QB	•	•	•	•	•	•	•
		183	•	•						•	•			OUT	OUT			•	CEN to QC	•	•	•	•	•	•	•
		184	•	•						•	•		ONI					•	CEN to QD	•	•	•	•	•	•	•
t_{PHL10}	3003 Fig. 6	185	•	8/	4.5 V	4.5 V	4.5 V	4.5 V		•	•						OUT	•	CEN to RCO	•	38	•	51	•	38	•
t_{TLH}	3003 Fig. 6	186	4.5 V	18/	GND	GND	GND	GND	4.5 V	•	•	•				OUT		•	QA	3	15	3	20	3	15	•
		187	•	•	4.5 V	4.5 V	•	•	•	•	•	•						•	QB	•	•	•	•	•	•	•
		188	•	•	•	4.5 V	4.5 V	•	•	•	•	•		OUT	OUT			•	QC	•	•	•	•	•	•	•
		189	•	•	•	GND	GND	4.5 V	•	•	•	•	ONI					•	QD	•	•	•	•	•	•	•
		190	•	•	GND				4.5 V	•	•	•					OUT	•	RCO	•	•	•	•	•	•	•
t_{THL}	3003 Fig. 6	191	•	•	4.5 V	GND	GND	GND	•	•	•	•				OUT		•	QA	•	•	•	•	•	•	•
		192	•	•	•	4.5 V	•	•	•	•	•	•						•	QB	•	•	•	•	•	•	•
		193	•	•	•	4.5 V	4.5 V	•	•	•	•	•		OUT	OUT			•	QC	•	•	•	•	•	•	•
		194	•	•	•	GND	GND	4.5 V	•	•	•	•	ONI					•	QD	•	•	•	•	•	•	•
		195	•	•	•	GND	GND	4.5 V	•	•	•	•					OUT	•	RCO	•	•	•	•	•	•	•

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TABLE III. Group A inspection for device type 32.

Symbol	MIL-STD-883 method	Case C, F, H	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLEAR	CLK	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}								
$V_{IC(pos)}$ 1/		1 2 3 4 5 6 7 8 9	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1/	1 mA	1 mA						GND	CLEAR CLK A B C D ENABLE P LOAD ENABLE T		1.5					V
$V_{IC(neg)}$		10 11 12 13 14 15 16 17 18	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND	-1 mA	-1 mA						1/	CLEAR CLK A B C D ENABLE P LOAD ENABLE T		1.5					V
I_{CCH}	3005	19	6.0 V	GND	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V		GND	6.0 V						6.0 V	V_{CC}		0.2		20			µA
I_{CCL}	3005	20	GND	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	GND		6.0 V	GND							V_{CC}		0.2		20			µA
V_{OH3}	3006 2/	21 22 23 24 25	6.0 V	TH	4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V			GND	1.2 V	20 µA	-20 µA	-20 µA	-20 µA	-20 µA		QA QB QC QD RCO	3.95		3.95		3.95		V
V_{OH5}	3006 2/	26 27 28 29 30			4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V				1.2 V	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA		QA QB QC QD RCO	3.48		3.7		3.48		V
V_{OL3}	3007 4/	31 32 33 34 35			1.2 V	1.2 V	1.2 V	1.2 V				1.2 V	20 µA	20 µA	20 µA	20 µA	20 µA		QA QB QC QD RCO		0.05		0.05		0.05	V
V_{OL5}	3007 4/	36 37 38 39 40			1.2 V	1.2 V	1.2 V	1.2 V		GND		1.2 V	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA		QA QB QC QD RCO		0.26		0.4		0.26	V
I_{OS4}	3011 5/	41 42 43 44 45	4.0 V		4.0 V 4.0 V	4.0 V 6/	4.0 V 6/	4.0 V 4.0 V	4.0 V	GND	GND	4.0 V	GND	GND	GND	GND	GND	4.0 V	QA QB QC QD RCO	-10	-120	-10	-120	-10	-120	mA

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLR	CLK	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}								
I_{IH}	3010	46	6.0 V	GND	GND	GND	GND	GND	GND	GND	GND	GND						6.0 V	CLR		0.05		0.1			mA
		47	GND	6.0 V	GND	•	•	•	•	•	•	•						•	CLK		•		•			
		48	•	GND	6.0 V	•	•	•	•	•	•	•						•	A		•		•			
		49	•	•	GND	6.0 V	•	•	•	•	•	•						•	B		•		•			
		50	•	•	•	GND	6.0 V	•	•	•	•	•						•	C		•		•			
		51	•	•	•	•	GND	6.0 V	•	•	•	•						•	D		•		•			
		52	•	•	•	•	•	GND	4.0 V	•	•	•						•	ENABLE P		•		•			
		53	•	•	•	•	•	•	GND	•	6.0 V	•						•	LOAD		•		•			
		54	•	•	•	•	•	•	GND	•	GND	6.0 V						•	ENABLE T		•		•			
I_{IL}	3009	55	•	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	4.0 V	•	6.0 V	•						•	CLR		-0.05		-0.1			mA
		56	6.0 V	•	GND	GND	•	•	•	•	•	•						•	CLK		•		•			
		57	•	GND	6.0 V	GND	•	•	•	•	•	•						•	A		•		•			
		58	•	6.0 V	GND	6.0 V	•	•	•	•	•	•						•	B		•		•			
		59	•	•	6.0 V	GND	•	•	•	•	•	•						•	C		•		•			
		60	•	•	•	6.0 V	GND	GND	•	•	•	•						•	D		•		•			
		61	•	•	•	•	6.0 V	6.0 V	GND	•	•	•						•	ENABLE P		•		•			
		62	•	•	•	•	•	•	4.0 V	•	•	•						•	LOAD		•		•			
		63	•	•	•	•	•	•	4.0 V	•	GND	GND	7/	7/	7/	7/		•	ENABLE T		•		•			
																				Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max					
C_C	3012	64	8/	8/						GND	8/							GND	CLK		15					pF
		65								•								•	CLR		•					
		66								•								•	LOAD		•					
		67							8/	•		8/						•	ENABLE T		•					
		68								•								•	ENABLE P		•					
C_{IN}	3012	69			8/	8/	8/	8/		•								•	A		10					pF
		70								•								•	B		•					
		71								•								•	C		•					
		72								•								•	D		•					
																			Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 9 T_C = -55°C Min	Subgroup 9 T_C = -55°C Max	
Truth table tests 10/ 11/	3014	73	B	A	A	A	A	A	A	GND	A	A	L	L	L	L	L	4.5 V	All outputs 12/ 13/							
		74	B	B	A	A	A	A	A	•	A	A	•	•	•	•	•	•								
		75	A	B	B	A	A	A	A	•	A	A	•	•	•	•	•	•								
		76	A	A	B	B	A	A	A	•	A	A	•	•	•	•	•	•								
		77	B	B	B	B	A	A	A	•	A	A	•	•	•	•	•	•								
		78	A	A	A	A	B	B	A	•	A	A	•	•	•	•	•	•								
		79	B	B	B	B	B	B	A	•	A	A	•	•	•	•	•	•								
		80	A	A	A	A	A	A	B	•	A	A	•	•	•	•	•	•								
		81	B	B	B	B	B	B	B	•	A	A	•	•	•	•	•	•								
		82	A	A	A	A	A	A	A	•	B	B	•	•	•	•	•	•								
		83	B	B	B	B	B	B	A	•	B	B	•	•	•	•	•	•								
		84	A	A	A	A	A	A	B	•	B	B	•	•	•	•	•	•								
		85	B	B	B	B	B	B	B	•	B	B	•	•	•	•	•	•								
		86	A	A	A	A	A	A	A	•	A	B	•	•	•	•	•	•								
		87	B	B	B	B	B	B	A	•	A	B	•	•	•	•	•	•								
		88	A	A	A	A	A	A	B	•	A	B	•	•	•	•	•	•								
		89	B	B	B	B	B	B	B	•	A	B	•	•	•	•	•	•								
		90	A	A	A	A	A	A	A	•	B	B	•	•	•	•	•	•								
		91	B	B	B	B	B	B	A	•	B	B	•	•	•	•	•	•								
		92	A	A	A	A	A	A	B	•	B	B	•	•	•	•	•	•								
		93	B	B	B	B	B	B	B	•	B	B	•	•	•	•	•	•								
		94	A	A	A	A	A	A	A	•	A	B	•	•	•	•	•	•								
		95	B	B	B	B	B	B	A	•	A	B	•	•	•	•	•	•								
		96	A	A	A	A	A	A	B	•	A	B	•	•	•	•	•	•								
		97	B	B	B	B	B	B	B	•	A	B	•	•	•	•	•	•								
		98	A	A	A	A	A	A	A	•	B	B	•	•	•	•	•	•								
		99	B	B	B	B	B	B	A	•	B	B	•	•	•	•	•	•								
		100	A	A	A	A	A	A	B	•	B	B	•	•	•	•	•	•								
		101	B	A	B	B	B	B	B	•	B	B	•	•	•	•	•	•								

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal 9/	Test limits Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLR	CLK	A	B	C	G	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}								
Truth table tests 10/ 11/	3014	102	A	0	A	A	A	A	A	GND	A	A	H	L	L	H	L	4.5 V	All outputs 12/ 13/							
		103	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		104	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		105	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		106	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		107	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		108	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		109	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		110	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		111	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		112	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		113	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		114	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		115	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		116	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		117	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		118	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		119	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		120	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		121	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		122	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		123	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		124	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		125	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		126	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		127	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		128	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		129	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		130	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		131	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		132	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		133	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		134	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		135	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		136	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		137	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		138	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		139	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		140	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		141	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		142	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		143	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		144	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		145	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		146	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		147	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		148	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		149	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		150	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		151	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		152	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		153	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		154	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		155	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		156	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		157	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		158	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		159	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		160	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		161	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		162	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		163	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		164	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		165	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		166	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		167	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		168	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		169	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		170	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		171	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								
		172	•	A	•	•	•	•	•	•	•	B	•	•	L	H	•	•								
		173	•	A	•	•	•	•	•	•	•	A	•	•	H	L	•	•								

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case E_2, F_2, K_2	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 9 T_C = +25°C 14/ Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C 14/ Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLR	CLK	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	$\overline{QD}$	$\overline{QC}$	QB	QA	RCO	V_{CC}								
f_{MAX} 15/	3003 Fig. 6	174	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		4.5 V	QA	28		21		28		MHz
t_{PLH1}	3003 Fig. 6	175	•	•					•	•	•	•				OUT		•	CLK to QA	8	36	8	48	8	36	ns
		176	•	•					•	•	•	•						•	CLK to QB	•	•	•	•	•	•	•
		177	•	•					•	•	•	•		OUT	OUT			•	CLK to QC	•	•	•	•	•	•	•
		178	•	•					•	•	•	•	OUT					•	CLK to QD	•	•	•	•	•	•	•
t_{PHL1}	3003 Fig. 6	179	•	•					•	•	•	•				OUT		•	CLK to QA	•	•	•	•	•	•	•
		180	•	•					•	•	•	•						•	CLK to QB	•	•	•	•	•	•	•
		181	•	•					•	•	•	•		OUT	OUT			•	CLK to QC	•	•	•	•	•	•	•
		182	•	•					•	•	•	•	OUT					•	CLK to QD	•	•	•	•	•	•	•
t_{PLH2}	3003 Fig. 6	183	•	•					•	•	•	•					OUT	•	CLK to RCO	•	38	•	50	•	38	•
t_{PHL2}	3003 Fig. 6	184	•	•					•	•	•	•					•	•	CLK to RCO	•	38	•	50	•	38	•
t_{PLH4}	3003 Fig. 6	186	•	7/					•	•	•	IN					•	•	ENABLE T to RCO	7	34	7	46	7	34	•
t_{PHL4}	3003 Fig. 6	188	•	7/					•	•	•	IN					•	•	ENABLE T to RCO	7	34	7	46	7	34	•
t_{PHL9}	3003 Fig. 6	187	IN	16/	4.5 V					•	GND					OUT		•	CLR to QA	8	40	8	53	8	40	•
		188	•	•		4.5 V				•	•							•	CLR to QB	•	•	•	•	•	•	•
		189	•	•			4.5 V			•	•			OUT	OUT			•	CLR to QC	•	•	•	•	•	•	•
		190	•	•				4.5 V		•	•		OUT					•	CLR to QD	•	•	•	•	•	•	•
t_{PHL10}	3003 Fig. 6	191	•	9/	4.5 V	4.5 V	4.5 V	4.5 V		•	•						OUT	•	CLR to RCO	•	38	•	51	•	38	•
t_{TLH}	3003 Fig. 6	192	4.5 V	IN/	GND	GND	GND	GND	4.5 V	•	IN/	4.5 V				OUT		•	QA	3	15	3	20	3	15	•
		193	•	•	4.5 V	GND	•	•	•	•	•	•						•	QB	•	•	•	•	•	•	•
		194	•	•	•	4.5 V	•	•	•	•	•	•		OUT	OUT			•	QC	•	•	•	•	•	•	•
		195	•	•	•	•	4.5 V	•	•	•	•	•	OUT					•	QD	•	•	•	•	•	•	•
		196	•	•	GND	•	4.5 V	4.5 V	•	•	•	•					OUT	•	RCO	•	•	•	•	•	•	•
t_{THL}	3003 Fig. 6	197	•	•	4.5 V	GND	GND	GND	•	•	•	•				OUT		•	QA	•	•	•	•	•	•	•
		198	•	•	•	4.5 V	GND	•	•	•	•	•						•	QB	•	•	•	•	•	•	•
		199	•	•	•	4.5 V	4.5 V	•	•	•	•	•		OUT	GND			•	QC	•	•	•	•	•	•	•
		200	•	•	•	GND	GND	4.5 V	•	•	•	•	OUT					•	QD	•	•	•	•	•	•	•
		201	•	•	•	4.5 V	4.5 V	4.5 V	•	•	•	•					OUT	•	RCO	•	•	•	•	•	•	•

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD 883 method	Case E, F, K	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CTR	CLK	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1 2 3 4 5 6 7 8 9	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1/	1 mA	1 mA						GND	CTR CLK A B C D ENABLE P LOAD ENABLE T		-1.5					V
$V_{IC(neg)}$		10 11 12 13 14 15 16 17 18	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND	-1 mA	-1 mA						1/	CTR CLK A B C D ENABLE P LOAD ENABLE T		-1.5					V
I_{CCH}	3005	19	6.0 V	GND	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V		GND	6.0 V						6.0 V	V_{CC}		0.2		20			µA
I_{CCL}	3005	20	GND	6.0 V	6.0 V	6.0 V	6.0 V	6.0 V	GND		6.0 V	GND							V_{CC}		0.2		20			µA
V_{OH3}	3006 2/	21 22 23 24 25	4.2 V	18	4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V			GND	4.2 V	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA		QA QB QC QD RCO	3.95		3.95		3.95		V
V_{OH5}	3006 2/	26 27 28 29 30			4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V				4.2 V	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA		QA QB QC QD RCO	3.48		3.7		3.48		V
V_{OL3}	3007 4/	31 32 33 34 35			1.2 V	1.2 V	1.2 V	1.2 V				1.2 V	20 µA	20 µA	20 µA	20 µA	20 µA		QA QB QC QD RCO		0.05		0.05		0.05	V
V_{OL5}	3007 4/	36 37 38 39 40			1.2 V	1.2 V	1.2 V	1.2 V		GND		1.2 V	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA		QA QB QC QD RCO		0.26		0.4		0.26	V
I_{OS4}	3011 5/	41 42 43 44 45	4.0 V		4.0 V 4.0 V	4.0 V 6/	4.0 V 6/	4.0 V 4.0 V	4.0 V		GND	4.0 V	GND	GND	GND	GND	GND	4.0 V	QA QB QC QD RCO	-10	-120	-10	-120	-10	-120	mA

See footnotes at end of table III.

MIL-M-38510/605B

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CEW	CLK	A	B	C	D	ENABLE P	GND	CCKW	ENABLE T	QB	QC	QD	QA	RCO	V_{CC}								
I_{IH}	3010	46	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	GND						5.0 V	CEW		0.05		0.1			mA
		47	GND	5.0 V	GND	•	•	•	•	•	•	•						•	CLK		•		•			•
		48	•	GND	5.0 V	•	•	•	•	•	•	•						•	A		•		•			•
		49	•	•	GND	•	•	•	•	•	•	•						•	B		•		•			•
		50	•	•	•	5.0 V	•	•	•	•	•	•						•	C		•		•			•
		51	•	•	•	GND	5.0 V	•	•	•	•	•						•	D		•		•			•
		52	•	•	•	•	GND	5.0 V	•	•	•	•						•	ENABLE P		•		•			•
		53	•	•	•	•	•	GND	5.0 V	•	GND	•						•	CCKW		•		•			•
		54	•	•	•	•	•	•	GND	•	5.0 V	5.0 V						•	ENABLE T		•		•			•
I_{IL}	3009	55	•	•	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	•	•	•						•	CEW		-0.05		-0.1			•
		56	5.0 V	GND	5.0 V	GND	•	•	•	•	•	•						•	CLK		•		•			•
		57	•	5.0 V	GND	5.0 V	GND	•	•	•	•	•						•	A		•		•			•
		58	•	•	5.0 V	•	5.0 V	GND	•	•	•	•						•	B		•		•			•
		59	•	•	•	GND	•	5.0 V	•	•	•	•						•	C		•		•			•
		60	•	•	•	•	GND	•	GND	•	•	•						•	D		•		•			•
		61	•	•	•	•	•	GND	5.0 V	•	GND	•						•	ENABLE P		•		•			•
		62	•	•	•	•	•	•	5.0 V	•	5.0 V	•						•	CCKW		•		•			•
		63	•	•	•	•	•	•	•	•	•	GND	1/	1/	1/	1/		•	ENABLE T		•		•			•
																				Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max					
C_C	3012	64	0/	0/					0/	GND	0/	0/						GND	CLK		15					pF
		65								•								•	CEW		•					•
		66								•								•	CCKW		•					•
		67								•								•	ENABLE T		•					•
		68								•								•	ENABLE P		•					•
C_{IN}	3012	69			0/	0/	0/	0/		•								•	A		10					•
		70								•								•	B		•					•
		71								•								•	C		•					•
		72								•								•	D		•					•
																			Measured terminal 0/	Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 9 T_C = -55°C Min	Subgroup 9 T_C = -55°C Max	
Truth table tests 10/ 11/	3014	73	B	A	B	B	B	B	A	GND	B	A	H	H	H	H	H	4.5 V	All outputs 12/ 13/							
		74	B	A	A	B	B	B	A	•	B	A	H	H	H	H	H	•								
		75	B	A	B	A	B	B	A	•	B	A	H	H	H	H	H	•								
		76	B	A	A	A	B	B	A	•	B	A	H	H	H	H	H	•								
		77	B	A	B	B	A	B	A	•	B	A	H	H	H	H	H	•								
		78	B	A	A	B	A	B	A	•	B	A	H	H	H	H	H	•								
		79	B	A	B	A	A	B	A	•	B	A	H	H	H	H	H	•								
		80	B	A	A	A	A	B	A	•	B	A	H	H	H	H	H	•								
		81	B	A	B	B	B	A	A	•	B	A	H	H	H	H	H	•								
		82	B	A	A	B	B	A	A	•	B	A	H	H	H	H	H	•								
		83	B	A	B	A	B	A	A	•	B	A	H	H	H	H	H	•								
		84	B	A	A	A	B	A	A	•	B	A	H	H	H	H	H	•								
		85	B	A	B	B	A	A	A	•	B	A	H	H	H	H	H	•								
		86	B	A	A	B	A	A	A	•	B	A	H	H	H	H	H	•								
		87	B	A	B	A	A	A	A	•	B	A	H	H	H	H	H	•								
		88	B	A	A	A	A	A	A	•	B	A	H	H	H	H	H	•								
		89	B	A	B	B	B	B	A	•	B	A	H	H	H	H	H	•								
		90	B	A	A	B	B	B	A	•	B	A	H	H	H	H	H	•								
		91	B	A	B	A	B	B	A	•	B	A	H	H	H	H	H	•								
		92	B	A	A	A	B	B	A	•	B	A	H	H	H	H	H	•								
		93	B	A	B	B	A	B	A	•	B	A	H	H	H	H	H	•								
		94	B	A	A	B	A	B	A	•	B	A	H	H	H	H	H	•								
		95	B	A	B	A	A	B	A	•	B	A	H	H	H	H	H	•								
		96	B	A	A	A	A	B	A	•	B	A	H	H	H	H	H	•								
		97	B	A	B	B	B	A	A	•	B	A	H	H	H	H	H	•								
		98	B	A	A	B	B	A	A	•	B	A	H	H	H	H	H	•								
		99	B	A	B	A	B	A	A	•	B	A	H	H	H	H	H	•								
		100	B	A	A	A	B	A	A	•	B	A	H	H	H	H	H	•								
		101	B	A	B	B	A	A	A	•	B	A	H	H	H	H	H	•								

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case E_1, F_2, K	Terminal conditions 17															Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 9 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20							
		Test no.	CCEN	CLK	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table tests 10/ 11/	3014	102	A	B	A	A	A	A	A	GND	A	A	L	H	H	H	L	4.5 V	All outputs 12/ 13/							
		103											H													
		104																								
		105																								
		106																								
		107																								
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		166																								
		167																								
		168																								
		169																								
		170																								

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case E, F, K	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 9 T_C = +25°C 16/ Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C 16/ Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	16	15	17	18	19	20								
		Test no.	CEN	CLK	A	B	C	D	ENABLE P	GND	CO/RCI	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}								
f_{MAX} 15/	JD03 Fig. 6	171	4.5 V	In					4.5 V	GND	4.5 V	4.5 V				OUT		4.5 V	QA	20		21		20		MHz
t_{PLH1}	JD03 Fig. 6	172 173 174 175	4.5 V										OUT	OUT	OUT	OUT			CLK to QA CLK to QB CLK to QC CLK to QD	8 " " "	36 " " "	8 " " "	40 " " "	8 " " "	36 " " "	ns " " "
t_{PHL1}	JD03 Fig. 6	176 177 178 179	4.5 V										OUT	OUT	OUT	OUT			CLK to QA CLK to QB CLK to QC CLK to QD	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "
t_{PLH2}	JD03 Fig. 6	180	"	"					"	"	"	"					OUT	"	CLK to RCO	"	38	"	50	"	38	"
t_{PHL2}	JD03 Fig. 6	181	"	"					"	"	"	"					"	"	CLK to RCO	"	38	"	50	"	38	"
t_{PLH3}	JD03 Fig. 6	182	"	7/					"	"	"	In					"	"	ENABLE T to RCO	7	34	7	46	7	34	"
t_{PHL3}	JD03 Fig. 6	183	"	7/					"	"	"	In					"	"	ENABLE T to RCO	7	34	7	46	7	34	"
t_{TLH}	JD03 Fig. 6	184 185 186 187 188	4.5 V " " " "	18/ " " " "	GND 4.5 V " " GND	GND GND 4.5 V 4.5 V GND	GND " 4.5 V GND	GND " " 6.5 V	" " " "	" " " "	19/ " " "	4.5 V " " "	OUT	OUT	OUT	OUT	OUT	" " " "	QA QB QC QD RCO	3 " " " "	15 " " " "	3 " " " "	20 " " " "	3 " " " "	15 " " " "	" " " " "
t_{THL}	JD03 Fig. 6	189 190 191 192 193	" " " " "	" " " " "	4.5 V " " " "	GND 4.5 V 4.5 V GND GND	" " 4.5 V GND GND	GND " " 6.5 V 4.5 V	" " " "	" " " "	" " " "	" " " "	OUT	ONT	OUT	OUT	ONT	" " " "	QA QB QC QD RCO	" " " " "	" " " " "	" " " " "	" " " " "	" " " " "	" " " " "	" " " " "

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Case outlines E, F, L 2	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Units
		2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	20								
		Test no.	CTR	CLR	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}								
$V_{IC(pos)}$ 2/		1 2 3 4 5 6 7 8 9	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1 mA	1/	1 mA	1 mA						GND	CTR CLR A B C D ENABLE P LOAD ENABLE T		1.5						
$V_{IC(neg)}$		10 11 12 13 14 15 16 17 18	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND	-1 mA	-1 mA						1/	CTR CLR A B C D ENABLE P LOAD ENABLE T		-1.5						
I_{CLR}	3009	19	0.0 V	GND	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		0.0 V	0.0 V						0.0 V	V_{CC}		0.2		80			μA
I_{CCL}	3009	20	GND	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	GND		0.0 V	GND							V_{CC}		0.2		80			μA
V_{OHJ}	3006 3/	21 22 23 24 25	0.0 V	10	4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V			GND	4.2 V	-20 μA	20 μA	-20 μA	-20 μA	-20 μA		RCO	3.99		3.95		3.99		V
V_{OHS}	3006 3/	26 27 28 29 30			4.2 V 4.2 V	4.2 V 3/	4.2 V 3/	4.2 V 4.2 V				4.2 V	-5.2 mA	-9.2 mA	-5.2 mA	-5.2 mA	-9.2 mA		RCO	3.48		3.2		3.48		V
V_{OLJ}	3007 4/	31 32 33 34 35			1.2 V	1.2 V	1.2 V	1.2 V				1.2 V	20 μA	20 μA	20 μA	20 μA	20 μA		RCO		0.05		0.05		0.05	V
V_{OLS}	3007 4/	36 37 38 39 40			1.2 V	1.2 V	1.2 V	1.2 V		GND		1.2 V	9.2 mA	9.2 mA	9.2 mA	9.2 mA	9.2 mA		RCO		0.26		0.4		0.26	V
I_{OSA}	3011 5/	41 42 43 44 45	4.0 V		4.0 V 4.0 V	4.0 V 0/	4.0 V 0/	4.0 V 4.0 V	4.0 V		GND	4.0 V	GND	GND	GND	GND	GND	4.0 V	RCO	-30	-120	30	-120	-30	-120	mA

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 01 - continued.

Symbol	MIL-STD-883 method	Case E, F, K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLR	CLK	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}								
I_{IH}	3010	46	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	GND						5.0 V	CLR		0.05		0.1			µA
		47	GND	5.0 V	GND	•	•	•	•	•	•	•						•	CLK		•		•			
		48	•	GND	5.0 V	5.0 V	•	•	•	•	•	•						•	A		•		•			
		49	•	•	GND	•	•	•	•	•	•	•						•	B		•		•			
		50	•	•	•	5.0 V	5.0 V	•	•	•	•	•						•	C		•		•			
		51	•	•	•	GND	GND	5.0 V	•	•	•	•						•	D		•		•			
		52	•	•	•	•	•	GND	5.0 V	•	•	•						•	ENABLE P		•		•			
		53	•	•	•	•	•	•	GND	•	5.0 V	•						•	LOAD		•		•			
		54	•	•	•	•	•	•	GND	•	GND	5.0 V						•	ENABLE T		•		•			
I_{IL}	3009	55	•	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	GND	5.0 V	•						•	CLR		-0.05		0.1			•
		56	5.0 V	GND	GND	•	•	•	•	•	•	•						•	CLK		•		•			
		57	•	5.0 V	GND	GND	GND	•	•	•	•	•						•	A		•		•			
		58	•	•	5.0 V	•	•	•	•	•	•	•						•	B		•		•			
		59	•	•	•	GND	GND	GND	•	•	•	•						•	C		•		•			
		60	•	•	•	•	•	GND	GND	•	•	•						•	D		•		•			
		61	•	•	•	•	•	•	GND	•	GND	•						•	ENABLE P		•		•			
		62	•	•	•	•	•	•	GND	•	GND	•						•	LOAD		•		•			
		63	•	•	•	•	•	•	GND	•	GND	GND	1/	1/	1/	1/		•	ENABLE T		•		•			
																				Subgroup 6 T_C = +25°C Min	Subgroup 6 T_C = +25°C Max					
C_C	3012	64	8/	8/					8/	•	8/							GND	CLK		15					pF
		65								•								•	CLR		•					
		66								•								•	LOAD		•					
		67								•								•	ENABLE T		•					
		68								•								•	ENABLE P		•					
C_{IN}	3012	69			8/	8/	8/	8/		•								•	A		10					pF
		70								•								•	B		•					
		71								•								•	C		•					
		72								•								•	D		•					
																			Measured terminal	Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	
Truth table tests 10/ 11/	3014	73	A	A	A	A	A	A	A	GND	A	A	L	L	L	L	L	L	9/ All outputs 12/ 13/							
		74	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		75	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		76	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		77	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		78	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		79	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		80	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		81	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		82	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		83	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		84	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		85	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		86	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		87	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		88	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		89	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		90	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		91	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		92	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		93	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		94	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		95	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		96	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		97	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		98	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		99	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		100	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								
		101	A	A	A	A	A	A	A	•	A	A	L	L	L	L	L	L								

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case C, P, X	Terminal conditions 1/																Measured terminal 9/	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CCK	CLK	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QD	QC	QB	QA	RCO	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table tests 10/ 11/	3014	102	A	B	A	A	A	A	A	GND	A	A	L	H	H	H	L	4.5 V	All outputs 12/ 13/							
		103	•	A	A	A	A	A	•	•	•	A	H	L	L	L	•	•								
		104	•	A	A	A	B	B	•	•	•	B	•	•	•	•	•	•								
		105	•	A	A	A	A	B	•	•	•	A	•	•	•	H	•	•								
		106	•	A	A	A	B	A	•	•	•	B	•	•	•	L	•	•								
		107	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		108	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		109	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		110	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		111	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		112	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		113	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		114	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		115	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		116	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		117	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		118	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		119	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		120	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		121	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		122	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		123	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		124	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		125	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		126	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		127	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		128	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		129	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		130	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		131	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		132	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		133	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		134	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		135	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		136	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		137	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		138	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		139	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		140	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		141	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		142	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		143	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		144	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		145	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		146	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		147	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		148	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		149	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		150	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		151	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		152	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		153	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		154	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		155	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		156	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		157	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		158	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		159	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		160	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		161	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		162	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		163	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		164	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		165	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		166	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								
		167	•	A	A	A	A	A	•	•	•	A	•	•	H	•	•	•								
		168	•	A	A	A	B	B	•	•	•	B	•	•	L	•	•	•								
		169	•	A	A	A	A	B	•	•	•	A	•	•	•	•	•	•								
		170	•	A	A	A	B	A	•	•	•	B	•	•	•	•	•	•								

See footnotes at end of table III.

MIL-M-38510/661

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case E,F,X	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal 9/	Test limits Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 9 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLK	CLR	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QB	QC	QD	QA	RCO	V_{CC}								
Truth table tests 10/ 11/	3014	171	A	A	B	A	B	A	A	GND	A	A	H	L	L	L	L	4.5 V								
		172	•	B	•	•	•	•	•	•	•	•	•	•	H	•	•	•								
		173	•	A	•	•	•	•	•	•	•	•	•	•	•	H	•	•								
		174	•	B	A	•	•	•	•	•	•	•	•	•	•	•	•	•								
		175	•	A	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		176	•	B	A	•	•	•	•	•	•	•	•	•	•	H	•	•								
		177	•	A	B	B	A	•	•	•	•	•	•	•	•	•	•	•								
		178	•	B	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		179	•	A	A	•	•	•	•	•	•	•	•	H	L	L	•	•								
		180	•	B	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		181	•	A	B	•	•	•	•	•	•	•	•	•	•	H	•	•								
		182	•	B	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		183	•	A	A	•	•	•	•	•	•	•	•	•	H	L	•	•								
		184	•	B	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		185	•	A	•	•	•	•	•	•	•	•	•	•	•	•	•	•								

Symbol	MIL-STD-883 method	Case E,F,X	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 9 T_C = +25°C 18/ Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C 18/ Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLK	CLR	A	B	C	D	ENABLE P	GND	LOAD	ENABLE T	QB	QC	QD	QA	RCO	V_{CC}								
f_{MAX} 11/	3003 Fig. 6	186	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		4.5 V	QA	20		21		20		MHz
t_{PLH1}	3003 Fig. 6	187	1.5 V	•					•	•	•	•						•	CLK to QA	8	36	8	48	8	36	ns
		188	•	•					•	•	•	•						•	CLK to QB	•	•	•	•	•	•	•
		189	•	•					•	•	•	•						•	CLK to QC	•	•	•	•	•	•	•
		190	•	•					•	•	•	•	OUT	OUT	OUT	OUT		•	CLK to QD	•	•	•	•	•	•	•
t_{PHL1}	3003 Fig. 6	191	4.5 V	•					•	•	•	•						•	CLK to QA	•	•	•	•	•	•	•
		192	•	•					•	•	•	•						•	CLK to QB	•	•	•	•	•	•	•
		193	•	•					•	•	•	•						•	CLK to QC	•	•	•	•	•	•	•
		194	•	•					•	•	•	•	OUT	OUT	OUT	OUT		•	CLK to QD	•	•	•	•	•	•	•
t_{PLH2}	3003 Fig. 6	195	•	•					•	•	•	•					OUT	•	CLK to RCO	•	38	•	90	•	38	•
t_{PHL2}	3003 Fig. 6	196	•	•					•	•	•	•					•	•	CLK to RCO	•	38	•	50	•	38	•
t_{PLH4}	3003 Fig. 6	197	•	3/					•	•	•	IN					•	•	ENABLE T to RCO	7	34	7	46	•	34	•
t_{PHL4}	3003 Fig. 6	198	•	3/					•	•	•	IN					•	•	ENABLE T to RCO	7	34	7	46	7	34	•
t_{TLH}	3003 Fig. 6	199	•	16/	GND	GND	GND	GND	•	•	19/	4.5 V						•	QA	3	15	3	20	3	15	•
		200	•	•	4.5 V	GND	•	•	•	•	•	•						•	QB	•	•	•	•	•	•	•
		201	•	•	•	4.5 V	4.5 V	•	•	•	•	•	OUT	OUT	OUT	OUT		•	QC	•	•	•	•	•	•	•
		202	•	•	GND	•	•	4.5 V	•	•	•	•					OUT	•	QD	•	•	•	•	•	•	•
		203	•	•		•	•	•	•	•	•	•						•	RCO	•	•	•	•	•	•	•
t_{THL}	3003 Fig. 6	204	•	•	4.5 V	GND	GND	GND	•	•	•	•						•	QA	•	•	•	•	•	•	•
		205	•	•	•	4.5 V	GND	•	•	•	•	•						•	QB	•	•	•	•	•	•	•
		206	•	•	•	4.5 V	4.5 V	4.5 V	•	•	•	•	OUT	OUT	OUT	OUT		•	QC	•	•	•	•	•	•	•
		207	•	•	•	GND	GND	•	•	•	•	•					OUT	•	QD	•	•	•	•	•	•	•
		208	•	•	•	4.5 V	4.5 V	4.5 V	•	•	•	•						•	RCO	•	•	•	•	•	•	•

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TABLE III. Group A inspection for device type 05.

Symbol	MIL-STD-883 method	Case B, F, X	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	OB	OA	ENABLE B	D/B	QC	QD	GND	DATA B	DATA C	LOAD	Max/Min	RCO	CLK	DATA A	V_{CC}								
$V_{IC(pos)}$ 2/		1 2 3 4 5 6 7 8	1 mA			1 mA	1 mA			1/	1 mA	1 mA	1 mA			1 mA	1 mA	GND	DATA B ENABLE B D/B DATA B DATA C LOAD CLK DATA A		1.5					V
$V_{IC(neg)}$		9 10 11 12 13 14 15 16	-1 mA			-1 mA	-1 mA			GND	-1 mA	-1 mA	-1 mA			-1 mA	-1 mA	1/	DATA B ENABLE B D/B DATA B DATA C LOAD CLK DATA A		-1.5					V
I_{CCH}	3005	17	6.0 V			GND	6.0 V				6.0 V	6.0 V	GND			6.0 V	6.0 V	6.0 V	V_{CC}		0.2		20			µA
I_{CCL}	3005	18	GND			GND	GND				GND	GND	GND			GND	GND		V_{CC}		0.2		20			µA
V_{OH3}	3006 2/	19 20 21 22 23 24	4.2 V 1.2 V	-20 µA	-20 µA	4.2 V	4.2 V	-20 µA	-20 µA		4.2 V 1.2 V	4.2 V 1.2 V	1.2 V	-20 µA	-20 µA	18	4.2 V 1.2 V		QB QA QC QD Max/Min RCO	5.95		5.95		5.95		V
V_{OH5}	3006 2/	25 26 27 28 29 30	4.2 V 1.2 V	-5.2 mA	-5.2 mA	4.2 V	4.2 V	-5.2 mA	-5.2 mA		4.2 V 1.2 V	4.2 V 1.2 V		-5.2 mA	-5.2 mA		4.2 V 1.2 V		QB QA QC QD Max/Min RCO	5.40		5.2		5.40		V
V_{OL3}	3007 4/	31 32 33 34 35 36	1.2 V 4.2 V 4.2 V	20 µA	20 µA	1.2 V	4.2 V 1.2 V	20 µA	[illegible] µA		1.2 V 1.2 V 4.2 V	1.2 V 1.2 V 4.2 V	1.2 V	20 µA	20 µA		1.2 V 1.2 V		QB QA QC QD Max/Min RCO		0.05		0.05		0.05	V
V_{OL5}	3007 4/	37 38 39 40 41 42	1.2 mA 4.2 V 4.2 V	5.2 mA	5.2 V	1.2 V	4.2 V 1.2 V	5.2 mA	5.2 mA		1.2 V 1.2 V 4.2 V	1.2 V 1.2 V 4.2 V		5.2 mA	5.2 mA	4.2 V	1.2 V 1.2 V		QB QA QC QD Max/Min RCO		0.26		0.4		0.26	V
I_{OS4}	3011 5/	43 44 45 46 47 48	4.0 V GND	GND	GND	4.0 V	4.0 V	GND	GND		4.0 V GND	1.0 V GND	GND	GND	GND	18	4.0 V GND	4.0 V	QB QA QC QD Max/Min RCO	-10	-120	-10	-120	-10	-120	mA

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Case C, F, K	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	OE	DA	ENABLE G	U/D	QC	QD	GND	DATA B	DATA C	LOAD	Max/Min	TCO	CLK	DATA A	V_{CC}								
I_{IH}	3010	49 50 51 52 53 54 55 56	6.0 V GND • • • • • •			GND 6.0 V GND • • • • •	GND GND 6.0 V GND • • • •			GND • • • • • • •	GND • • 6.0 V GND • • •	GND • • 6.0 V GND • • •	GND • • • • 6.0 V GND GND				GND • • • • • 6.0 V GND	GND • • • • • 6.0 V	6.0 V • • • • • •	DATA B ENABLE G U/D DATA B DATA C LOAD CLK DATA A		0.05 • • • • • •		0.1 • • • • • •			μA • • • • • •
I_{IL}	3009	57 58 59 60 61 62 63 64	• 6.0 V • • • • • •			6.0 V GND 6.0 V • • • • •	6.0 V 6.0 V GND 6.0 V • • • •			• • • • • • • •	6.0 V • • GND 6.0 V • • •	6.0 V • • • • GND 6.0 V •	6.0 V • • • • GND 6.0 V 6.0 V				6.0 V • • • • • GND 6.0 V	• • • • • • GND •	• • • • • • • •	DATA B ENABLE G U/D DATA B DATA C LOAD CLK DATA A		-0.05 • • • • • •		-0.1 • • • • • •			• • • • • • •
																				Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max					
C_{IN}	3012	65 66 67 68	4/							GND • •	4/	4/					4/	GND • •	DATA B DATA B DATA C DATA A		10 • •					pF • •
C_C	3012	69 70 71 72				4/	4/			• • • •			4/			4/		• • • •	CLK ENABLE G U/D LOAD		15 • •					pF • •
																			Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	
Truth table tests 10/ 11/	3014	73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101	A • B • A • • B A • • • B A • B	H • H	H • • • • • • • L L H H L L H H L L H H L L H H L L H H L	B A •	B •	H •	L •	GND •	A • B • A • • B A • • • B A • B A • • B A • • • B A • B A	A • B • A • • B A • • • B A • B A • • B A • • • B A • B A	A • B • A • • B A • • • B A • B A • • B A • • • B A • B A	H •	H •	A • B • A • • B A • • • B A • B A • • B A • • • B A • B A	4.5 V •	All outputs 12/ 13/							

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TABLE III. Group A inspection for device type 08 - Continued.

Symbol	MIL-STD-883 method	Case E, F, K	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal 9/	Test limits: Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test No.	DATA B	$\overline{CB}$	UA	ENABLE G	$\overline{E}/\overline{D}$	QC	$\overline{QD}$	GND	DATA B	DATA C	$\overline{CLEAR}$	Run/Min	$\overline{RCO}$	CLK	DATA A	V_{CC}								
Truth table tests 10/ 11/	3014	102	B	H	H	B	B	L	H	GND	A	A	A	L	H	A	A	4.5 V	All outputs 12/ 13/							
		103														B	B									
		104		L	L			H				B				A										
		105			L							B				B										
		106	A		H							A				A	A									
		107	B		H							B				B										
		108	A	H	L							A				A	A									
		109	B		L							B				B										
		110	A		H							A				A	A									
		111	B		H							B				B										
		112	A	L	L				L			A				A	A									
		113	B		L		A					B				B										
		114	A		L							A				A	A									
		115	B		L							B				B										
		116	A		L							A				A	A									
		117	B		L		B					B				B										
		118	A		H							A				A	A									
		119	B	H	H				H			B				B										
		120	A		H							A				A	A									
		121	B		L		A					B				B										
		122	A		L							A				A	A									
		123	B		L		B					B				B										
		124	A		H							A				A	A									
		125	B		H							B				B										
		126	A		H							A				A	A									
		127	B	H	H							B				B										
		128	A		H							A				A	A									
		129	B		L		A					B				B										
		130	A		L							A				A	A									
		131	B		L		B					B				B										
		132	A		H							A				A	A									
		133	B		H							B				B										
		134	A		H							A				A	A									
		135	B	H	H				H			B				B										
		136	A		H							A				A	A									
		137	B		L		A					B				B										
		138	A		L							A				A	A									
		139	B		L		B					B				B										
		140	A		H							A				A	A									
		141	B		H							B				B										
		142	A		H							A				A	A									
		143	B	H	H				H			B				B										
		144	A		H							A				A	A									
		145	B		L		A					B				B										
		146	A		L							A				A	A									
		147	B		L		B					B				B										
		148	A		H							A				A	A									
		149	B		H							B				B										
		150	A		H							A				A	A									
		151	B	H	H				H			B				B										
		152	A		H							A				A	A									
		153	B		L		A					B				B										
		154	A		L							A				A	A									
		155	B		L		B					B				B										
		156	A		H							A				A	A									
		157	B	H	H				H			B				B										
		158	A		H							A				A	A									
		159	B		L		A					B				B										
		160	A		L							A				A	A									
		161	B		L		B					B				B										
		162	A		H							A				A	A									
		163	B	H	H				H			B				B										
		164	A		H							A				A	A									
		165	B		L		A					B				B										
		166	A		L							A				A	A									
		167	B		L		B					B				B										
		168	A		H							A				A	A									
		169	B	H	H				H			B				B										
		170	A		H							A				A	A									
		171	B		L		A					B				B										
		172	A		L							A				A	A									

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TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Case C, F, 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 9 T_C = +25°C 14/ Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C 14/ Min	Max	Unit
		2	2	3	4	6	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	QB	QA	ENABLE G	D/D	QC	QD	GND	DATA B	DATA C	LOAD	Max/Min	RCO	CLK	DATA A	V_{CC}								
f_{MAX} 15/	3003 Fig. 6	173			OUT	GND	GND			GND			4.5 V			IN		4.5 V	CLK to QA	23		17		23		MHz
t_{PLH6}	3003 Fig. 6	174	4.5 V	OUT	OUT			OUT	OUT		4.5 V	4.5 V	IN				4.5 V		LOAD to QA	8	51	8	60	8	51	ns
		175																	LOAD to QB							
		176																	LOAD to QC							
		177																	LOAD to QD							
t_{PHL5}	3003 Fig. 6	178	GND	OUT	OUT			OUT	OUT		GND	GND					GND		LOAD to QA							
		179																	LOAD to QB							
		180																	LOAD to QC							
		181																	LOAD to QD							
t_{PLH1}	3003 Fig. 6	182	GND	OUT	OUT	GND	GND	OUT	OUT			GND	16/			IN	GND		CLK to QA	7	39	7	52	7	39	
		183	GND														4.5 V		CLK to QB							
		184	4.5 V									4.5 V							CLK to QC							
		185	4.5 V																CLK to QD							
t_{PHL1}	3003 Fig. 6	186		OUT	OUT			OUT	OUT				4.5 V						CLK to QA							
		187																	CLK to QB							
		188																	CLK to QC							
		189																	CLK to QD							
t_{PHL6}	3003 Fig. 6	190	IN	OUT	OUT			OUT	OUT		IN	IN					IN		DATA A to QA		42		56		42	
		191																	DATA B to QB							
		192																	DATA C to QC							
		193																	DATA D to QD							
t_{PLH3}	3003 Fig. 6	194					GND							OUT					CLK to Max/Min	0	45	0	60	0	45	
t_{PHL3}	3003 Fig. 6	195												OUT					CLK to Max/Min	0	45	0	60	0	45	
t_{PLH2}	3003 Fig. 6	196													OUT				CLK to RCO	5	26	5	35	5	26	
t_{PHL2}	3003 Fig. 6	197																	CLK to RCO	5	26	5	35	5	26	
t_{PLH4}	3003 Fig. 6	198				IN													ENABLE G to RCO	0	35	0	47	0	35	
t_{PHL4}	3003 Fig. 6	199				IN													ENABLE G to RCO		35		47		35	
t_{PLH7}	3003 Fig. 6	200				GND	IN							OUT		IN			D/D to Max/Min		33		44		33	
t_{PHL7}	3003 Fig. 6	201												OUT					D/D to Max/Min		33		44		33	
t_{PLH8}	3003 Fig. 6	202													OUT				D/D to RCO	0	41	0	54	0	41	
t_{PHL8}	3003 Fig. 6	203													OUT				D/D to RCO	0	41	0	54	0	41	

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case outlines	Test no.	Terminal conditions 1/																Measured terminal	Test limits				Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_C = +25°C 14/ T_A = +25°C 14/		Subgroup 10 T_C = +125°C 14/ T_A = +125°C 14/		Subgroup 11 T_C = -55°C 14/ T_A = -55°C 14/		
																					Min	Max	Min	Max	Min	Max	
I_{CCLH}	3003 Fig. 6		204	4.5 V	OUT	OUT	GND	GND	OUT	OUT	GND	GND 4.5 V	GND 4.5 V	20/	OUT	OUT	20/	GND 4.5 V	4.5 V	V_{CC}	3	15	3	20	3	15	mA
			205									4.5 V		21/			21/	GND 4.5 V		1/2 I_{CCH}							
			206										4.5 V	19/			19/	GND		1/2 I_{CCL}							
			207																								
			208																								
			209																								
I_{CCLL}	3003 Fig. 6		210	GND 4.5 V	OUT	OUT	GND	GND	OUT	OUT	GND	GND 4.5 V	GND 4.5 V	20/	OUT	OUT	22/	GND 4.5 V	4.5 V	V_{CC}	3	15	3	20	3	15	mA
			211									4.5 V		21/			21/	GND 4.5 V		1/2 I_{CCH}							
			212										4.5 V	19/			19/	GND		1/2 I_{CCL}							
			213																								
			214																								
			215																								

See footnotes at end of table III

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TABLE III. Group A inspection for device type 06.

Symbol	MIL-STD-883 method	Case E, F, K	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	QB	QA	CLK count down	CLK count up	QC	QD	GND	DATA D	DATA C	LOAD	CARRY	BORROW	CLR	DATA A	V_{CC}								
$I_{IC(pos)}$		1 2 3 4 5 6 7 8	1 mA			1 mA	1 mA			1/	1 mA	1 mA	1 mA			1 mA	1 mA	GND	DATA B CLK count down CLK count up DATA B DATA C LOAD CLR DATA A		1.5					V
$I_{IC(neg)}$		9 10 11 12 13 14 15 16	-1 mA			-1 mA	-1 mA			GND	-1 mA	-1 mA	-1 mA			-1 mA	-1 mA	1/	DATA B CLK count down CLK count up DATA B DATA C LOAD CLR DATA A		-1.5					V
I_{CCH}	3005	17	6.0 V								6.0 V	6.0 V	GND			GND	6.0 V	6.0 V	V_{CC}		0.2		20			µA
I_{CCL}	3005	18	GND								GND	GND	GND			GND	6.0 V		V_{CC}		0.2		20			µA
V_{OH3} 2/	3006	19 20 21 22 23 24	4.2 V	-20 µA	-20 µA	4.2 V	4.2 V	-20 µA	-20 µA		4.2 V	4.2 V	1.2 V		-20 µA	-20 µA	1.2 V 4.2 V		QB QA QC QD CARRY BORROW	3.95		3.95		3.95		V
V_{OH5} 2/	3006	25 26 27 28 29 30	4.2 V	-5.2 mA	-5.2 mA	4.2 V	4.2 V	-5.2 mA	-5.2 mA		4.2 V	4.2 V	1.2 V		-5.2 mA	-5.2 mA	4.2 V		QB QA QC QD CARRY BORROW	3.48		3.2		3.48		V
V_{OL3} 2/	3007	31 32 33 34 35 36	1.2 V 1.2 V	20 µA	20 µA	1.2 V	1.2 V	20 µA	20 µA		1.2 V 4.2 V	1.2 V 1.2 V	1.2 V	20 µA		1.2 V 4.2 V	1.2 V 4.2 V		QB QA QC QD CARRY BORROW		0.05		0.05		0.05	V
V_{OL5} 2/	3007	37 38 39 40 41 42	1.2 V 1.2 V	5.2 mA	5.2 mA	1.2 V	1.2 V	5.2 mA	5.2 mA		1.2 V 4.2 V	1.2 V 1.2 V	1.2 V	5.2 mA		1.2 V 4.2 V	1.2 V 4.2 V		QB QA QC QD CARRY BORROW		0.25		0.4		0.25	V
I_{OS} 2/	3011	43 44 45 46 47 48	4.0 V	GND	GND	4.0 V	4.0 V	GND	GND		6.0 V	4.0 V	GND	GND	GND	GND	4.0 V	4.0 V	QB QA QC QD CARRY BORROW	-30	-120	-30	-130	-30	-120	mA

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case E,F,I	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Units
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	QB	QA	CLK count down	CLK count up	QC	QD	GND	DATA D	DATA C	COUT	CONTY	BORROW	CLR	DATA A	V_{CC}								
I_{IH}	3010	49	6.0 V			GND	GND			GND	GND	GND	GND			GND	GND	6.0 V	DATA B		0.05		0.1			µA
		50	GND			6.0 V	GND			•	•	•	•			•	•	•	CLK count down		•		•			•
		51	•			GND	6.0 V			•	•	•	•			•	•	•	CLK count up		•		•			•
		52	•			•	•			•	6.0 V	•	•			•	•	•	DATA B		•		•			•
		53	•			•	•			•	GND	6.0 V	•			•	•	•	DATA C		•		•			•
		54	•			•	•			•	•	GND	6.0 f			•	•	•	COUT		•		•			•
		55	•			•	•			•	•	•	GND			6.0 V	•	•	CLR		•		•			•
		56	•			•	•			•	•	•	GND			GND	6.0 V	•	DATA A		•		•			•
I_{IL}	3009	57	•			6.0 V	6.0 V			•	6.0 V	6.0 V	6.0 f			6.0 V	•	•	DATA B		-0.05		-0.1			•
		58	6.0 V			GND	6.0 V			•	•	•	•			•	•	•	CLK count down		•		•			•
		59	•			6.0 V	GND			•	•	•	•			•	•	•	CLK count up		•		•			•
		60	•			•	6.0 V			•	GND	GND	•			•	•	•	DATA B		•		•			•
		61	•			•	•			•	6.0 V	6.0 V	GND			•	•	•	DATA C		•		•			•
		62	•			•	•			•	•	•	6.0 f			•	•	•	COUT		•		•			•
		63	•			•	•			•	•	•	6.0 f			GND	GND	•	CLR		•		•			•
		64	•			•	•			•	•	•	•			6.0 V	GND	•	DATA A		•		•			•
																				Subgroup 4 T_C = +25°C Min	Max					
C_{IN}	3012	65	8/							GND	8/	8/					8/	GND	DATA B		10					pF
		66								•								•	DATA B		•					•
		67								•								•	DATA C		•					•
		68								•								•	DATA A		•					•
C_C	3012	69				8/	8/			•						8/		•	CLK count down		15					•
		70								•								•	CLK count up		•					•
		71								•			8/					•	CLR		•					•
		72								•								•	COUT		•					•
																			Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	
Truth table tests 10/ 11/	3014	73	A	L	L	A	A	L	L	GND	A	A	A	H	H	A	A	4.5 V	All outputs 12/ 13/							
		74	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		75	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		76	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		77	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		78	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		79	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		80	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		81	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		82	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		83	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		84	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								
		85	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•								

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Case E,F,K	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal 9/	Test limits Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	CB	QA	CLK count down	CLK count up	QC	QB	GND	DATA D	DATA C	LOAD	CARRY	BORROW	CLR	DATA A	V_{CC}								
Truth table tests 11/ 19/	3014	86	A	H	L	A	A	H	L	GND	A	A	A	H	H	D	A	4.5 V	All outputs 12/ 13/							
		87			H		B																			
		88					A																			
		89			L		B	L	H																	
		90		L			A																			
		91					B																			
		92			H		A																			
		93			L		B		L					L												
		94			H		A							H												
		95			L	B			L						L											
		96			H	A			H						H											
		97			L	B																				
		98			H	A			L																	
		99			L	B		H																		
		100		H		A			L																	
		101			H	B																				
		102			L	A			H																	
		103			H	B																				
		104		L		A			L																	
		105			L	B																				
		106			H	A			H																	
		107			L	B																				
		108		H		A			L																	
		109			H	B																				
		110			L	A			H																	
		111			H	B																				
		112		L		A			H																	
		113			L	B																				
		114			H	A			L																	
		115			L	B																				
		116			H	A			H						H											
		117			L	B		H																		
		118			H	A			L																	
		119			L	B																				
		120			H	A			H																	
		121			L	B																				
		122			H	A			L																	
		123			L	B																				
		124			H	A			H						H											
		125			L	B		H																		
		126			H	A			L																	
		127			L	B																				
		128			H	A			H																	
		129			L	B																				
		130			H	A			L																	
		131			L	B																				
		132			H	A			H						H											
		133			L	B		H																		
		134			H	A			L																	
		135			L	B																				
		136			H	A			H						H											

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TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal conditions 17																Measured terminal	Test limits						Unit
			1	2	3	4	5	7	8	9	10	11	12	13	14	15	16			Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		2	2	3	4	5	7	8	9	10	13	14	15	17	18	19	20									
		Test no.	DATA B	$\overline{LD}$	QA	CLK count down	CLK count up	QC	QD	GND	DATA D	DATA C	$\overline{LOAD}$	$\overline{COUNT}$	$\overline{BORROW}$	CLR	DATA A	V_{CC}		Min	Max	Min	Max	Min	Max	
f_{MAX} 11/	3003 Fig. 6	137			OUT	4.5 V	IN			GND			4.5 V			GND		4.5 V	CLK count up to QA	19		14		19		MHz
		138			OUT	IN	4.5 V			•			4.5 V			•		•	CLK count down to QA	19		14		19		MHz
t_{PLH5}	3003 Fig. 6	139	4.5 V	OUT	OUT			OUT	OUT	•	4.5 V	4.5 V	IN			•	4.5 V	•	$\overline{LOAD}$ to QB	12	51	12	60	12	51	ns
		140								•			•			•		•	$\overline{LOAD}$ to QA	•	•	•	•	•	•	•
		141								•			•			•		•	$\overline{LOAD}$ to QC	•	•	•	•	•	•	•
		142								•			•			•		•	$\overline{LOAD}$ to QD	•	•	•	•	•	•	•
t_{PHL5}	3003 Fig. 6	143	GND	OUT	OUT			OUT	OUT	•	GND	GND	•			•	GND	•	$\overline{LOAD}$ to QB	•	•	•	•	•	•	•
		144								•			•			•		•	$\overline{LOAD}$ to QA	•	•	•	•	•	•	•
		145								•			•			•		•	$\overline{LOAD}$ to QC	•	•	•	•	•	•	•
		146								GND			•			•		•	$\overline{LOAD}$ to QD	•	•	•	•	•	•	•
t_{PLH1}	3003 Fig. 6	147		OUT		4.5 V	IN			•			4.5 V			•		•	CLK count up to QB	11	48	12	64	11	48	ns
		148			OUT	•	•			•			•			•		•	CLK count up to QA	•	•	•	•	•	•	•
		149				•	•	OUT		•			•			•		•	CLK count up to QC	•	•	•	•	•	•	•
		150				•	•		OUT	•			•			•		•	CLK count up to QD	•	•	•	•	•	•	•
		151		OUT		IN	4.5 V			•			•			•		•	CLK count down to QB	•	•	•	•	•	•	•
		152			OUT	•	•			•			•			•		•	CLK count down to QA	•	•	•	•	•	•	•
		153				•	•	OUT		•			•			•		•	CLK count down to QC	•	•	•	•	•	•	•
		154				•	•		OUT	•			•			•		•	CLK count down to QD	•	•	•	•	•	•	•
t_{PHL1}	3003 Fig. 6	155		OUT		4.5 V	IN			•			•			•		•	CLK count up to QB	•	•	•	•	•	•	•
		156			OUT	•	•			•			•			•		•	CLK count up to QA	•	•	•	•	•	•	•
		157				•	•	OUT		•			•			•		•	CLK count up to QC	•	•	•	•	•	•	•
		158				•	•		OUT	•			•			•		•	CLK count up to QD	•	•	•	•	•	•	•
		159		OUT		IN	4.5 V			•			•			•		•	CLK count down to QB	•	•	•	•	•	•	•
		160			OUT	•	•			•			•			•		•	CLK count down to QA	•	•	•	•	•	•	•
		161				•	•	OUT		•			•			•		•	CLK count down to QC	•	•	•	•	•	•	•
		162				•	•		OUT	•			•			•		•	CLK count down to QD	•	•	•	•	•	•	•
t_{PHL9}	3003 Fig. 6	163	4.5 V	OUT	OUT			OUT	OUT	•	4.5 V	4.5 V	19/			IN	4.5 V	•	CLK to QB	•	47	•	62	•	47	ns
		164								•			•			•		•	CLK to QA	•	•	•	•	•	•	•
		165								•			•			•		•	CLK to QC	•	•	•	•	•	•	•
		166								4.5 V			•			•		•	CLK to QD	•	•	•	•	•	•	•
t_{PLH2}	3003 Fig. 6	167				4.5 V	IN			•	4.5 V		•	OUT		GND	4.5 V	•	CLK count up to carry	4	29	4	39	4	29	ns
		168	GND			IN	4.5 V			•	GND	GND	•		OUT	•	GHT	•	CLK count down to borrow	•	•	•	•	•	•	•
t_{PHL2}	3003 Fig. 6	169				4.5 V	IN			•	4.5 V		•	OUT		•	4.5 V	•	CLK count up to carry	•	•	•	•	•	•	ns
		170	GND			IN	4.5 V			•	GND	GND	•		OUT	•	GHT	•	CLK count down to borrow	•	•	•	•	•	•	•

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TABLE III. Group A inspection for device type 06 - Continued.

Symbol	MIL-STD-883 method	Case outlines	Test number	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits: Subgroup 9 T_C = +25°C 10/ Min	Subgroup 9 T_C = +25°C 10/ Max	Subgroup 10 T_C = +125°C 10/ Min	Subgroup 10 T_C = +125°C 10/ Max	Subgroup 11 T_C = -55°C 10/ Min	Subgroup 11 T_C = -55°C 10/ Max	Unit
			Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
t_{PLH}	3003 Fig. 6	C, E, F	171	4.5 V	$\overline{CLR}$	OUT		CLK count H	OUT	OUT	GND		DATA C	CENT	CUSBY		GND	DATA A	V_{CC}	$\overline{Q_B}$							ns
			172																	Q_A							
			173																	Q_C							
			174																	Q_D							
			175																	CARRY BORROW							
			176	GND			4.5 V	20/ 4.5 V				4.5 V				OUT		4.5 V GND									
			177	GND			20/					GND															
t_{PHL}	3003 Fig. 6	C, E, F	177	4.5 V	$\overline{CLR}$	OUT		CLK count H	OUT	OUT	GND		DATA C	CENT	CUSBY		GND	DATA A	V_{CC}	$\overline{Q_B}$							ns
			178																	Q_A							
			179																	Q_C							
			180																	Q_D							
			181																	TRIST BORROW							
			182	GND			4.5 V	20/ 4.5 V				4.5 V				OUT		4.5 V GND									
			183	GND			20/					GND															

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TABLE III. Group A inspection for device type 07.

Symbol	MIL-STD-883 method	Case C, F, X	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	QB	QA	CLK count down	CLK count up	QC	QD	GND	DATA B	DATA C	LOAD	CARRY	BORROW	CLR	DATA A	V_{CC}								
$V_{IC(pos)}$ 1/		1	1 mA			1 mA	1 mA			1/	1 mA	1 mA	1 mA			1 mA	1 mA	GND	DATA B		1.5					V
		2																	CLK count down							
		3																	CLK count up							
		4																	DATA B							
		5																	DATA C							
		6																	LOAD							
		7																	CLR							
		8																	DATA A							
$V_{IC(neg)}$		9	-1 mA			-1 mA	-1 mA			GND	-1 mA	-1 mA	-1 mA			-1 mA	-1 mA	1/	DATA B		-1.5					V
		10																	CLK count down							
		11																	CLK count up							
		12																	DATA B							
		13																	DATA C							
		14																	LOAD							
		15																	CLR							
		16																	DATA A							
I_{CCH}	3006	17	6.0 V								6.0 V	6.0 V	GND			GND	6.0 V	6.0 V	V_{CC}		0.2		20			µA
I_{CCL}	3006	18	GND								GND	GND	GND			GND	GND		V_{CC}		0.2		20			µA
V_{OH3} 2/	3006	19	4.2 V	-20 µA	-20 µA			-20 µA	-20 µA		4.2 V	4.2 V	1.2 V	-20 µA	-20 µA	1.2 V	4.2 V		QB	3.95		3.95		3.95		V
		20																	QA							
		21																	QC							
		22																	QD							
		23				4.2 V	4.2 V												CARRY							
		24																	BORROW							
V_{OH5} 2/	3006	25	4.2 V	-5.2 mA	-5.2 mA			-5.2 mA	-5.2 mA		4.2 V	4.2 V	1.2 V	-5.2 mA	-5.2 mA		4.2 V		QB	3.48		3.2		3.48		V
		26																	QA							
		27																	QC							
		28																	QD							
		29				4.2 V	4.2 V												CARRY							
		30																	BORROW							
V_{OL3} 3/	3007	31	1.2 V	20 µA	20 µA			20 µA	20 µA		1.2 V	1.2 V	1.2 V	20 µA	20 µA		1.2 V		QB		0.05		0.05		0.05	V
		32																	QA							
		33																	QC							
		34									1.2 V								QD							
		35	1.2 V			1.2 V	1.2 V				4.2 V	1.2 V				4.2 V	4.2 V		CARRY							
		36																	BORROW							
V_{OL5} 3/	3007	37	1.2 V	5.2 mA	5.2 mA			5.2 mA	5.2 mA		1.2 V	1.2 V	1.2 V	5.2 mA	5.2 mA	1.2 V	1.2 V		QB		0.76		0.4		0.76	V
		38																	QA							
		39																	QC							
		40									1.2 V								QD							
		41	1.2 V			1.2 V	1.2 V				4.2 V	1.2 V				4.2 V	4.2 V		CARRY							
		42																	BORROW							
I_{OS4} 3/	3011	43	6.0 V	GND	GND			GND	GND		6.0 V	6.0 V	GND	GND	GND	GND	6.0 V	6.0 V	QB	-10	-120	-10	-120	-10	-120	mA
		44																	QA							
		45																	QC							
		46																	QD							
		47				4.0 V	4.0 V				4.0 V								CARRY							
		48																	BORROW							

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 07 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	QB	QA	CLK count down	CLK count up	QC	QD	GND	DATA B	DATA C	LOAD	COUNT	BORROW	CLR	DATA A	V_{CC}								
I_{IH}	3010	49	6.0 V			GND	GND			GND	GND	GND	GND			GND	GND	6.0 V	DATA A		0.05		0.1			µA
		50	GND			6.0 V	GND			"	"	"	"			"	"	"	CLK count down		"		"			"
		51	"			GND	6.0 V			"	"	"	"			"	"	"	CLK count up		"		"			"
		52	"			"	GND			"	6.0 V	"	"			"	"	"	DATA B		"		"			"
		53	"			"	"			"	GND	0.0 V	"			"	"	"	DATA C		"		"			"
		54	"			"	"			"	"	"	6.0 V			"	"	"	LOAD		"		"			"
		55	"			"	"			"	"	"	GND			6.0 V	"	"	CLR		"		"			"
		56	"			"	"			"	"	"	GND			GND	6.0 V	"	DATA A		"		"			"
I_{IL}	3009	57	"			6.0 V	6.0 V			"	6.0 V	"	6.0 V			6.0 V	"	"	DATA A		-0.05		-0.1			"
		58	6.0 V			GND	6.0 V			"	"	"	"			"	"	"	CLK count down		"		"			"
		59	"			6.0 V	GND			"	"	"	"			"	"	"	CLK count up		"		"			"
		60	"			"	6.0 V			"	GND	"	"			"	"	"	DATA B		"		"			"
		61	"			"	"			"	6.0 V	GND	"			"	"	"	DATA C		"		"			"
		62	"			"	"			"	"	6.0 V	GND			"	"	"	LOAD		"		"			"
		63	"			"	"			"	"	"	6.0 V			GND	"	"	CLR		"		"			"
		64	"			"	"			"	"	"	6.0 V			6.0 V	GND	"	DATA A		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_{IN}	3012	65	8/							GND	8/	8/					8/	GND	DATA B		10					pF
		66								"								"	DATA B		"					"
		67								"								"	DATA C		"					"
		68								"								"	DATA A		"					"
C_C	3012	69				8/				"								"	CLK count down		15					"
		70					8/			"								"	CLK count up		"					"
		71								"			8/					"	LOAD		"					"
		72								"						8/		"	CLR		"					"
																			Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	
Truth table tests 10/ 11/	3014	73	B	L	L	A	A	L	L	GND	B	B	B	H	H	B	B	4.5 V	All outputs 12/ 13/							
		74	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"										
		75	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"										
		76	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"										
		77	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"										
		78	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"										
		79	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"										
		80	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B										
		81	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"										
		82	"	H	"	"	"	"	"	"	"	"	"	"	"	"	"	"										
		83	"	"	L	"	"	"	"	"	"	"	"	"	"	"	"	"										
		84	"	"	"	H	"	"	"	"	"	"	"	"	"	"	"	"										
		85	"	"	"	"	L	"	"	"	"	"	"	"	"	"	"	"										
		86	"	"	"	"	"	L	"	"	"	"	"	"	"	"	"	"										
		87	"	"	"	"	"	"	H	"	"	"	"	"	"	"	"	"										
		88	B	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"										
		89	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"										
		90	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"										
		91	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"										
		92	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"										
		93	A	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"										
		94	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A										
		95	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"										
		96	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"										

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 01/ - Continued.

Symbol	MIL-STD-883 method	Case E, F, K	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Reasoned terminal 9/	Test limits Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	$\overline{CB}$	$\overline{QA}$	CLK count down	CLK count up	$\overline{QC}$	$\overline{QD}$	GND	DATA B	DATA C	$\overline{LOAD}$	$\overline{CARRY}$	$\overline{BORROW}$	CLR	DATA A	V_{CC}								
Truth table tests 10/ 11/	3014	97	B	L	L	A	A	H	H	GND	A	A	B	H	H	B	B	6.5 V	All outputs 12/ 13/							
		98			H								A				A									
		99			L								B				B									
		100		H	H								A				A									
		101	A		L								B				B									
		102			H								A				A									
		103			L								B				B									
		104		L	H				L				A				A									
		105			L								B				B									
		106			H								A				A									
		107			L								B			L										
		108		H	H				H				A			H										
		109			L								B													
		110			H								A													
		111			L								B													
		112			H								A													
		113			L								B													
		114			H								A													
		115			L								B													
		116			H				L				A													
		117			L								B													
		118			H								A													
		119			L								B													
		120			H								A													
		121			L								B													
		122			H								A													
		123			L								B													
		124			H				H				A													
		125			L								B													
		126			H								A													
		127			L								B													
		128			H								A													
		129			L								B													
		130			H								A													
		131			L								B													
		132			H								A													
		133			L								B													
		134			H								A													
		135			L								B													
		136			H								A													
		137			L								B													
		138			H								A													
		139			L								B													
		140			H								A													
		141			L								B													
		142			H								A													
		143			L								B													
		144			H								A													
		145			L								B													
		146			H								A													
		147			L								B													
		148			H								A													
		149			L								B													
		150			H								A													
		151			L								B													
		152			H								A													
		153			L								B													
		154			H								A													
		155			L								B													
		156			H								A													
		157			L								B													
		158			H								A													
		159			L								B													
		160			H								A													
		161			L								B													
		162			H								A													
		163			L								B													
		164			H								A													
		165			L								B													
		166			H								A													
		167			L								B													
		168			H								A													
		169			L								B													

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TABLE III. Group A inspection for device type 07 - Continued.

Symbol	MIL-STD-883 method	Case C, P, Y	Terminal conditions 17/																Measured terminal 9/	Test limits						Unit
			1	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 14/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 14/		
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	QB	QA	CLK count down	CLK count up	QC	QD	GND	DATA D	DATA C	LOAD	CARRY	BORROW	CLR	DATA A	V_{CC}		Min	Max	Min	Max	Min	Max	
Truth table tests 10/ 11/	3014	170 171 172 173 174 175 176 177 178 179 180 181	A • • • • • • • • • • •	L • • • • • • H • • • •	L • • • • • • H • • • •	A • • • B A • • B A • B A	A A B A • • • • B A A A	L • • • • • • H • • • •	L • • • • • • H • • • •	GND • • • • • • • • • • •	A • • • • • • • • • • •	A • • • • • • • • • • •	A • • • • • • B • • • •	H • • • • • • • L H • •	H • • • • L H • • • • •	B • • • • • B • • • • •	A • • • • • • • • • • •	4.5 V • • • • • • • • • • •	All outputs 12/ 13/							
t_{PHL1}	3003 Fig. 6	182 183 184 185 186 187 188 189		OUT • • • OUT • • •	OUT • • • OUT • • •	4.5 V • • • IH • • •	IH • • • 4.5 V • • •	OUT • OUT • OUT • OUT •	GND • • OUT • • • OUT •				4.5 V • • • • • • • •			GND • • • • • • • •		4.5 V • • • • • • • •	CLK count up to QB CLK count up to QA CLK count up to QC CLK count up to QD CLK count down to QB CLK count down to QA CLK count down to QC CLK count down to QD	11 • • • • • • •	48 • • • • • • •	11 • • • • • • •	64 • • • • • • •	11 • • • • • • •	48 • • • • • • •	ns • • • • • • •
t_{PHL9}	3003 Fig. 6	190 191 192 193	4.5 V	OUT	OUT				OUT	• • • OUT	4.5 V	4.5 V	19/ • • •				18 • • •	4.5 V	•	CLR to QB CLR to QA CLR to QC CLR to QD	• • • •	42 • • •	• • • •	62 • • •	• • • •	42 • • •	• • • •
t_{PLH2}	3003 Fig. 6	194 195	4.5 V GND			4.5 V IH	IH 4.5 V			• •	4.5 V GND	4.5 V GND	• •	OUT	OUT	GND •	4.5 V GND	•	CLK count up to carry CLK count down to borrow	4 •	29 •	4 •	39 •	4 •	29 •	• •
t_{PHL2}	3003 Fig. 6	196 197	4.5 V GND			4.5 V IH	IH 4.5 V			• •	4.5 V GND	4.5 V GND	• •	OUT	OUT	• •	4.5 V GND	•	CLR count up to carry CLR count down to borrow	• •	• •	• •	• •	• •	• •	• •
f_{MAX} 15/	3003 Fig. 6	198 199			OUT OUT	4.5 V IH	IH 4.5 V			• •			4.5 V 4.5 V			• •		•	CLK count up to QA CLK count down to QB	19 19		14 14		19 19		MHz MHz
t_{PLH5}	3003 Fig. 6	200 201 202 203	4.5 V	OUT	OUT				OUT	• • • OUT	4.5 V	4.5 V	IH • • •			• • • •	4.5 V	•	LOAD to QB LOAD to QA LOAD to QC LOAD to QD	12 • • •	51 • • •	12 • • •	51 • • •	12 • • •	51 • • •	ns • • •
t_{PHL5}	3003 Fig. 6	204 205 206 207	GND	OUT	OUT				OUT	• • • OUT	GND	GND	• • • •			• • • •	GND	•	LOAD to QB LOAD to QA LOAD to QC LOAD to QD	• • • •	• • • •	• • • •	• • • •	• • • •	• • • •	• • • •

MIL-M-38510/606

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TABLE III. Group A inspection for device type 07 - Continued.

Symbol	Mil-STD-883 method	Case E,F,X	Terminal conditions 17																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	DATA B	$\overline{LD}$	QA	CLK count down	CLK count up	QC	QD	GND	DATA D	DATA C	$\overline{LOAD}$	CU/DN	BORROW	CLR	DATA A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 Fig. 6	208		OUT		4.5 V	IH			GND			4.5 V			GND		4.5 V	CLK count up to QB	11	48	11	64	11	48	ns
		209			OUT	"	"			"			"			"		"	CLK count up to QA	"	"	"	"	"	"	"
		210				"	"	OUT		"			"			"		"	CLK count up to QC	"	"	"	"	"	"	"
		211				"	"		OUT	"			"			"		"	CLK count up to QD	"	"	"	"	"	"	"
		212		OUT		IH	4.5 V			"			"			"		"	CLK count down to QB	"	"	"	"	"	"	"
		213			OUT	"	"			"			"			"		"	CLK count down to QA	"	"	"	"	"	"	"
		214				"	"	OUT		"			"			"		"	CLK count down to QC	"	"	"	"	"	"	"
		215				"	"		OUT	"			"			"		"	CLK count down to QD	"	"	"	"	"	"	"
t_{THL}	3003 Fig. 6	216	4.5 V	OUT	OUT			OUT	OUT		4.5 V	4.5 V	20/				4.5 V		QB	3	18	3	20	3	15	ns
		217											"						QA	"	"	"	"	"	"	"
		218											"						QC	"	"	"	"	"	"	"
		219											"						QD	"	"	"	"	"	"	"
		220	4.5 V			4.5 V	20/				4.5 V	4.5 V	"	CNT	OUT		4.5 V		CNT BORROW	"	"	"	"	"	"	"
		221	GND			20/	4.5 V				GND	GND	19/				GND									
t_{THL}	3003 Fig. 6	222	4.5 V	OUT	OUT			OUT	OUT		4.5 V	4.5 V				21/	4.5 V		QB							
		223														"			QA							
		224														"			QC							
		225														GND	4.5 V		CNT							
		226	4.5 V			4.5 V	20/				4.5 V	4.5 V		OUT	OUT	GND	4.5 V		BORROW							
		227	GND			21/	4.5 V				GND	GND														

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TABLE III. Group A inspection for device type 08.

Symbol	MIL-STD-883 method	Case E, F, K	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		2	2	3	6	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLK 1A	CLR1	1QA	CLK 1B	1QB	1QC	1QD	GND	2QD	2QC	2QB	CLK 2B	2QA	CLR2	CLK 2A	V_{CC}								
$V_{IC(pos)}$		1 2 3 4 5 6	1 mA	1 mA		1 mA				1/				1 mA		1 mA	1 mA	GND	CLK 1A CLR1 CLK 1B CLK 2B CLR2 CLK2A		1.5					V
$V_{IC(neg)}$		7 8 9 10 11 12	-1 mA	-1 mA		-1 mA				GND				-1 mA		-1 mA	-1 mA	1/	CLK 1A CLR1 CLK 1B CLK 2B CLR2 CLK2A		-1.5					V
I_{CCH}	3005	13	2/			2/								2/			2/	6.0 V	V_{CC}		0.2		20			µA
I_{CCL}	3005	14	GND	24/		GND								GND		24/	GND		V_{CC}		0.2		20			µA
V_{OH3}	3007	15 16 17 18 19 20 21 22	24/	25/	-20 µA	24/ 26/ 27/	-20 µA	-20 µA	-20 µA		-20 µA	-20 µA	-20 µA	24/ 26/ 27/	-20 µA	25/	24/		1QA 1QB 1QC 1QD 2QA 2QB 2QC 2QD	5.95		5.95		5.95		V
V_{OH5}	3007	23 24 25 26 27 28 29 30	24/	25/	-5.2 mA	24/ 26/ 27/	-5.2 mA	-5.2 mA	-5.2 mA		-5.2 mA	-5.2 mA	-5.2 mA	24/ 26/ 27/	-5.2 mA	25/	24/		1QA 1QB 1QC 1QD 2QA 2QB 2QC 2QD	5.48		5.2		5.48		
V_{OL3}	3007	31 32 33 34 35 36 37 38	24/	25/	20 µA	24/ 26/ 27/	20 µA	20 µA	20 µA		20 µA	20 µA	20 µA	24/ 26/ 27/	20 µA	25/	24/		1QA 1QB 1QC 1QD 2QA 2QB 2QC 2QD		0.05		0.05		0.05	
V_{OL5}	3007	39 40 41 42 43 44 45 46	24/	25/	5.2 mA	24/ 26/ 27/	5.2 mA	5.2 mA	5.2 mA		5.2 mA	5.2 mA	5.2 mA	24/ 26/ 27/	5.2 mA	25/	24/		1QA 1QB 1QC 1QD 2QA 2QB 2QC 2QD		0.26		0.4		0.26	
I_{OS4} 3/	3011	47 48 49 50 51 52 53 54	IN	IN	GND	IN	GND	GND	GND		GND	GND	GND	IN	GND	IN	IN	4.0 V	1QA 1QB 1QC 1QD 2QA 2QB 2QC 2QD	-10	-120	-10	-120	-10	-120	mA

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TABLE III. Group A inspection for device type 08 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal additions 17																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	Min	Max	Min	Max	
		Test no.	CLK 1A	CLR1	1QA	CLK 1B	1QB	1QC	1QD	GND	2QB	2QC	2QB	CLK 2B	2QA	CLR2	CLK 2A	V_{CC}								
I_{IH}	3010	55 56 57 58 59 60	GND 6.0 V GND • • •	6.0 V GND • • • •		GND GND 6.0 V GND • •				GND • • • • •				GND • • • 6.0 V		GND • • 6.0 V GND GND	GND • • 6.0 V GND	6.0 V • • • • •	CLR1 CLK 1A CLK 1B CLR2 CLK 2A CLK 2B		0.05 • • • • •		0.1 • • • • •			nA • • • • •
I_{IL}	3009	61 62 63 64 65 66	6.0 V GND 6.0 V • • •	• 6.0 V • • • •		6.0 V 6.0 V GND 6.0 V • •				• • • • • •				• • • • • GND		6.0 V • • GND 6.0 V 6.0 V	6.0 V • • • GND	• • • • • •	CLR1 CLK 1A CLK 1B CLR2 CLK 2A CLK 2B		-0.05 • • • • •		-0.1 • • • • •			- • • • • •
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_C	3012	67 68 69 70 71 72	8/	8/	8/					GND • • • • •				8/		8/	8/	GND • • • • •	CLR1 CLR2 CLK 1A CLK 1B CLK 2B CLK 2A		15 • • • • •						pF • • • • •
																			Measured terminals 9/	Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests 10/ 11/	3014	73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88	A B • • • • • • • • • • • • • • A B	A • B • • • • • • • • • • • • • •	L • • • • • • • • • • • • • • • H	A B A B A B A B A B A B A B A B •	L • • • H L H L H L H L H L H L •	L • • • • H L H L H L H L H L H •	L • • • • • H L H L H L H L H L •	L • • • • • • H L H L H L H L H •	GND • • • • • • • • • • • • • • • •	L • • • • • • • • H L H L H L H •	L • • • H L H L H L H L H L H L •	L • • • • H L H L H L H L H L H •	A B A B A B A B A B A B A B A B •	L • • • • • • • • • • • • • • • H	A • B • • • • • • • • • • • • • •	A B • • • • • • • • • • • • • • A B	4.5 V • • • • • • • • • • • • • • • •	All outputs 12/ 13/							

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 08 - Continued.

Symbol	MIL-STD-883 method	Case C_1F_2	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits: Subgroup 9 T_C = +25°C 14/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 14/ Min	Subgroup 11 Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	CLK 1A	CLR1	1QA	CLK 1B	1QB	1QC	1QD	GND	2QD	2QC	2QB	2CK 2B	2QA	CLR2	CLK 2A	V_{CC}								
f_{MAX} 15/ 30/	3003 Fig. 6	89	IN	GND	OUT					GND						GND		4.5 V	1CK 1A to 1QA	31		23		31		MHz
		90																	1CK 2A to 2QA							
		91				IN			OUT								IN		1CK 1B to 1QD							
		92									OUT			IN					1CK 2B to 2QD							
t_{PLH1}	3003 Fig. 6	93	IN	GND	OUT														1CK1A to 1QA	4	26	4	34	4	26	ns
		94						OUT											1CK1A to 1QC	9	51	9	68	9	51	
		95				IN	OUT												1CK1B to 1QB	4	27	4	36	4	27	
		96						OUT											1CK1B to 1QC	5	37	5	49	5	37	
		97							OUT						OUT				1CK1B to 1QD	4	27	4	36	4	27	
		98														GND	IN		1CK2A to 2QA	4	26	4	34	4	26	
		99										OUT							1CK2A to 2QC	9	51	9	68	9	51	
		100											OUT	IN					1CK2B to 2QB	4	27	4	38	4	27	
		101										OUT							1CK2B to 2QC	5	37	5	49	5	37	
		102									OUT								1CK2B to 2QD	4	27	4	36	4	27	
t_{PHL1}	3003 Fig. 6	103	IN	GND	OUT														CLK1A to 1QA	4	26	4	34	4	26	ns
		104						OUT											CLK1A to 1QC	9	51	9	68	9	51	
		105				IN	OUT												CLR1B to 1QB	4	27	4	36	4	27	
		106						OUT											CLK1B to 1QC	5	37	5	49	5	37	
		107							OUT						OUT				CLK1B to 1QD	4	27	4	36	4	27	
		108														GND	IN		CLK2A to 2QA	4	26	4	34	4	26	
		109										OUT							CLK2A to 2QC	9	51	9	68	9	51	
		110											OUT	IN					CLK2B to 2QB	4	27	4	36	4	27	
		111										OUT							CLK2B to 2QC	5	37	5	49	5	37	
		112									OUT								CLK2B to 2QD	4	27	4	36	4	27	
t_{PHL3}	3003 Fig. 6	113	20/	IN	OUT														CLR1 to 1QA	5	29	5	39	5	29	
		114				20/	OUT	OUT											CLR1 to 1QB							
		115							OUT										CLR1 to 1QC							
		116											OUT	5/	OUT	IN	5/		CLR1 to 1QD							
		117										OUT							CLR2 to 2QA							
		118																	CLR2 to 2QB							
		119																	CLR2 to 2QC							
		120									OUT								CLR2 to 2QD							
	3003 Fig. 6	121	23/	GND	OUT														1QA	3	15	3	20	3	15	
		122																	1QB							
		123																	1QC							
		124						OUT											1QD							
		125				23/	OUT												2QA							
		126							OUT						OUT	GND	23/		2QB							
		127										0UT	OUT	23/					2QC							
		128									OUT								2QD							
t_{THL}	3003 Fig. 6	129	5/	IN	OUT														1QA							
		130																	1QB							
		131																	1QC							
		132						OUT											1QD							
		133				5/	OUT												2QA							
		134							OUT						OUT	IN	5/		2QB							
		135										OUT	OUT	5/					2QC							
		136									OUT								2QD							

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TABLE III. Group A inspection for device type 09.

Symbol	MIL-STD-883 method	Case C_x 0 / 2 / Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	6	8	9	10	12	13	14	16	18	19	20								
		Test no.	CLK 1A	CLR1	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	CLR2	CLK 2A	V_{CC}								
$V_{IC(pos)}$ 1/		1 2 3 4	1 mA	1 mA					1/					1 mA	1 mA	GND	CLK 1A CLR1 CLR2 CLK 2A		1.5					V
$V_{IC(neg)}$		5 6 7 8	-1 mA	-1 mA					GND					-1 mA	-1 mA	1/	CLK 1A CLR1 CLR2 CLK 2A		-1.5					V
I_{CCH}	3005	9	5/												5/	6.0 V	V_{CC}		0.2		20			µA
I_{CCL}	3005	10	GND	25/										25/	GND		V_{CC}		0.2		20			µA
V_{OH1}	3006	11 12 13 14 15 16 17 18	5/	25/	-20 µA	-20 µA	-20 µA	-20 µA		-20 µA	-20 µA	-20 µA	-20 µA	25/	5/		1QA 1QB 1QC 1QD 2QD 2QC 2QB 2QA	5.95		5.95		5.95		V
V_{OH5}	3006	19 20 21 22 23 24 25 26	5/	25/	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA		-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	25/	5/		1QA 1QB 1QC 1QD 2QD 2QC 2QB 2QA	5.48		5.2		5.48		V
V_{OL3}	3007	27 28 29 30 31 32 33 34	36/	1.2 V 12/	20 µA	20 µA	20 µA	20 µA		20 µA	20 µA	20 µA	20 µA		1.2 V		1QA 1QB 1QC 1QD 2QD 2QC 2QB 2QA		0.05		0.05		0.05	V
V_{OL5}	3007	35 36 37 38 39 40 41 42		1.2 V 12/	5.2 mA	5.2 mA	5.2 mA	5.2 mA		5.2 mA	5.2 mA	5.2 mA	5.2 mA	1.2 V			1QA 1QB 1QC 1QD 2QD 2QC 2QB 2QA		0.26		0.6		0.26	V
I_{OS4} 5/	3011	43 44 45 46 47 48 49 50	1B	1B	GND	GND	GND	GND		GND	GND	GND	GND	1B	1B	6.0 V	1QA 1QB 1QC 1QD 2QD 2QC 2QB 2QA	-10	-120	-10	-180	-10	-120	mA

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD-883 method	Case C_I	Case Z	Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
I_{IH}	3010			51	CTR 1A 6.0 V	CLR1 GND	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	CLR2	CTR 2A	V_{CC} 5.0 V	CTR 1A CLR1 CLR2 CTR 2A		0.05		-0.1				µA
				52	GND	6.0 V																				
				53	6.0 V																					
				54	GND																					
I_{IL}	3009			55	GND	6.0 V					GND						CTR 2A		CTR 1A CLR1 CLR2 CTR 2A		-0.05		-0.1				
				56	6.0 V	GND																				
				57	GND																					
				58	6.0 V																					
					Subgroup 4 T_C = +25°C															Min	Max						
I_{CC}	3012			59	5.5 V	2/					GND					2/		GND	CLR1 CLR2 CTR 1A CTR 2A		15						µA
				60	GND																					
				61	5.5 V																					
				62	GND												2/									
					Measured terminal 3/															Min	Max	Min	Max	Min	Max	
Truth table tests 10/ 11/	3014			63	All outputs 12/ 13/	4.5 V	L	L	L	L	GND	L	L	L	L	A	A	5.0 V								
				64			L	L	L	L		L	L	L	L	A	A									
				65			L	L	L	L		L	L	L	L	A	A									
				66			L	L	L	L		L	L	L	L	A	A									
				67			L	L	L	L		L	L	L	L	A	A									
				68			L	L	L	L		L	L	L	L	A	A									
				69			L	L	L	L		L	L	L	L	A	A									
				70			L	L	L	L		L	L	L	L	A	A									
				71			L	L	L	L		L	L	L	L	A	A									
				72			L	L	L	L		L	L	L	L	A	A									
				73			L	L	L	L		L	L	L	L	A	A									
				74			L	L	L	L		L	L	L	L	A	A									
				75			L	L	L	L		L	L	L	L	A	A									
				76			L	L	L	L		L	L	L	L	A	A									
				77			L	L	L	L		L	L	L	L	A	A									
				78			L	L	L	L		L	L	L	L	A	A									

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TABLE III. Group A inspection for device type 09 - Continued.

Symbol	MIL-STD 883 method	Case C, D	Terminal conditions 1/															Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		2	2	3	4	6	8	9	10	12	13	14	16	18	19	20			Min	Max	Min	Max	Min	Max	
		Test no.	CLK 1A	CLR1	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	CLR2	CLK 2A	V_{CC}									
f_{MAX} 15/	3003	97 98	IN	GND	OUT				GND				OUT	GND	IN	4.5 V		CLK 1A to 1QA CLK 2A to 2QA	28 28		21 21		28 28		MHz MHz
t_{PLH1}	3003 Fig. 6	99 100 101 102 103 104 105 106	IN	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	GND	IN			CLK 1A to 1QA CLK 1A to 1QB CLK 1A to 1QC CLK 1A to 1QD CLK 2A to 2QA CLK 2A to 2QB CLK 2A to 2QC CLK 2A to 2QD	6 6 8 9 6 6 8 9	24 34 42 55 24 34 42 55	6 6 8 9 6 6 8 9	32 45 56 73 32 46 56 73	6 6 8 9 6 6 8 9	24 34 42 55 24 34 42 55	ns
t_{PHL1}	3003 Fig. 6	107 108 109 110 111 112 113 114	IN	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	GND	IN			CLK 1A to 1QA CLK 1A to 1QB CLK 1A to 1QC CLK 1A to 1QD CLK 2A to 2QA CLK 2A to 2QB CLK 2A to 2QC CLK 2A to 2QD	6 6 8 9 6 6 8 9	24 34 42 55 24 34 42 55	6 6 8 9 6 6 8 9	32 45 56 73 32 46 56 73	6 6 8 9 6 6 8 9	24 34 42 55 24 34 42 55	ns
t_{PHL2}	3003 Fig. 6	115 116 117 118 119 120 121 122	5/	IN	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	IN	5/			CLR1 to 1QA CLR1 to 1QB CLR1 to 1QC CLR1 to 1QD CLR2 to 2QA CLR2 to 2QB CLR2 to 2QC CLR2 to 2QD	5	29	5	39	5	29	ns
t_{TLH}	3003 Fig. 6	123 124 125 126 127 128 129 130	IN	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	GND	IN			CLR1 to 1QA CLR1 to 1QB CLR1 to 1QC CLR1 to 1QD CLR2 to 2QA CLR2 to 2QB CLR2 to 2QC CLR2 to 2QD	3	15	3	20	3	15	ns
t_{THL}	3003 Fig. 6	131 132 133 134 135 136 137 138	5/	IN	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	IN	34/			1QC 1QB 1QC 1QD 2QA 2QB 2QC 2QD							ns

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MIL-M-38510/66J

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TABLE III. Group A inspection for device type 10.

Symbol	MIL-STD-883 method	Case E, F, K	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	5 OUT	1 OUT	0 OUT	2 OUT	6 OUT	7 OUT	3 OUT	GND	8 OUT	4 OUT	9 OUT	CARRY OUT	CLOCK ENABLE	CLK	CLR	V_{CC}								
$V_{IC(pos)}$		1 2 3								1/ "					1 mA	1 mA	1 mA	GND " "	CLR CLK CLK ENABLE	1/	1.5 " "					V " "
$V_{IC(neg)}$		4 5 6								GND " "					-1 mA	-1 mA	-1 mA	1/ " "	CLR CLK CLK ENABLE		-1.5 " "					" " "
I_{CCH} I_{CCL} I_{CCH} I_{CCH}	3005 2/	7 8 9 10								" " " "					GND " " "	GND PA PF 6.0 V	6.0 V GND " "	6.0 V " " "	I_{CC} " " "		0.2 " " "		20 " " "			mA " " "
V_{OH3}	3006 2/	11 12 13 14 15 16 17 18 19 20 21	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	-20 µA	" " " " " " " " " " "	-20 µA	-20 µA	-20 µA	-20 µA	" " " " " " " " " " "	HI " " " " " " " " " "	GND " " " " " " " " " "	6.0 V " " " " " " " " " "	0 OUT 1 OUT 2 OUT 3 OUT 4 OUT 5 OUT 6 OUT 7 OUT 8 OUT 9 OUT CARRY OUT	5.95 " " " " " " " " " "		5.95 " " " " " " " " " "		5.95 " " " " " " " " " "		V " " " " " " " " " "
V_{OH5}	3006 2/	22 23 24 25 26 27 28 29 30 31 32	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	" " " " " " " " " " "	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	" " " " " " " " " " "	" " " " " " " " " " "	" " " " " " " " " " "	" " " " " " " " " " "	0 OUT 1 OUT 2 OUT 3 OUT 4 OUT 5 OUT 6 OUT 7 OUT 8 OUT 9 OUT CARRY OUT	5.40 " " " " " " " " " "		5.2 " " " " " " " " " "		5.40 " " " " " " " " " "		V " " " " " " " " " "
V_{OL3}	3007 4/	33 34 35 36 37 38 39 40 41 42 43	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	20 µA	" " " " " " " " " " "	20 µA	20 µA	20 µA	20 µA	" " " " " " " " " " "	" " " " " " " " " " "	" " " " " " " " " " "	" " " " " " " " " " "	0 OUT 1 OUT 2 OUT 3 OUT 4 OUT 5 OUT 6 OUT 7 OUT 8 OUT 9 OUT CARRY OUT		0.05 " " " " " " " " " "		0.05 " " " " " " " " " "		0.05 " " " " " " " " " "	V " " " " " " " " " "
V_{OL5}	3007 4/	44 45 46 47 48 49 50 51 52 53 54	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA	" " " " " " " " " " "	5.2 mA	5.2 mA	5.2 mA	5.2 mA	" " " " " " " " " " "	" " " " " " " " " " "	" " " " " " " " " " "	" " " " " " " " " " "	0 OUT 1 OUT 2 OUT 3 OUT 4 OUT 5 OUT 6 OUT 7 OUT 8 OUT 9 OUT CARRY OUT		0.26 " " " " " " " " " "		0.4 " " " " " " " " " "		0.26 " " " " " " " " " "	V " " " " " " " " " "

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TABLE III Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	S OUT	1 OUT	0 OUT	2 OUT	6 OUT	7 OUT	3 OUT	GND	8 OUT	4 OUT	9 OUT	CARRY OUT	CLOCK ENABLE	CLR	CLR	V_{CC}								
I_{IH}	3010	55 56 57								GND "					GND GND 6.0 V	GND 6.0 V GND	6.0 V GND GND	6.0 V " "	CLR CLK CLK ENABLE		0.05 " "		0.1 " "			mA " "
I_{IL}	3009	58 59 60								" "					" " GND	6.0 V GND 6.0 V	GND 6.0 V 6.0 V	" " "	CLR CLK CLK ENABLE		-0.05 " "		-0.1 " "			" " "
I_{OSD} 8/	3011	61 62 63 64 65 66 67 68 69 70 71	GND	GND	GND	GND	GND	GND	GND	" " " " " " " " " "	GND	GND	GND	GND	GND " " " " " " " " "	18 " " " " " " " " "	GND " " " " " " " " "	4.0 V " " " " " " " " "	0 OUT 1 OUT 2 OUT 3 OUT 4 OUT 5 OUT 6 OUT 7 OUT 8 OUT 9 OUT CARRY OUT	-18 " " " " " " " " "	-120 " " " " " " " " "	-18 " " " " " " " " "	-120 " " " " " " " " "	-18 " " " " " " " " "	-120 " " " " " " " " "	mA " " " " " " " " " "

																				Subgroup 4 T_C = +25°C Min	Max					
C_C	3012	72 73 74								GND "					8/	8/	8/	GND "	CLR CLK CLK ENABLE		15 " "					pF " "

																			Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	
Truth table tests 17/	3014	75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100	L "	L L H "	H L "	L H "	H L "	L H "	L H "	L H "	L H "	GND "	H "	L "	L "	H "	GND "	4.5 V 4.5 V PA "	4.5 V GND "	4.5 V "	Each output							

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TABLE III. Group A inspection for device type 10 - Continued.

Symbol	MIL-STD-883 method	Case E,F,X	Terminal conditions 17															Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_C = +25°C 14/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 14/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20							
		Test no.	5 OUT	1 OUT	0 OUT	2 OUT	6 OUT	7 OUT	3 OUT	GND	8 OUT	4 OUT	9 OUT	CARRY OUT	CLOCK ENABLE	CLR	CLR	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 Fig. 6	101	OUT		OUT					GND					GND	IN	GND	4.5 V	CLK to 0 OUT	6	40	6	55	6	48	ns
		102		OUT		OUT													CLK to 1 OUT							
		103							IN										CLK to 2 OUT							
		104																	CLK to 3 OUT							
		105										OUT							CLK to 4 OUT							
		106	OUT																CLK to 5 OUT							
		107					OUT												CLK to 6 OUT							
		108						OUT											CLK to 7 OUT							
		109									OUT								CLK to 8 OUT							
		110											OUT						CLK to 9 OUT							
		111												OUT					CLK to CARRY OUT							
t_{PLH1}	3003 Fig. 6	112			OUT														CLK to 0 OUT							
		113		OUT		OUT													CLK to 1 OUT							
		114							IN										CLK to 2 OUT							
		115										OUT							CLK to 3 OUT							
		116	OUT																CLK to 4 OUT							
		117					OUT												CLK to 5 OUT							
		118						OUT											CLK to 6 OUT							
		119									OUT								CLK to 7 OUT							
		120											OUT						CLK to 8 OUT							
		121												OUT					CLK to 9 OUT							
		122																	CLK to CARRY OUT							
t_{PHL9}	3003 Fig. 6	123			OUT														CLR to 0 OUT							
		124		OUT		OUT													CLR to 1 OUT							
		125							IN										CLR to 2 OUT							
		126										OUT							CLR to 3 OUT							
		127																	CLR to 4 OUT							
		128	OUT																CLR to 5 OUT							
		129					OUT												CLR to 6 OUT							
		130						OUT											CLR to 7 OUT							
		131									OUT								CLR to 8 OUT							
		132											OUT						CLR to 9 OUT							
		133												OUT					CLR to CARRY OUT							
t_{PLH9}		134			OUT														CLR to 0 OUT							
		135		OUT		OUT													CLR to 1 OUT							
		136							IN										CLR to 2 OUT							
		137										OUT							CLR to 3 OUT							
		138																	CLR to 4 OUT							
		139	OUT																CLR to 5 OUT							
		140					OUT												CLR to 6 OUT							
		141						OUT											CLR to 7 OUT							
		142									OUT								CLR to 8 OUT							
		143											OUT						CLR to 9 OUT							
		144												OUT					CLR to CARRY OUT							

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 10 - Continued.

Symbol	MIL-STD-883 method	Case E, F, X	Terminal conditions 17																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 14/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	5 OUT	1 OUT	0 OUT	2 OUT	6 OUT	7 OUT	3 OUT	GND	8 OUT	4 OUT	9 OUT	CARRY OUT	CLOCK ENABLE	CLK	CLR	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL4}	3003 Fig. 6	145																								
		146																								
		147																								
		148																								
		149																								
		150																								
		151																								
		152																								
		153																								
		154																								
		155																								
t_{PLH4}		156																								
		157																								
		158																								
		159																								
		160																								
		161																								
		162																								
		163																								
		164																								
		165																								
		166																								
f_{MAX} 15/	3003 Fig. 6	167																								
t_{THL}	3004 Fig. 6	168																								
		169																								
		170																								
		171																								
		172																								
		173																								
		174																								
		175																								
		176																								
		177																								
		178																								
t_{TLH}	3004 Fig. 6	179																								
		180																								
		181																								
		182																								
		183																								
		184																								
		185																								
		186																								
		187																								
		188																								
		189																								
		190																								

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TABLE III. Group A inspection for device type 11.

Symbol	MIL-STD-883 method	Case	Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 TC = +25°C Min	Subgroup 1 TC = +25°C Max	Subgroup 2 TC = +125°C Min	Subgroup 2 TC = +125°C Max	Subgroup 3 TC = -55°C Min	Subgroup 3 TC = -55°C Max	Unit
$V_{IC(pos)}$		1 2												1 mA					CLK CLR	CLK CLR	1.5 1.5						V
$V_{IC(neg)}$		3 4												-1 mA					CLK CLR	CLK CLR	-1.5 -1.5						V
I_{CCH} I_{CCL} I_{CCZ} I_{CCD} I_{CCR}	3005	6 7 8 9 10												6.0 V GND 1/ GND 6.0 V					Vcc Vcc Vcc Vcc Vcc	Vcc Vcc Vcc Vcc Vcc	0.2 0.2 0.2 0.2 0.2		20 20 20 20 20				μA μA μA μA μA
V_{OH3}	3006 1/	11 12 13 14 15 16 17 18 19 20 21 22				-20 μA			-20 μA			-20 μA		1/			-20 μA		-20 μA	Q12 Q13 Q14 Q5 Q6 Q7 Q4 Q1 CLK Q9 Q8 Q10 Q11	2.96		2.96		2.96		V
V_{OH5}	3006 1/	23 24 25 26 27 28 29 30 31 32 33 34				-5.2 mA			-5.2 mA			-5.2 mA		1/			-5.2 mA		-5.2 mA	Q12 Q13 Q14 Q5 Q6 Q7 Q4 Q1 CLK Q9 Q8 Q10 Q11	2.48		2.2		2.48		V
V_{OL3}	3007 1/	35 36 37 38 39 40 41 42 43 44 45 46				20 μA			20 μA			20 μA		1/			20 μA		20 μA	Q12 Q13 Q14 Q5 Q6 Q7 Q4 Q1 CLK Q9 Q8 Q10 Q11		0.05		0.05		0.25	V
V_{OH5}	3007 1/	47 48 49 50 51 52 53 54 55 56 57 58				5.2 mA			5.2 mA			5.2 mA		1/			5.2 mA		5.2 mA	Q12 Q13 Q14 Q5 Q6 Q7 Q4 Q1 CLK Q9 Q8 Q10 Q11		0.25		0.4		0.25	V

See footnotes at end of table III.

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FIGURE 2-3. Acceptance criteria for radiographic inspection - continued.

TABLE III. Group A inspection for device type 11 - Continued.

Symbol	MIL-STD-883 method	Case C, F, K	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Max	Subgroup 2 T_C = +125°C Min	Max	Subgroup 3 T_C = -55°C Min	Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	Q12	Q13	Q14	Q6	Q5	Q7	Q4	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}								
I_{IH}	3010	59 60								GND -		6.0 V GND	GND 6.0 V					6.0 V -	CLK CLR		0.05 0.05	0	3.2 0.1			mA -
I_{IL}	3009	61 62								- -		GND 0.0 V	6.0 V GND					- -	CLK CLR		-0.05 -0.05		-0.1 -0.1			- -
I_{OSD}	3011 9/	63 64 65 66 67 68 69 70 71 72 73 74	GND	GND	GND	GND	GND	GND	GND	- - - - - - - - - - - -	GND	15 - - - - - - - - - - -	15 - - - - - - - - - - -	GND	GND	GND	GND	4.5 V - - - - - - - - - - -	61 62 63 64 65 66 67 68 69 610 611 612 613 614	-10 - - - - - - - - - - -	-120 - - - - - - - - - - -	-10 - - - - - - - - - - -	-120 - - - - - - - - - - -	-10 - - - - - - - - - - -	-120 - - - - - - - - - - -	mA - - - - - - - - - - -
																				Subgroup 4 T_C = +25°C Min	Max					
C_C	3012	75 76								GND GND		9/	9/					GND GND	CLK CLR		15 15					pF pF
																			Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 8 T_C = -55°C Min	Max	
Truth table tests 29/	3014	77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100	L H L H L H L H L H L H L H L H L H L H L H L H	L L H H L L H H L L H H L L H H L L H H L L H H	L L L L H H H H L L L L H H H H L L L L H H H H	L L L L L L L L H H H H H H H H L L L L L L L L	L L L L L L L L L L L L L L L L H H H H H H H H	L L L L L L L L L L L L L L L L L L L L H H H H	L L L L L L L L L L L L L L L L L L L L L L L L	L L L L L L L L L L L L L L L L L L L L L L L L	GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND GND	L L L L L L L L L L L L L L L L L L L L L L L L	GND GND 4.5 V 4.5 V GND GND 4.5 V 4.5 V GND GND 4.5 V 4.5 V GND GND 4.5 V 4.5 V GND GND 4.5 V 4.5 V GND GND 4.5 V 4.5 V	4.5 V GND -	L L H H L L H H L L H H L L H H L L H H L L H H	L L L L H H H H L L L L H H H H L L L L H H H H	L L L L L L L L H H H H H H H H L L L L L L L L	L L L L L L L L L L L L L L L L H H H H H H H H	L L L L L L L L L L L L L L L L L L L L L L L L	4.5 V -	Each output							

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TABLE III. Group A inspection for device type 11 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions	Measured terminal	Test limits						Unit
					Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
					Min	Max	Min	Max	Min	Max	
Truth table tests 2/	3014			Each output							
I_{CCH1}	3003 Fig. 6										
I_{CCH11}	3003 Fig. 6										

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TABLE III. Group A inspection for device type 11 - Continued.

Symbol	MIL-STD-883 method	Case E,F,K	Case 2	Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 9 T_C = +25°C 14/ Min	Max	Subgroup 10 T_C = +125°C Min	Max	Subgroup 11 T_C = -55°C 14/ Min	Max	Units
t_{PHL1}	3003 Fig. 6			162	IN	Q12	Q13	Q14	Q6	Q5	Q7	GND	Q4	GND	CLR	Q1	Q9	Q8	Q10	Q11	V_{CC}	Q1 to Q4 Q5 to Q8 Q9 to Q12 Q13 to Q16	4	22	4	22	6	22	ns
				163																									
				164																									
				165																									
				166																									
				167																									
				168																									
				169																									
				170																									
				171																									
				172																									
t_{PHL2}	3003 Fig. 6			173	IN	Q12	Q13	Q14	Q6	Q5	Q7	GND	Q4	GND	CLR	Q1	Q9	Q8	Q10	Q11	V_{CC}	Q1 to Q4 Q5 to Q8 Q9 to Q12 Q13 to Q16	6	42	6	42	6	42	ns
				174																									
				175																									
				176																									
				177																									
				178																									
				179																									
				180																									
				181																									
				182																									
				183																									
				184																									
t_{PHL3}	3004 Fig. 6			185	IN	Q12	Q13	Q14	Q6	Q5	Q7	GND	Q4	GND	CLR	Q1	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14	3	15	3	15	3	15	ns
				186																									
				187																									
				188																									
				189																									
				190																									
				191																									
				192																									
				193																									
				194																									
				195																									
				196																									
t_{PLH}	3005 Fig. 6			197	IN	Q12	Q13	Q14	Q6	Q5	Q7	GND	Q4	GND	CLR	Q1	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14	4	22	4	22	6	22	ns
				198																									
				199																									
				200																									
				201																									
				202																									
				203																									
				204																									
				205																									
				206																									
				207																									
				208																									
t_{PLH}	3003 Fig. 6			209	IN	Q12	Q13	Q14	Q6	Q5	Q7	GND	Q4	GND	CLR	Q1	Q9	Q8	Q10	Q11	V_{CC}	Q1 to Q4 Q5 to Q8 Q9 to Q12 Q13 to Q16	21		21		20		ns

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TABLE III. Group A inspection for device type 12.

Symbol	MIL-STD-883 method	Case C, 9	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	6	8	9	10	11	13	14	16	18	19	20								
		Test no.	CLK	CLR	Q7	Q6	Q5	Q4	GND	NC	Q3	NC	Q2	Q1	NC	V_{CC}								
V_{IC}(pos)		1 2	1 mA	1 mA					1/ 1/							GND GND	CLK CLR		1.5 1.5					V
V_{IC}(neg)		3 4	-1 mA	-1 mA					GND							1/ 1/	CLK CLR		-1.5 -1.5					V
I_{CCH} I_{CCL} I_{CCL} 2/ I_{CCL} I_{CCL} 2/ I_{CCH}	3005	5 6 7 8 9 10	GND GND PI GND PI 6.0 V	6.0 V GND												6.0 V	V_{CC} V_{CC} GND V_{CC} GND V_{CC}		0.2 0.2 0.2 0.2		20 20 20 20			µA µA µA µA
V_{OH3}	3006 3/	11 12 13 14 15 16 17	IN	IN	-20 µA	-20 µA	-20 µA	-20 µA			-20 µA		-20 µA	-20 µA			Q1 Q2 Q3 Q4 Q5 Q6 Q7	5.95		5.95		5.95		V
V_{OH6}	3006 3/	18 19 20 21 22 23 24			-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA			-5.2 mA		-5.2 mA	-5.2 mA			Q1 Q2 Q3 Q4 Q5 Q6 Q7	5.48		5.2		5.48		V
V_{OL3}	3007 4/	25 26 27 28 29 30 31			20 µA	20 µA	20 µA	20 µA			20 µA		20 µA	20 µA			Q1 Q2 Q3 Q4 Q5 Q6 Q7		0.05		0.05		0.05	V
V_{OL6}	3007 4/	32 33 34 35 36 37 38			5.2 mA	5.2 mA	5.2 mA	5.2 mA			5.2 mA		5.2 mA	5.2 mA			Q1 Q2 Q3 Q4 Q5 Q6 Q7		0.26		0.4		0.26	V
I_{IH}	3010	39 40	6.0 V GND	GND 6.0 V													CLK CLR		0.05 0.05		0.1 0.1			µA
I_{IL}	3009	41 42	GND 6.0 V	6.0 V GND													CLK CLR		-0.05 -0.05		-0.1 -0.1			mA
I_{OS4} 5/	3011	43 44 45 46 47 48 49	IN	IN	GND	GND	GND	GND			GND		GND	GND		4.0 V	Q1 Q2 Q3 Q4 Q5 Q6 Q7	-10	-120	-10	-120	-10	-120	mA

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TABLE III. Group A inspection for device type 12 - Continued.

Symbol	MIL-STD-883 method	Case C_s 8	Terminal conditions 17/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgroup 4 T_C = +25°C Min	Max	Unit
		2	2	3	4	6	8	9	10	11	13	14	16	18	19	20				
		Test No.	CLK	CLR	Q7	Q6	Q5	Q4	GND	NC	Q3	NC	Q2	Q1	NC	V_{CC}				
C_C	3012	80 11	8/	8/					GND GND							GND GND	CLK CLR		18 18	pF pF

																	Measured terminal 9/	Subgroup 7 T_C = +25°C Min	Max	Subgroup 8 T_C = +125°C Min	Max	Subgroup 9 T_C = -55°C Min	Max	
Truth table tests 17/	3014	52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84	GND GND 4.5 V GND GND 4.5 V GND GND 4.5 V

TABLE III. Group A inspection for device type 12 - Continued.

Symbol	MIL-STD-883 method	Case C, D	Case 2	Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgroup 9 T_C = +25°C Min	Subgroup 9 T_C = +25°C Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 T_C = +125°C Max	Subgroup 11 T_C = -55°C Min	Subgroup 11 T_C = -55°C Max	Unit
t_{PHL1}	3003				CLK		$\overline{Q7}$	$\overline{Q4}$	$\overline{Q5}$	$\overline{Q6}$	GND		$\overline{Q3}$		$\overline{Q2}$	Q1				V_{CC} = 4.5 V						ns
t_{PHL2}	3003																									
t_{PHL11}	3003																									
t_{PHL111}	3003																									
t_{PHL9}	3003																									
t_{TLH}	3004																									
t_{THL}	3004																									
t_{PHL} 11/	3003																									

11/ See footnotes at end of table III.

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TABLE III. Group A inspection for device type 13.

Symbol	MIL-STD-883 method	Case E_2F_2	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}								
$V_{IC(pos)}$		1 2								1/ 1/								GND GND	CLK CLR		1.5 1.5					V V
$V_{IC(neg)}$		3 4								GND								1/ 1/	CLK CLR		-1.5 -1.5					V V
I_{CCH} I_{CCL} I_{CCL} 29/ I_{CCL} I_{CCL} 29/ I_{CCH}	3005	5 6 7 8 9 10										GND GND P3 GND P3 6.0 V	6.0 V GND					6.0 V	V_{CC} V_{CC} None V_{CC} None V_{CC}		0.2 0.2 0.2 0.2		20 20 20 20			μA μA μA μA
V_{OH3}	3006 2/	11 12 13 14 15 16 17 18 19 20 21 22	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA	-20 μA		-20 μA	18	18	-20 μA	-20 μA	-20 μA	-20 μA		01 02 03 04 05 06 07 08 09 010 011 012	3.95		3.95		3.95		V
V_{OH5}	3006 2/	23 24 25 26 27 28 29 30 31 32 33 34	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA		-5.2 mA			-5.2 mA	-5.2 mA	-5.2 mA	-5.2 mA		01 02 03 04 05 06 07 08 09 010 011 012	3.48		3.2		3.48		V
V_{OL3}	3007 2/	35 36 37 38 39 40 41 42 43 44 45 46	20 μA	20 μA	20 μA	20 μA	20 μA	20 μA	20 μA		20 μA			20 μA	20 μA	20 μA	20 μA		01 02 03 04 05 06 07 08 09 010 011 012		0.05		0.05		0.05	V
V_{OL5}	3007 2/	47 48 49 50 51 52 53 54 55 56 57 58	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA	5.2 mA		5.2 mA			5.2 mA	5.2 mA	5.2 mA	5.2 mA		01 02 03 04 05 06 07 08 09 010 011 012		0.26		0.4		0.26	V

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TABLE III. Group A inspection for device type 11 - Continued.

Symbol	MIL-STD-883 method	Case E,F,X	Case 2	Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgroups 1, 7 T_C = +25°C Min	Max	Subgroups 2, 8 T_C = +125°C Min	Max	Subgroups 3, 9 T_C = -55°C Min	Max	Unit
I_{IH}	3010	E0	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}									µA
I_{IL}	3009	E1	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}									µA
I_{OS4}	3011	E0	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E1	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E2	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E3	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E4	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E5	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E6	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E7	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E8	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	E9	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	EA	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	EB	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	EC	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	ED	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	EE	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	EF	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{OS4}	3011	EX	2	1	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	-10 -120	0.05 0.05	-10 -120	-0.1 -0.1	-10 -120	-0.1 -0.1	mA
I_{CC}	3012	E0	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E1	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E2	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E3	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E4	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E5	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E6	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E7	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E8	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	E9	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	EA	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	EB	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	EC	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	ED	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	EE	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	EF	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
I_{CC}	3012	EX	2	77	H	H	H	H	H	H	H	GND	H	GND	GND	H	H	H	H	4.5 V	Each output								V
Truth table tests 2/	3014	E0	2	101	L	L	L	L	L	L	L	GND	L	GND	GND	L	L	L	L	4.5 V	Each output								V
Truth table tests 2/	3014	E1	2	102	L	L	L	L	L	L	L	GND	L	GND	GND	L	L	L	L	4.5 V	Each output								V
Truth table tests 2/	3014	E2	2	103	L	L	L	L	L	L	L	GND	L	GND	GND	L	L	L	L	4.5 V	Each output								V
Truth table tests 2/	3014	E3	2	104	L	L	L	L	L	L	L	GND	L	GND	GND	L	L	L	L	4.5 V	Each output								V

See footnotes at end of table III.

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Table III. Group A inspection for device type 13 - Continued.

Symbol	MIL-STD-883 method	Case E,F,X	Case 2	Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits: Subgroup 9 T_C = +25°C Min	Subgroup 9 T_C = +25°C Max	Subgroup 10/11 T_C = +125°C Min	Subgroup 10/11 T_C = +125°C Max	Subgroup 11 T_C = -55°C 10/ Min	Subgroup 11 T_C = -55°C 10/ Max	Unit
				Truth table tests 2/																								
Test 1	3014			105																	Each output							
				106																								
				107																								
				108																								
				109																								
				110																								
				111																								
				112																								
				113																								
				114																								
				115																								
				116																								
				117																								
				118																								
				119																								
				120																								
				121																								
				122																								
				123																								
				124																								
				125																								
Level I	3003 Fig. 6			126																	Each output							
				127																								
				128																								
				129																								
				130																								
				131																								
				132																								
				133																								
				134																								
				135																								
				136																								
Level II	3003 Fig. 6			137																	Each output							
				138																								
				139																								
				140																								
				141																								
				142																								
				143																								
				144																								
				145																								
				146																								
				147																								
				148																								
				149																								
				150																								
Level III	3003 Fig. 6			151																	Each output							
				152																								
				153																								
				154																								
				155																								
				156																								
				157																								
				158																								
				159																								
				160																								
				161																								
				162																								

See footnotes at end of table III.

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TABLE III. Group A inspection for device type 13 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, I	Terminal conditions 17																Reserved terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_C = +25°C 16/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 16/		
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	Q12	Q6	Q5	Q7	Q4	Q3	Q2	GND	Q1	CLK	CLR	Q9	Q8	Q10	Q11	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLH11}	3003 Fig. 6	159 160 161 162 163 164 165 166 167 168 169	OUT	OUT 19	OUT 18	OUT 17	OUT 16	OUT 15	OUT 14	•	IN	IN	•	OUT 19	OUT 18	OUT 19	OUT 19	4.5 V	Q1 to Q2 Q2 to Q3 Q3 to Q4 Q4 to Q5 Q5 to Q6 Q7 to Q8 Q8 to Q9 Q9 to Q10 Q10 to Q11 Q11 to Q12	4	22	4	29	4	22	ns
t_{PHL9}	3003 Fig. 6	170 171 172 173 174 175 176 177 178 179 180 181	OUT	OUT	OUT	OUT	OUT	OUT	OUT	•	OUT	•	IN	OUT	OUT	OUT	OUT	•	CLR to Q1 CLR to Q2 CLR to Q3 CLR to Q4 CLR to Q5 CLR to Q6 CLR to Q7 CLR to Q8 CLR to Q10 CLR to Q11 CLR to Q12	6	42	6	56	6	42	ns
t_{THL}	3004 Fig. 6	182 183 184 185 186 187 188 189 190 191 192 193	OUT	OUT	OUT	OUT	OUT	OUT	OUT	•	OUT	•	GND	OUT	OUT	OUT	OUT	•	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12	3	18	3	28	3	18	ns
t_{TLH}	3004 Fig. 6	194 195 196 197 198 199 200 201 202 203 204 205	OUT	OUT	OUT	OUT	OUT	OUT	OUT	•	OUT	•	•	OUT	OUT	OUT	OUT	•	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12							ns
f_{MAX} 11/	3003 Fig. 6	206								•	OUT	•	•					•	μA	40		21		20		MHz

See footnotes at top of next page.

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1/ Input pins not designated shall be "high" level logic or "low" level logic, or may be left open provided they do not influence the outcome of the measurement. Output pins not designated shall be tied to the loads or left open provided they do not influence the outcome of the measurement. Exceptions are as follows:

a. $V_{IC(pos)}$ tests, the GND terminal shall be open and the minimum limit of 0.4 V applies only where test equipment limitations do not allow for the GND pin to be open during test.

b. $V_{IC(neg)}$ tests, the V_{CC} terminal shall be open.

c. I_{CC} tests, the output terminals shall be open.

2/ Apply sufficient clock pulses (see 28/) to set outputs 'high'.

3/ Apply 1.2 V for device types 01 and 03; apply 4.2 V for device types 02 and 04.

4/ Apply sufficient  pulses to set outputs 'low' prior to test.

5/ Apply sufficient  pulses to set outputs 'high' prior to test.

6/ Apply GND for device types 01 and 03; apply 4.0 V for device types 02 and 04.

7/ For device types 01 and 03 set outputs to count of nine (QA = 1, QD = 1, QB = 0, QC = 0) prior to measurement. For device types 02 and 04, set outputs to count of fifteen (QA = QB = QC = QD =1) prior to test.

8/ See 4.4.1c.

9/ Only summary attributes data is required.

10/ Tests performed in sequence.

11/ A = 3.7 V; B = 0.4 V.

12/ X = indeterminate output voltage.

13/ Output shall be: high $\geq$ 2.5 V; low < 2.5 V; $|I_{OUT}|$ = 20 µA.

14/ See 4.4.1d.

15/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half the input frequency. The f_{MAX} requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.

16/ For device types 01 and 02, apply one clock pulse prior to test. For device types 03 and 04, apply one clock pulse prior to test and another clock pulse during test.

17/ Apply a positive going pulse,  4.2 V minimum/6.0 V maximum

18/ Apply  one pulse while load is 'low' and another pulse after load has gone 'high' (during test).

19/ Apply  pulse prior to test.

20/ Apply  pulse during test.

21/ Apply  pulse during test.

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22/ Apply a positive going pulse, 3.7 V minimum/4.5 V maximum / 0.0 V to 0.9 V low

23/ Apply sufficient pulses (see 5/) to cause output to transition.

24/ Apply 4.2 V minimum/6.0 V maximum / 0.0 V to 0.9 V prior to test after clear pulse.

25/ Apply 4.2 V minimum/6.0 V maximum / 0.0 V to 0.9 V pulse prior to test.

26/ Apply 2 pulses prior to test after clear pulse (see 24/).

27/ Apply 4 pulses prior to test after clear pulse (see 24/).

28/ Apply sufficient 4.2 V minimum/6.0 V maximum / 0.0 V to 1.2 V pulses to set output 'high' prior to test.

29/ This I_{CC} and functional tests shall be performed in the test number sequence shown with no intervening changes to terminal conditions. During the functional test, input terminals designated "PA", "PB", etc., shall have applied thereto a specified number of single pulses with the following parameters: Pulse amplitude = V_{CC} maximum to V_{CC}, 4 percent minimum. For I_{CC} tests, input terminal is as designated. These pulses are enumerated as follows:

Symbol	Pulses	Symbol	Pulses	Symbol	Pulses	Symbol	Pulses
PA	1	PF	7	PK	85	PS	2047
PB	2	PG	15	PL	127	PT	4095
PC	3	PH	31	PM	255	PU	5461
PD	4	PI	42	PN	511	PV	8191
PE	5	PJ	63	PR	1023	PY	10922

During the functional tests, device output voltages are: don't care "X", high "H", or low "L" as specified in the terminal conditions columns. The output voltage limits over the specified temperature range are "H" $\geq$ 2.5 V; "L" $\leq$ 2.5 V $|I_{OUT}|$ = 20 µA maximum. Only a summary of attribute data is required.

30/ f_{MAX} minimum limits specified are the frequency at the inputs. The output frequency shall be one-half the input frequency when measured at Q_A. The output frequency shall be one-fifth the input frequency when measured at Q_D.

31/ Apply input test parameters such that t_{su}, t_{hold}, t_{rem}, and t_w values are not greater than the recommended operating minimums (see 1.4). Preset outputs to required state if necessary, prior to test.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 herein and as shown on figure 3 (note 3), or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within +10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at +25°C.

Parameter 1/	Device types
	All
I_{CC}	±30 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5.3 Quiescent supply current (I_{CC} test). When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be applied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.5).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2c).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.4), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_{IN}	Input terminal-to-GND capacitance.
GND	Ground zero voltage potential.
I_{CC}	Quiescent supply current.
T_A	Free air temperature.
V_{CC}	Positive supply voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54HC160
02	54HC161
03	54HC162
04	54HC163
05	54HC191
06	54HC192
07	54HC193
08	54HC390
09	54HC393
10	54HC4017
11	54HC4020
12	54HC4024
13	54HC4040

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended.

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and operator.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OC, SH

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0846)

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(See Instructions – Reverse Side)

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MIL-M-38510/663	MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS, COUNTERS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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